

Deep Learning for Search and Matching Models*

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Abstract

We develop a new method to globally solve and estimate search and matching models with aggregate shocks and heterogeneous agents. We characterize general equilibrium as a high-dimensional partial differential equation with the distribution as a state variable. We then use deep learning to solve the model and estimate economic parameters using the simulated method of moments. This allows us to study a wide class of search markets where the distribution affects agent decisions and compute variables (e.g. wages and prices) that were previously unattainable. In applications to labor search models, we show that distributional feedback plays an important role in amplification and that positive assortative matching weakens in prolonged expansions, disproportionately benefiting low-wage workers.

Keywords: Search and Matching, Distributional Feedback, Two-sided Heterogeneity, Business Cycles, Sorting, Over-the-Counter Financial Markets, Deep learning.

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1 Introduction

Heterogeneity in markets with search frictions is important for explaining economic phenomena such as wage dispersion, sorting, job ladders, and asset liquidity (e.g. [Burdett and Mortensen, 1998](#); [Shimer and Smith, 2000](#); [Postel-Vinay and Robin, 2002](#); [Duffie et al., 2005](#)). These markets feature volatile distribution dynamics over business cycles, during major crises, and in response to policy reforms (e.g. [Moscarini and Postel-Vinay, 2023](#); [Chen et al., 2017](#)). However, modeling equilibrium in these markets has proven technically challenging because optimizing agents need to understand and forecast time-varying distributions. To ensure tractability, many researchers have simplified the problem by limiting how distributions impact agent decisions (e.g. the “block-recursive” equilibria in [Lagos and Rocheteau, 2009](#); [Menzio and Shi, 2011](#); [Lise and Robin, 2017](#)).¹ Developments in the deep learning literature have opened up the possibility of relaxing these restrictions. In this paper, we present a general formulation of heterogeneous agent search and matching models with aggregate shocks as a collection of high-dimensional partial differential equations (PDEs) with the agent distribution explicitly as a state variable. We then develop the first deep learning method for solving these models and estimating economic parameters using the simulated method of moments. We use our method to study foundational models in labor search that have never been solved in their general form. This allows us to understand the interactions between aggregate shocks and heterogeneity during crises and business cycles.

We focus on models with the following features. The economy is populated by heterogeneous workers and firms that can be matched or unmatched. Matches generate utility that depends on the idiosyncratic types of the matched worker and firm, as well as on an exogenous aggregate variable that follows a continuous-time Markov chain. Unmatched agents engage in random search to meet each other. Upon meeting, they choose whether to accept the match and then bargain over the division of the matching surplus. We show that the equilibrium for this economy can be characterized recursively with a state space consisting of the exogenous aggregate variable and the endogenous distribution of matches across types in the economy. The match distribution impacts agent decisions because the opportunity cost of ac-

¹Following [Menzio and Shi \(2011\)](#) and [Lise and Robin \(2017\)](#), we call an equilibrium “block-recursive” if the agents’ value and policy functions are independent of the endogenous distribution of agents across their idiosyncratic states.

cepting a match depends upon which other agents are looking for matches and which agents are available to match. The equilibrium is then characterized by a “master” PDE for the surplus function, which includes high dimensional derivatives capturing the impact of distributional changes on surplus.

In Section 2, we propose a deep learning method for solving this class of PDEs and estimating economic parameters using the simulated method of moments, which we refer to as DeepSAM. To solve the model, we approximate the surplus function with a neural network (NN) and then train the neural network parameters to minimize the average loss in the PDE on a collection of sample points. We estimate the model using the simulated method of moments by building on the state-of-the-art practices in the deep learning literature. Instead of solving the model repeatedly across different economic parameter values, we introduce these parameters as pseudo-state variables and solve the resulting “extended master” equations using deep learning. This provides an explicit solution to the problem over a large range of structural parameter values. We then use this solution to build a surrogate model mapping structural parameters to simulated moments that we utilize for estimation.

To our knowledge, we are the first to apply deep learning to study search and matching models, which requires resolving a number of technical challenges. First, unlike some Walrasian market models (e.g. [Krusell and Smith, 1998](#)), the distribution does not enter the agent optimization problem only through aggregate prices. Instead, there is complicated feedback between the shape of the distribution and agent decisions because agents need to forecast who they will meet. Second, the free entry condition in search models typically requires solving a non-linear fixed point problem at each training step. Third, the algorithm must be stable across unusually shaped surplus functions and the economic parameter space. We resolve these problems by developing novel sampling techniques that help train the model in the economically meaningful state space, deriving an explicit representation of the firm distribution, and gradually introducing curvature into the solution algorithm.

In the main text we deploy our method in two “canonical” labor search models. In each case, we relax assumptions previously imposed for tractability and study the general environment.

In Section 3, we solve a labor market search model with two-sided heterogeneity that has been extended to include aggregate crisis shocks. This model can be thought of as the [Shimer and Smith \(2000\)](#) model with aggregate shocks. Even though this

environment is a key building block in the literature, our method is the first to be able to solve the model without simplifying assumptions. We verify the accuracy of our solution in several ways. In our full model with aggregate shocks, we show that the out-of-sample numerical root mean square error (RMSE) in the master equation is on the order of 10^{-3} . In the simplified version of our model without aggregate shocks that can be solved with traditional methods, we show that the RMSE between our solution and the fixed point solution is also on the order of 10^{-3} .

We use our solution to study the role of distributional feedback during the COVID-19 pandemic. We calibrate the model to include a crisis state that generates the heterogeneous employment decline across worker skill groups and firm industries that occurred during COVID, as estimated by [Cajner et al. \(2020\)](#). We then compute the impulse response following a COVID shock and decompose the time path by comparing the results to the “restricted” dynamics when agents make matching decisions under the “myopic” belief that they are always at the ergodic mean of the match distribution. We find that in the full model, unemployment falls approximately 25% faster during the recovery than in the restricted model. This is because, in the full model, firms understand that COVID disproportionately increases unemployment among low-skilled workers, which leads them to forecast a higher opportunity cost of waiting for a high-skilled worker and so offer jobs to a wider range of workers. We also consider a counterfactual crisis where unemployment increases symmetrically across all agent types and show that for this case the restricted dynamics closely approximate the full dynamics. This illustrates that solving the model globally across the distribution is particularly important for understanding large, asymmetric shocks.

In Section 4, we study a labor market with on-the-job search (OJS) and endogenous separation. Our environment builds on the recent literature (e.g. [Lise and Robin, 2017](#) and subsequent papers) but relaxes the assumptions commonly imposed to achieve block recursivity. In particular, we consider a general class of OJS bargaining protocols that allow workers to possess positive bargaining power during all meetings (initial and during OJS) and vary the workers’ freedom to move and renegotiate. This nests commonly used OJS bargaining protocols such as [Postel-Vinay and Robin \(2002\)](#) and [Cahuc et al. \(2006\)](#) as well as continuous renegotiation protocols such as [Pissarides \(1994\)](#). We also explore a range of firm entry conditions from “short-lived” vacancies to slow firm entry.

We use our deep learning method to estimate the model to match empirical mo-

ments of the US labor market. The entire solution and estimation process takes approximately 3 hours, where the model is solved over the economic parameter space and simulated across 10,000 parameter combinations to build the surrogate model deployed for the simulated method of moments. Our estimated model finds support for the [Okun \(1973\)](#) hypothesis that low-wage workers benefit disproportionately from longer expansions. This occurs because there is counter-cyclical sorting in our model that is more pronounced as expansions and recessions are prolonged. During an expansion (recession) high-type workers become increasingly scarce (plentiful) so high-type firms become less (more) picky in their job offers. Consequently, longer expansions lead to greater weakening of positive assortative matching, which benefits low-skilled workers. This offers an explanation for Okun’s hypothesis through the feedback between distribution changes and acceptance decisions, a connection that is shut down in other models.

Our method allows us to study business cycle dynamics that have previously been too difficult to examine. One important example is the dynamics of the wage distribution. As pointed out by [Lentz et al. \(2017\)](#), even though the surplus function is block recursive in the commonly used [Lise and Robin \(2017\)](#) framework, the division of the surplus, i.e. the wage, is not. This means that wages necessarily depend upon the distribution and so cannot be computed with traditional methods. In our general OJS bargaining environment, the details of the bargaining protocol have a large impact on the wage process. At one extreme (e.g. [Pissarides, 1994](#)), every match with the particular worker and firm productivity has the same wage. At the other extreme (e.g. [Postel-Vinay and Robin, 2002](#)), wages are history dependent and so there is a distribution of wages across matches with a particular worker and firm productivity. We use DeepSAM to overcome these challenges and solve for the high dimensional wage function for different OJS bargaining protocols. Our solution finds that wages of low-skilled workers are more procyclical over the business cycle. We also use our method to revisit the implications of the assumptions required to generate block-recursivity, e.g. that unemployed workers have no bargaining power when meeting with firms. By solving the model over a wide range of worker bargaining power values, we show that unemployment and vacancies are more responsive to business cycle shocks when worker bargaining power is small, indicating that the block recursive assumptions are quantitatively important.

In the Online Appendix, we study a collection of extensions to our core models that

illustrate the breadth of our solution technique. In Appendix B.5, we solve a model with binary acceptance decisions that introduce kinks into the surplus function. This is a characterization that would be very challenging for perturbation approaches that rely on differentiability (e.g. [Reiter, 2009](#); [Bilal, 2023](#)). In Appendix D, we solve a model with stochastic idiosyncratic productivity similar to [Mortensen and Pissarides \(1994\)](#). This allows us to characterize how the endogenous separation rate moves with aggregate shocks. In Appendix E, we solve a model with slow firm entry so the mass of firms becomes an additional state variable. Finally, in a companion note, we solve an over-the-counter (OTC) bond market model with heterogeneous investors, different bond maturities, and aggregate default risk. This can be thought of as an extension to [Duffie et al. \(2005\)](#) and [Weill \(2008\)](#) that expands investor and asset heterogeneity and allows for aggregate risk.

Literature Review. Over the past three decades, there have been major advances in solving search and matching models with heterogeneity and aggregate risk. One advance is the Bertrand competition model of wage setting introduced in [Postel-Vinay and Robin \(2002\)](#) and deployed in many papers (e.g. [Cahuc et al., 2006](#); [Lise and Robin, 2017](#)). Another is the directed search block recursive structure introduced by [Moen \(1997\)](#); [Menzio and Shi \(2010, 2011\)](#). Both these approaches impose contracting and entry assumptions that ensure agent decisions are independent of the distribution of matches. Our paper relaxes these constraints and solves a general class of models where the distribution may impact agents’ decisions.

Our labor market models (Sections 3 and 4) are connected to recent papers studying business cycle dynamics in heterogeneous agent labor search models (e.g. [Schaal, 2017](#); [Mueller, 2017](#); [Krusell et al., 2017](#); [Moscarini and Postel-Vinay, 2018](#); [Fukui, 2020](#); [Baley et al., 2022](#); [Qiu, 2023](#); [Moscarini and Postel-Vinay, 2023](#); [Gregory et al., 2024](#)). Our DeepSAM approach offers a way to expand the range of models used in this literature by enriching agent heterogeneity, relaxing block recursivity, departing from perfect foresight, studying non-linear crisis dynamics, and potentially other extensions. We also show how to deploy deep learning to estimate these models.

We are part of a growing literature using deep learning methods to solve economic models. These techniques attempt to overcome the limitations of traditional solution techniques that typically use low dimensional non-linear projections (e.g. [Krusell and Smith, 1998](#)) or rely on perturbations that primarily capture local dynamics

and may not accurately capture the impact of kinks in the equilibrium functions (e.g. [Reiter, 2009](#); [Bilal, 2023](#)). These papers have focused on solving heterogeneous agent macroeconomic models with incomplete but Walrasian markets (e.g. [Azinovic et al., 2022](#); [Maliar et al., 2021](#); [Han et al., 2021](#); [Kahou et al., 2021](#); [Fernández-Villaverde et al., 2023](#); [Gopalakrishna, 2021](#); [Huang, 2022](#); [Gu et al., 2023](#); [Azinovic and Žemlička, 2023](#); [Duarte et al., 2024](#), among others, see the recent review by [Fernández-Villaverde, 2025](#); [Fernández-Villaverde et al., 2024](#)). Our contribution is to show how to undertake deep learning to solve search and matching models, which are workhorse models for a large literature in macroeconomics, monetary economics, and finance. What makes all macroeconomic models difficult to solve is that they involve a fixed point problem where agent optimization (the HJB equation) is connected to distribution evolution (the KF equation). For the Walrasian models that have typically been solved by the existing literature (e.g. [Krusell and Smith, 1998](#)), the fixed point problem is simplified because the distribution only affects agent decisions through prices. Many existing solution techniques have exploited this model structure to build tractable deep learning algorithms that use dimension reduction. What makes search and matching models particularly difficult to solve compared to many competitive incomplete market models (e.g. [Krusell and Smith, 1998](#)) is that the fixed point problem is particularly complicated because the shape of the distribution matters directly for the equilibrium. This is because, as summarized in Table 1, the distribution impacts agents’ decisions via the matching rates with other types rather than through aggregate prices. This imposes greater challenges on how we develop our numerical and sampling schemes to get an accurate solution.

	Distribution	How distribution affect agents’ decisions
HAM	Asset wealth and income	Via aggregate prices
SAM	Type (productivity) of agents in two sides of matching	Via matching process with other types

Table 1: Distribution impact in heterogeneous agent models (HAM) vs search and matching (SAM) models.

Our estimation approach builds on the literature integrating deep learning and structural estimation (particularly the introduction of structural parameters as pseudo-state variables), proposed by [Norets \(2012\)](#) and deployed in macroeconomics by [Fernández-Villaverde et al. \(2023\)](#); [Chen et al. \(2023\)](#); [Kase et al. \(2024\)](#); [Friedl et al.](#)

(2023); Duarte and Fonseca (2024). Our PDE formulation is related to the master equations in Bilal (2023); Alvarez et al. (2023) but we do not linearize the PDE. Finally, our training approach draws on the “Physics-informed neural networks” (PINN) literature (e.g. Raissi et al., 2019) in computational science.

The paper is structured as follows. Section 2 describes our DeepSAM methodology for solving and estimating a general class of search and matching models. Section 3 applies DeepSAM to solve a canonical labor market search model with two-sided heterogeneity and aggregate shocks, and studies the impact of the COVID-19 shock. Section 4 applies DeepSAM to estimate a model with on-the-job search and endogenous separation. Section 5 concludes.

2 Methodology

In this section, we outline an illustrative labor search and matching model with two-sided heterogeneity and aggregate shocks. Conceptually, this model extends the Shimer and Smith (2000) model by incorporating aggregate shocks. We use this environment to introduce our deep learning method for solving search and matching models. We then outline how to use deep learning to undertake efficient estimation to match simulated moments (a deep learning based “simulated method of moments”). Although we focus on a relatively stylized model to keep the exposition straightforward, in subsequent sections and appendices we develop a collection of extensions to demonstrate the generality of our techniques.

2.1 Environment

Setting: The economy is in continuous time with an infinite horizon. It is populated by a continuum of infinitely lived, heterogeneous agents. A unit measure of agents are workers indexed by type $x \in \mathcal{X}$ and a measure are firms indexed by type $y \in \mathcal{Y}$. We use workers and firms for notational convenience, but we could equivalently label the different agents as investors and financial dealers, or other market participants. Workers are either employed (e) in a match or unmatched (u). Firms are either producing (p) in a match or vacant (v). The distribution of matches between workers and firms is endogenous and determined by agent decisions. All agents have a discount rate ρ . The exogenous aggregate state of the economy is indexed by $z_t \in \mathcal{Z}$, which

follows a continuous time Markov chain with transition intensities $\lambda(z, \check{z})$ for all $z, \check{z}$.

Match utility: If a worker is unmatched (unemployed), they get flow utility b . If a firm is unmatched (vacant), then they must pay a flow cost c to operate. Workers match with firms but not with each other. If a worker of type x is matched with a firm of type y , then they generate utility $F(x, y, z)$, where F is increasing in each variable and twice differentiable with uniformly bounded first partial derivatives on $\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$.

Distributions: Let $g_t^w(x)$ denote the population measure function of workers, and let $g_t^f(y)$ denote the population measure function of firms. For expositional simplicity, here we focus on the case with an exogenous and time-invariant g^w and an entry condition below that determines g_t^f . Let $g_t(x, y)$ denote the measure function of matched workers and firms, $g_t^e(x)$ denote the measure function of matched workers, $g_t^u(x)$ denote the measure function of unmatched workers, $g_t^p(y)$ denote the measure function of producing firms, and $g_t^v(y)$ denote the measure function of vacant firms. The relationships between the functions are given in Table 2 below. We follow the literature and refer to g , g^e , g^u , g^p , g^v , g^w , and g^f as the “distributions” in the economy, even though they are technically neither densities nor cumulative distribution functions. We define the aggregate worker employment by $\mathcal{E}_t := \int g_t^e(x)dx$, aggregate worker unemployment by $\mathcal{U}_t := \int g_t^u(x)dx$, aggregate producing firms by $\mathcal{P}_t := \int g_t^p(y)dy$, aggregate vacant firms by $\mathcal{V}_t := \int g_t^v(y)dy$, and aggregate firms by $\mathcal{F}_t := \int g_t^f(y)dy$.

Description	Measure Function	Conditional Density
Matches	$g_t(x, y)$	
Employed workers	$g_t^e(x) = \int g_t(x, y)dy$	$g_t^e(x)/\mathcal{E}_t$
Unemployed workers	$g_t^u(x) = g_t^w(x) - g_t^e(x)$	$g_t^u(x)/\mathcal{U}_t$
Producing firms	$g_t^p(y) = \int g_t(x, y)dx$	$g_t^p(y)/\mathcal{P}_t$
Vacant firms	$g_t^v(y) = g_t^f(y) - g_t^p(y)$	$g_t^v(y)/\mathcal{V}_t$
Total firms	$g_t^f(y)$	$g_t^f(y)/\mathcal{F}_t$

Table 2: Summary of distributions.

Search and Matching Technology: Only and all unmatched workers engage in random search. A function $m : \mathbb{R}^+ \times \mathbb{R}^+ \rightarrow \mathbb{R}^+$, $(\mathcal{U}_t, \mathcal{V}_t) \mapsto m(\mathcal{U}_t, \mathcal{V}_t)$ takes the current level of unemployment and vacancies and generates meetings. The rate at which a worker meets a potential firm is $\mathcal{M}_t^u := m(\mathcal{U}_t, \mathcal{V}_t)/\mathcal{U}_t$, while the rate at which a vacant firm

meets a potential hire is $\mathcal{M}_t^v := m(\mathcal{U}_t, \mathcal{V}_t)/\mathcal{V}_t$. The rate at which a worker meets any firm in $Y \subset \mathcal{Y}$ equals $\mathcal{M}_t^u \int_Y (g_t^v(y)/\mathcal{V}_t) dy$, where $g_t^v(y)/\mathcal{V}_t$ is the firm density conditional on being vacant. The rate at which a firm meets any worker $x \in X \subset \mathcal{X}$ equals $\mathcal{M}_t^v \int_X (g_t^u(x)/\mathcal{U}_t) dx$, where $g_t^u(x)/\mathcal{U}_t$ is the worker density conditional on being unemployed. Matches are broken at exogenous rate $\delta(x, y, z)$ that potentially depends upon the idiosyncratic type and the aggregate state.

Surplus division: We impose that, upon matching, agents divide surplus using a bilateral bargaining protocol that allocates a fraction β of the surplus to workers. For example, this could be the worker’s weight in Nash bargaining, the inverse of the slope of the Pareto frontier in Kalai-Smorodinsky bargaining, or the relative rate of time preference in a Rubinstein alternating offer game.² The contract is implemented by providing a flow “wage” $w(x, y, z, g)$ to the worker and flow “profit” $F(x, y, z) - w(x, y, z, g)$ to the firm. However, as we will show, the wage is not allocative in the sense that we can determine equilibrium employment without solving for the wage.

Firm exit and entry: For our baseline formulation of the model, we allow firms to freely and instantaneously enter or exit the economy. Following [Hagedorn et al. \(2017\)](#), new firms enter the economy with a draw of y from the uniform distribution $U(0, 1)$. Once a firm enters the economy, they stay (either vacant or producing) until they exit. In addition to freely chosen exit, we assume that firms (producing and vacant) exit at an exogenous flow rate $\varsigma(z)$ and a fraction $\sigma(z; \check{z})$ of firms exit when z changes to $\check{z}$. For notational simplicity, in this Section we assume that firms and workers are fully compensated when firm exit occurs. We relax this assumption in Appendix A.1 and subsequent sections.

2.2 Model Extensions

We have attempted to keep our baseline model simple to help streamline the exposition of our algorithm. However, throughout the paper and the Appendix, we show

²Technically, we have the following mappings between bargaining problems. For Kalai-Smorodinsky bargaining $\beta = 1/(1 + \gamma)$ where $\gamma V^e + V^p = 1$ is the (weighted) Pareto frontier between the employed worker and producing firm value functions. For a Rubinstein alternating offer game, $\beta = r^W/(r^W + r^F)$ where r^W and r^F are the rate of time preference for workers and firms respectively and we have taken the limit as offers are made infinitely quickly.

that our approach can handle many variations. In Section 4, we extend our model to incorporate on-the-job-search (OJS) and endogenous match separation in the manner of [Lise and Robin \(2017\)](#). When solving the OJS model, we remove compensation for exit and consider two alternative specifications of firm entry/exit: instantaneous vacancies like in the original [Lise and Robin \(2017\)](#) paper (in Section 4.1) and slow firm entry (in Appendix E). In Appendix D, we introduce idiosyncratic productivity risk and endogenous match separation in a similar manner to [Mortensen and Pissarides \(1994\)](#). In Appendix B.5, we compare solutions for smoothed and binary agent decisions. In Appendix A.3, we present a discrete time version of the model to show that the equations look very similar.

Although we focus on labor models in this paper, our approach can be used to solve a wide range of search and matching environments. In a companion note, we solve an over-the-counter (OTC) bond market model with heterogeneous investors, different bond maturities, and aggregate default risk (an extension of [Duffie et al., 2005](#); [Weill, 2008](#)). Our approach could also be applied to solve models in the money-search literature (e.g. [Lagos and Wright, 2005](#)). Finally, we have used our approach to solve models with non-transferable utility (e.g. the marriage model in [Smith, 2006](#)).

2.3 Recursive Characterization of Equilibrium

We now define and characterize a recursive equilibrium. The aggregate states are (z, g) , where z is the aggregate productivity and g is the match measure function. We let $V^u(x, z, g)$ denote the value of an unemployed worker of type x , $V^e(x, y, z, g)$ denote the value of a worker of type x employed at a firm of type y , $V^v(y, z, g)$ denote the value of a vacancy for firm y , and $V^p(x, y, z, g)$ denote the value of a producing firm y employing a worker of type x . The surplus of a match is defined as:

$$S(x, y, z, g) := V^p(x, y, z, g) - V^v(y, z, g) + V^e(x, y, z, g) - V^u(x, z, g).$$

We let $\alpha(x, y, z, g)$ be an indicator function for whether a type- x worker and a type- y firm choose to accept a match at aggregate state (z, g) .

In principle, aggregate shocks pose technical challenges for free entry and exit in search models because they potentially lead to mathematically complex endogenous exit rates from among vacant firms. In this section, for notational simplicity, we describe the model assuming that $\varsigma(z)$ and $\sigma(z, \check{z})$ are sufficiently large that there

is no endogenous firm exit. In our quantitative application in Section 3 we show that this assumption holds under our calibration. In Appendix A.1 we describe the more mathematically complicated master equation with endogenous firm exit and parametrized compensation for exit.

2.3.1 Match Distribution Evolution

Given an agent match acceptance policy α , the match measure function g evolves by:

$$dg_t(x, y) = \mu_t^g(x, y)dt - \sum_{\check{z}_t \neq z_t} \sigma(z_t, \check{z}_t) g_t(x, y) dN_t(z_t; \check{z}_t) \quad (2.1)$$

where the endogenous drift of g is:

$$\mu_t^g(x, y) = -(\delta(x, y, z_t) + \varsigma(z_t))g_t(x, y) + \alpha(x, y, z_t, g_t)m(\mathcal{U}_t, \mathcal{V}_t)\frac{g_t^u(x)}{\mathcal{U}_t}\frac{g_t^v(y)}{\mathcal{V}_t}.$$

and $N_t(z_t; \check{z}_t)$ is a Poisson random variable such that $dN_t(z_t; \check{z}_t) = 1$ when z_t changes to $\check{z}_t$. Equation (2.1) is a Kolmogorov Forward (KF) equation written from the perspective of worker movements (it can be written equivalently from the point of view of the firms). The first term is the drift $\mu_t^g(x, y)$, which accounts for the outflow from the breakup of matches, the outflow from the exit of firms, and the inflow from the creation of new worker matches. The second term in equation (2.1) is a jump term that captures the exogenous exit of firms when $dN_t(z_t; \check{z}_t) = 1$, i.e., the aggregate state changes. It follows that the jumps in the match distribution are exogenous, while the drift is endogenous. Given g_t and the firm measure function g_t^f , we can recover the other measures from Table 2. So, the endogenous drift of the match distribution can be written as $\mu_t^g(x, y) = \mu^g(x, y, z_t, g_t)$, where:

$$\begin{aligned} \mu^g(x, y, z_t, g_t) = & -(\delta(x, y, z) + \varsigma(z_t))g_t(x, y) \\ & + \frac{m(\mathcal{U}_t, \mathcal{V}_t)}{\mathcal{U}_t \mathcal{V}_t} \alpha(x, y, z_t, g_t) \left(g^w(x) - \int g_t(x, y) dy \right) \left(g_t^f(y) - \int g_t(x, y) dx \right) \end{aligned} \quad (2.2)$$

2.3.2 Agent Hamilton-Jacobi-Bellman (HJB) Equations

We denote agents' belief about the endogenous drift of the match measure g by $\check{\mu}^g$, which the agents take as given but equals μ^g in equilibrium.

Surplus division and match acceptance: The Nash Bargaining protocol implies that the division of surplus is given by:

$$\begin{aligned}\beta S(x, y, z, g) &= V^e(x, y, z, g) - V^u(x, z, g) \\ (1 - \beta)S(x, y, z, g) &= V^p(x, y, z, g) - V^v(y, z, g).\end{aligned}\tag{2.3}$$

Optimizing workers and firms accept the match when the surplus is positive. So the matching acceptance function α is given by:

$$\alpha(x, y, z, g) := \begin{cases} 1, & \text{if } S(x, y, z, g) \geq 0 \\ 0, & \text{otherwise.} \end{cases}\tag{2.4}$$

The bargaining problem implicitly determines the wage $w(x, y, z, g)$. Following the literature, we assume that contract terms are continuously renegotiated to track the contracts of new hires so that $V^e(x, y, z, g) - V^u(x, z, g)$ is the same for all workers with a particular (x, y) , and $V^p(x, y, z, g) - V^v(y, z, g)$ is the same for all firms with a particular (x, y) . We discuss alternative arrangements in Subsection 4.1.

Workers: Given beliefs and optimal decisions, the worker value functions V^u and V^e satisfy the Hamilton Jacobi Bellman (HJB) Equations:

$$\begin{aligned}\rho V^u(x, z, g) &= b + \mathcal{M}^u \int \alpha(x, \tilde{y}, z, g) (V^e(x, \tilde{y}, z, g) - V^u(x, z, g)) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\ &+ \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (V^u(x, \tilde{z}, (1 - \sigma(z, \tilde{z}))g) - V^u(x, z, g)) + \langle D_g V^u, \check{\mu}^g \rangle\end{aligned}\tag{2.5}$$

$$\begin{aligned}\rho V^e(x, y, z, g) &= w(x, y, z, g) + \delta(x, y, z) (V^u(x, z, g) - V^e(x, y, z, g)) \\ &+ \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (V^e(x, y, \tilde{z}, (1 - \sigma(z, \tilde{z}))g) - V^e(x, y, z, g)) + \langle D_g V^e, \check{\mu}^g \rangle\end{aligned}\tag{2.6}$$

where for $j \in \{u, e\}$, $D_g V^j$ is the Frechet derivative of V^j with respect to g , and $\langle D_g V^e, \check{\mu}^g \rangle = \int \int [D_g V^j(x, y, z, g)](a, b) \check{\mu}^g(a, b, z, g) da db$ is the inner product.³⁴

The HJB equation for unemployed agents can be interpreted in the following way.

³We setup the model imposing Frechet differentiability. However, for numerical purposes, Gateaux derivatives are likely to be sufficient.

⁴The mean field game literature has studied the mathematical difficulties involved in defining a Frechet derivative for HJB equations with an infinite dimensional state (Cardaliaguet et al., 2019). It has not been established whether these characterizations hold for economic models. However, our numerical examples will always work with a finite type space where the derivatives are well defined.

The left-hand side of (2.5) is the flow value of being unemployed. On the right-hand-side, the first term is the flow utility unemployment benefit, the second term is the meeting rate multiplied by the expected gain in a meeting, the third term is the value function shift when the exogenous aggregate state changes (incorporating the distribution shift), and the final term governs how the value function is impacted by distribution changes. The HJB equation for an employed worker can be interpreted in a similar way. The first term on the right-hand-side of (2.6) is their wage from the match, the second term is the change in value when the match breaks up because the match dissolves, and the third term is the change in value when the aggregate shock occurs (incorporating the distribution shift to $(1 - \sigma(z, \tilde{z}))g$).

Firms: Given beliefs and optimal decisions, the firm value functions V^v and V^p satisfy the Hamilton Jacobi Bellman (HJB) Equations:

$$\begin{aligned} \rho V^v(y, z, g) = & -c + \mathcal{M}^v \int \alpha(\tilde{x}, y, z, g)(V^p(\tilde{x}, y, z, g) - V^v(y, z, g)) \frac{g^u(\tilde{x})}{\mathcal{U}} d\tilde{x} \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (V^v(y, \tilde{z}, (1 - \sigma(z, \tilde{z}))g) - V^v(y, z, g)) + \langle D_g V^v, \check{\mu}^g \rangle \end{aligned} \quad (2.7)$$

$$\begin{aligned} \rho V^p(x, y, z, g) = & F(x, y, z) - w(x, y, z, g) + \delta(x, y, z)(V^v(y, z, g) - V^p(x, y, z, g)) \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (V^p(x, y, \tilde{z}, (1 - \sigma(z, \tilde{z}))g) - V^p(x, y, z, g)) + \langle D_g V^p, \check{\mu}^g \rangle \end{aligned} \quad (2.8)$$

where again $D_g V^j$ is the Frechet derivative of V^j with respect to the measure function g for $j \in \{v, p\}$. The HJB equations for firms have a similar interpretation to those for workers. The direct impact of exogenous firm exit from ς and σ does not appear in equations (2.6), (2.7), or (2.8) because we assumed that workers and firms receive compensation upon exogenous firm exit. Equations (A.5), (A.6), and (A.7) in Appendix A.1 show the general HJB equations.

2.3.3 Free Entry and the Firm Distribution

For our baseline model, as in [Hagedorn et al. \(2017\)](#), we allow firms to freely enter or exit, which determines g_t^f as a jump variable that adjusts continuously so that free entry holds at all reachable aggregate states (z, g) . The free entry condition requires

that it is not profitable to post a vacancy:

$$0 = \mathbb{E}_{\tilde{y}}[V^v(\tilde{y}, z, g)] = \int_0^1 V^v(\tilde{y}, z, g) \bar{h}(\tilde{y}) d\tilde{y}, \quad (2.9)$$

where $\bar{h}(\tilde{y}) = 1$ is the density for firm type entry draws from $U(0, 1)$. At a jump in z , we use the timing convention that z changes, exogenous firm exit occurs, and then g_t^f adjusts to restore (2.9).

Combining the free-entry condition (2.9) with the surplus division (2.3) and the vacancy HJB equation (2.7) gives:

$$\frac{m(\mathcal{U}_t, \mathcal{V}_t)}{\mathcal{V}_t} = \frac{c}{\int \int \alpha(\tilde{x}, \tilde{y}, z_t, g_t) \frac{g_t^u(\tilde{x})}{\mathcal{U}_t} (1 - \beta) S(\tilde{x}, \tilde{y}, z_t, g_t) d\tilde{x} d\tilde{y}} \quad (2.10)$$

where $(\mathcal{U}_t, \mathcal{V}_t, g_t^u)$ are computed from g_t via Table 2. In the case of a Poisson jump in z , $(\mathcal{U}_t, \mathcal{V}_t, g_t^u)$ are computed after any exogenous exit has occurred. Conceptually, the matching rate depends upon the average surplus because new firms enter until it is no longer profitable to do so. We impose matching functions that are homothetic in $\mathcal{U}$ and $\mathcal{V}$ (as is common in the literature), so $\frac{m(\mathcal{U}, \mathcal{V})}{\mathcal{V}} = \widehat{m}\left(\frac{\mathcal{V}}{\mathcal{U}}\right)$ and $\mathcal{V}_t$ can be solved for explicitly from equation (2.10).⁵ Since firm y draws are uniformly distributed and firms cannot withdraw their vacancies after a bad draw of y , the firm type distribution g_t^f is uniform with mass $\mathcal{V}_t + \mathcal{P}_t$. That is: $g_t^f(y) = \mathcal{V}_t + \mathcal{P}_t$ for all y .

2.3.4 Equilibrium and Master Equation

Definition 1. A **recursive rational expectations equilibrium** is a collection of functions $\{V^u, V^e, V^v, V^p, w, \alpha, g^f\}$ of the state variables such that: (i) given beliefs about the evolution of g_t , $(V^u, V^e, V^v, V^p, \alpha)$ solve the HJB equations (2.5)-(2.8), (ii) the division of surplus satisfies (2.3), (iii) the firm distribution g^f adjusts at every (z, g) so the free entry condition (2.9) holds, and (iv) agent beliefs about the evolution of g_t are consistent in the sense that $\check{\mu}^g = \mu^g$, where μ^g is given by equation (2.2).

We follow the approach of [Lions \(2007-2011\)](#) and characterize the equilibrium in a combined PDE, which is often referred to as the “master equation” or “equilibrium PDE” for the economic system. Conceptually, a master equation imposes equilibrium belief consistency (and other equilibrium conditions) on the agent optimization equa-

⁵Otherwise, we can deploy a non-linear fixed point solver for $\mathcal{V}_t$ in the loop of our algorithm.

tions.⁶ For our specific problem, we can combine the equilibrium HJB equations to get the following self-contained master equation for the equilibrium surplus function:

$$\begin{aligned}
0 = \mathcal{L}^S S = & -\rho S(x, y, z, g) + F(x, y, z) - \delta(x, y, z)S(x, y, z, g) \\
& - b - \beta \frac{m(\mathcal{U}, \mathcal{V})}{\mathcal{U}} \int \alpha(x, \tilde{y}, z, g) S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\
& + c - (1 - \beta) \frac{m(\mathcal{U}, \mathcal{V})}{\mathcal{V}} \int \alpha(\tilde{x}, y, z, g) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}} d\tilde{x} \\
& + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) \left(S(x, y, \tilde{z}, (1 - \sigma(z, \tilde{z}))g) - S(x, y, z, g) \right) + \langle D_g S, \mu^g \rangle \quad (2.11)
\end{aligned}$$

where μ^g is given by (2.2), $(\mathcal{U}, \mathcal{V}, g^u, g^v)$ can be calculated by Table 2, g^f is pinned down by (2.9), and α is given by equation (2.4). Conceptually, the master equation says that the “flow surplus” $\rho S(x, y, z, g)$ from a match (x, y) is equal to the match output $F(x, y, z)$ minus the average rate at which surplus is lost from match break ups (the term $\delta S(x, y, z, g)$), minus the opportunity cost of matching for the workers (line 2), minus the opportunity cost of matching for firms (line 3), and plus the change in surplus when the aggregate states change (line 4).

Once we obtain $S(x, y, z, g)$ and $\alpha(x, y, z, g)$ by solving Equation (2.11), we can obtain $\{V^u, V^e, V^v, V^p, w\}$ by solving equations (2.5), (2.6), (2.7), (2.8) together with the surplus division equation (2.3).

To complete the recursive characterization, in Appendix A.2 we prove a collection of transversality conditions and show that the solution to a recursive equilibrium from Definition 1 is a sequential equilibrium.⁷ The essential feature of the proof is that the equilibrium surplus function satisfying (2.11) is bounded. We comment on, but do not prove, uniqueness in Appendix A.4.

2.4 Approximation with Finite Types

Our goal is to solve the master equation (2.11) numerically to obtain $S(x, y, z, g)$ and $\alpha(x, y, z, g)$. We can then solve for the value and wage functions using Equations

⁶Formally, let $V^j(s, z, g; \check{\mu}^g)$ denote the solution to the HJB equation for agent $j \in \{u, e, v, p\}$ with individual or match state s under belief $\check{\mu}^g$. Then, the equilibrium value function is $V^j(s, z, g; \mu^g)$, where μ^g is the equilibrium evolution of g solving the KF equation under equilibrium agent choices. The master equation is the PDE that characterizes the equilibrium value function $V^j(s, z, g; \mu^g)$.

⁷Kahou et al. (2025) explores the importance of transversality conditions for machine learning in macroeconomics and discusses conditions under which they are automatically satisfied by neural network solutions.

(2.3) to (2.8). Solving Equation (2.11) is difficult because the state space contains an infinite-dimensional variable, g , and so the master equation contains Frechet derivatives with respect to g . To make progress on this problem, we approximate g on a discretized type space so that equation (2.11) becomes a high, but finite-dimensional, partial differential equation that can be solved using deep learning.⁸

Discrete type space and KF equation: To represent g , we restrict the possible types to finite collections: $x \in \mathcal{X} = \{x_1, \dots, x_{n_x}\}$ and $y \in \mathcal{Y} = \{y_1, \dots, y_{n_y}\}$. We let $\underline{g}_t$ denote the vector of measures of matched agents at the points $(\mathcal{X}, \mathcal{Y})$, where $g_{t,ij} = g_t(x_i, y_j)$ denotes g evaluated at type (x_i, y_j) . We also let $g_{t,i}^w := g_t^w(x_i)$ and $g_{t,j}^f := g_t^f(y_j)$. The discretized aggregate state variables are $\{z, \underline{g}_t\}$: the aggregate productivity, and the vector of measures of agent matches. The Riemann approximation to the drift of the KF equation on the discretized type space is given by:

$$\begin{aligned} \mu^g(x_i, y_j, z_t, \underline{g}_t) = & -(\delta(x_i, y_j, z) + \varsigma(z_t))g_{t,ij} \\ & + \frac{m(z_t, \underline{g}_t)\alpha(x_i, y_j, z_t, \underline{g}_t)}{\mathcal{U}(z_t, \underline{g}_t)\mathcal{V}(z_t, \underline{g}_t)} \left(g_{t,i}^w - \frac{1}{n_y} \sum_{k=1}^{n_y} g_{t,ik} \right) \left(g_{t,j}^f - \frac{1}{n_x} \sum_{l=1}^{n_x} g_{t,lj} \right), \end{aligned} \quad (2.12)$$

where $m(z_t, \underline{g}_t) := m(\mathcal{U}(z_t, \underline{g}_t), \mathcal{V}(z_t, \underline{g}_t))$ and the approximation to the jump term is $-\sigma(z_t; \tilde{z}_t)g_{t,ij}$ for all $i \leq n_x, j \leq n_y$.

⁸There are alternative discretization approaches that could also be deployed. Previous work by some of the authors (Gu et al., 2023) outlines different approaches to reducing the dimension of the density including: a finite agent approximation, an approximation to g on a discretized type space, and a projection onto a finite set of basis functions. We find that a finite type space is particularly convenient for this problem because the shape of the distribution is important for the HJB equations and the approximation to the KF equation (2.12) is simple. Gu et al. (2023) shows that simpler approximations can work for competitive market models in which the distribution only enters the HJB equations through prices.

Master equation: The discretized Master equation for the Surplus is given by:

$$\begin{aligned}
0 = \mathcal{L}^S S = & F(x, y, z) - (\rho + \delta(x, y, z))S(x, y, z, \underline{g}) \\
& - b - \beta \frac{m(z, \underline{g})}{\mathcal{U}(z, \underline{g})\mathcal{V}(z, \underline{g})} \frac{1}{n_y} \sum_{l=1}^{n_y} \alpha(x, y_l, z, \underline{g}) S(x, y_l, z, \underline{g}) g_l^v \\
& + c - (1 - \beta) \frac{m(z, \underline{g})}{\mathcal{U}(z, \underline{g})\mathcal{V}(z, \underline{g})} \frac{1}{n_x} \sum_{k=1}^{n_x} \alpha(x_k, y, z, \underline{g}) S(x_k, y, z, \underline{g}) g_k^u \\
& + \sum_{\check{z} \neq z} \lambda(z, \check{z}) \left(S(x, y, \check{z}, \underline{g}^+) - S(x, y, z, \underline{g}) \right) + \sum_{k=1}^{n_x} \sum_{l=1}^{n_y} \partial_{g_{kl}} S(x, y, z, \underline{g}) \mu^g(x_k, y_l, z, \underline{g})
\end{aligned}$$

where $\underline{g}_{ij}^+ = (1 - \sigma(z, \check{z}))g_{ij}$ is the post-jump match distribution, μ^g is given by the discretized KF equation (2.12), $(g^u, g^v, \mathcal{U}, \mathcal{V})$ are calculated using Table 2 (after appropriate discretization), and g^f comes from $g_t^f(y) = \mathcal{V}_t + \mathcal{P}_t$ for all y . The differential equations for the value functions, (2.5) to (2.8), have analogous discretized forms.

After discretization, we have a finite, but high-dimensional, system. The discretized measure function g is a finite collection of continuous masses at each point on the $n_x \times n_y$ grid approximating the x and y dimensions. The surplus function $S(x, y, z, \underline{g})$ is then defined on a finite support with dimension $3 + (n_x \times n_y)$. In this sense, the discretized problem becomes a standard, but high-dimensional, dynamic programming problem.

Soft acceptance: For our main exercises, we approximate $\alpha(x, y, z, \underline{g})$ by $\alpha(x, y, z, \underline{g}) = (1 + e^{-\xi S(x, y, z, \underline{g})})^{-1}$ to ensure differentiability of the value function when there is a finite number of types. This “softened” α function can be interpreted as a logit choice model where utility shocks come from an extreme value distribution with parameter ξ . Although working with a differentiable value function helps make the gradient training efficient, it is not essential for our algorithm. In Appendix B.5 we show how our DeepSAM technique can be adapted to solve for models with the exact $\alpha(x, y, z, \underline{g}) = \mathbb{1}\{S(x, y, z, \underline{g}) \geq 0\}$ choice rule.⁹ We demonstrate that the soft and hard problems lead to very similar equilibrium surplus functions for high ξ .

⁹Formally, the combination of finite types and the exact binary choice rule means that the surplus function is potentially not smooth on a zero measure set and so the solution to the master equation should be interpreted as a Viscosity solution.

2.5 The DeepSAM Method

In this section, we present the DeepSAM method for using deep learning to solve and estimate the general class of search and matching models constructed in Subsections 2.1 to 2.4. We first present, for given structural parameters, how to solve the high dimensional master equation (2.13). We then present how to estimate the model by introducing structural parameters as pseudo-state variables in the neural network and using a surrogate model to find the parameters that match simulated moments to the empirical data.

2.5.1 Solution Algorithm for Given Structural Parameters

Let $\omega = (x, y, z, \underline{g}) \in \Omega^\omega$ denote the state space. We approximate the surplus function, S , by a neural network, $\hat{S} : \Omega^\omega \times \Omega^\Theta \rightarrow \mathbb{R}$, $(\omega, \Theta) \mapsto \hat{S}(\omega; \Theta)$, of the form:

$$\begin{aligned} \mathbf{h}^{(1)} &= \phi^{(1)}(W^{(1)}\omega + \mathbf{b}^{(1)}) && \dots \text{Hidden layer 1} \\ \mathbf{h}^{(2)} &= \phi^{(2)}(W^{(2)}\mathbf{h}^{(1)} + \mathbf{b}^{(2)}) && \dots \text{Hidden layer 2} \\ &\vdots && \\ \hat{S} &= \phi^{(H)}(W^{(H)}\mathbf{h}^{(H-1)} + \mathbf{b}^{(H)}) && \dots \text{Layer } H, \text{ Surplus} \end{aligned} \tag{2.13}$$

where, using the terminology of the deep learning literature, H is the number of layers, the length of vector $\mathbf{h}^{(i)}$ is the number of neurons in layer i , $\phi^{(i)}$ is the activation function for layer i , and the collection $\Theta = (W^1, \dots, W^{(H)}, b^{(1)}, \dots, b^{(H)}) \in \Omega^\Theta$ are the weights of the neural network.¹⁰

Our goal is to find the weights for the neural network that approximately solve the master differential equation (2.13) globally across the state space.¹¹ Our approach is summarized in Algorithm 1. Essentially, we use stochastic gradient descent algorithms or their variants to train the neural network to minimize the average loss in the master equation on a collection of sample points. As with other neural network approaches, there are many implementation details involved with these generic steps. We discuss two: sampling and stability.

¹⁰Sometimes Θ is called the parameter set for the neural network but we use the term “weight” set to distinguish from the economic parameters. Sometimes the W -terms are referred to specifically as weights and the b -terms as biases but for convenience we refer to all the terms as weights.

¹¹We follow the computational economics literature and refer to our approach as a “global solution”, in contrast to “local solution” approaches which typically perturb around a steady state.

Algorithm 1: Generic Solution Algorithm

1. Approximate the surplus function by the neural network (NN)
 $S(x, y, z, g) \approx \hat{S}(x, y, z, g; \Theta)$ with weights Θ .
 2. Start with initial NN weight guess Θ^0 .
 3. At iteration n with NN weight vector Θ^n :
 - (a) Generate K sample points, $Q^n = \left\{ (x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y}) \right\}_{k \leq K}$.
 - (b) Calculate the average mean squared error of the surplus master equation (2.13) on the sample points:
$$L(\Theta^n, Q^n) := \frac{1}{K} \sum_{k \leq K} \left| \mathcal{L}^S \hat{S}(x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y}) \right|^2$$
 - (c) Update NN weights using stochastic gradient descent (SGD) or a variant.
For example by $\Theta^{n+1} = \Theta^n - \zeta^n \nabla_{\Theta} L(\Theta^n, Q^n)$, for $\zeta^n > 0$
 - (d) Repeat until $L(\Theta^n, Q^n) \leq \epsilon$ with precision threshold ϵ .
 4. Once S has converged, we can train neural network approximations to the value and wage functions ($\hat{V}^u(x, z, g; \Theta^u)$, $\hat{V}^e(x, y, z, g; \Theta^e)$, $\hat{V}^v(y, z, g; \Theta^v)$, $\hat{V}^p(x, y, z, g; \Theta^p)$, $\hat{w}(x, y, z, g; \Theta^w)$) with NN weights $(\Theta^u, \Theta^e, \Theta^v, \Theta^p, \Theta^w)$ that solve the discretized versions of equations (2.3) to (2.8).
-

Procedure to construct the sample points Q^n : It is trivial to sample across different values of x, y, z because they are low dimensional. We can draw values of (x, y, z) using a Latin hypercube sampler or a multinomial sampler on discretized grids. Sampling across the discretized distribution $(g_{i,j})_{i \leq n_x, j \leq n_y}$ is more complicated because it is high dimensional. We first solve the model at the steady state for the different fixed values of z . We then draw distributions $((g_{ij,k})_{i \leq n_x, j \leq n_y})_{k \leq K}$ that are perturbed random combinations of the steady state distributions for the different z . This approach samples values of g that lie in regions of the state space that the economy is most likely to visit in equilibrium, thereby improving solution accuracy in economically relevant areas. Once the error is small, we use our current solution guess to simulate the economy and then use the simulated distribution path as the sample points. We describe sampling in more detail in Appendix B.2.

Algorithm stability: It is more difficult to stabilize the algorithm when $\hat{S}(x, y, z, g; \Theta)$ and/or $\hat{\alpha}(x, y, z, g; \Theta)$ has sharp curvature. When these problems appear, we use the following “homotopy”-style procedure. We start by solving the problem for economic parameters that give low curvature in $\hat{S}$ and $\hat{\alpha}(x, y, z, g; \Theta)$. We then change parameters to move closer to the desired values and retrain the neural network starting from the previous solution. We keep changing parameters and retraining until we have reached a solution at the desired parameters. We use this approach to solve the model with binary acceptance choices $\alpha \in \{0, 1\}$ in Appendix B.5.

2.5.2 Solution and Estimation

A key goal of quantitative macroeconomics is to estimate structural model parameters so that the model generates output that is consistent with observed data moments. Traditionally, this requires solving the model repeatedly for different parameter values, which can be computationally intensive. A number of papers have pointed out that deep learning based solution methods offer an advantage for model estimation (e.g., [Fernández-Villaverde et al., 2023](#); [Chen et al., 2023](#); [Kase et al., 2024](#)). By introducing economic parameters as pseudo-state variables in the neural network, we can efficiently solve models simultaneously across both the state space and the economic parameter space. We can then estimate the model parameters using the method of simulated moments or maximum likelihood.

In this section, we outline how we solve and estimate the model. Let $\Psi \in \Omega^\Psi$ denote the structural parameters that will be estimated (or calibrated internally). Let $\tilde{\varphi} = (\tilde{\varphi}_1, \dots, \tilde{\varphi}_N)$ denote the $N \times 1$ data moments that we want to match. Let $\varphi(\Psi) = (\varphi_1(\Psi), \dots, \varphi_N(\Psi))$ denote the corresponding model moments for given structural parameters Ψ , that will be generated by simulating the model. For fixed $\{\gamma_i\}_{i \leq N}$, our goal is to estimate the model by choosing parameters Ψ to solve:

$$\hat{\Psi} = \arg \min_{\Psi} \sum_{i=1}^N \gamma_i \left(\frac{\tilde{\varphi}_i - \varphi_i(\Psi)}{\tilde{\varphi}_i} \right)^2. \quad (2.14)$$

The main computational challenge of estimation is to solve $\varphi_i(\Psi)$ for many different values of the structural parameters. We overcome this challenge by introducing Ψ as a pseudo-state vector for the surplus function, and solving the resulting extended

master equation, which in its discretized form is given by:

$$\begin{aligned}
0 = \mathcal{L}^S S = & F(x, y, z) - (\rho + \delta(x, y, z))S(x, y, z, \underline{\mathbf{g}}, \Psi) \\
& - b - \beta \frac{m(z, \underline{\mathbf{g}})}{\mathcal{U}(z, \underline{\mathbf{g}})\mathcal{V}(z, \underline{\mathbf{g}})} \frac{1}{n_y} \sum_{l=1}^{n_y} \alpha(x, y_l, z, \underline{\mathbf{g}}, \Psi) S(x, y_l, z, \underline{\mathbf{g}}, \Psi) g_l^v \\
& + c - (1 - \beta) \frac{m(z, \underline{\mathbf{g}})}{\mathcal{U}(z, \underline{\mathbf{g}})\mathcal{V}(z, \underline{\mathbf{g}})} \frac{1}{n_x} \sum_{k=1}^{n_x} \alpha(x_k, y, z, \underline{\mathbf{g}}, \Psi) S(x_k, y, z, \underline{\mathbf{g}}, \Psi) g_k^u \\
& + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) \left(S(x, y, \tilde{z}, \underline{\mathbf{g}}^+, \Psi) - S(x, y, z, \underline{\mathbf{g}}, \Psi) \right) \\
& + \sum_{k=1}^{n_x} \sum_{l=1}^{n_y} \partial_{g_{kl}} S(x, y, z, \underline{\mathbf{g}}, \Psi) \mu^g(x_k, y_l, z, \underline{\mathbf{g}}, \Psi),
\end{aligned}$$

where $\underline{\mathbf{g}}_{ij}^+ = (1 - \sigma(z, \tilde{z}))g_{ij}$ is the post-jump match distribution and μ^g is given by the discretized KFE (2.12) (with Ψ as an additional argument wherever relevant). We approximate the extended surplus function S by a neural network $\hat{S} : \Omega^\omega \times \Omega^\Psi \times \Omega^\Theta \rightarrow \mathbb{R}$, $(\omega, \Psi, \Theta) \mapsto \hat{S}(\omega, \Psi; \Theta)$ and solve it in a similar way as for equation (2.13). The main additional complexity is that the structural parameter vector Ψ is now also part of the state space. This means that we need to sample from the parameter space and expand the range of sampled distributions to incorporate the variation in the parameter space. For the former, we draw economic parameters uniformly from across a prespecified range. For the latter, we first solve the model at the steady state for different fixed values of z and economic parameters. Then, as before, we draw distributions that are perturbed random combinations of these steady state distributions.

We then use the solution to the extended master equation to build a surrogate model that maps economic parameters to simulated moments: $\hat{\varphi} : \Omega^\Psi \times \Omega^\Theta \rightarrow \mathbb{R}^N$, $(\Psi, \Theta^\varphi) \mapsto \hat{\varphi}(\Psi; \Theta^\varphi)$. We do this by simulating the model under N_l different structural parameter vectors $\{\Psi^l\}_{l \leq N_l}$ (in parallel) and computing the model moments $\{\varphi(\Psi^l) = (\varphi_1(\Psi^l), \dots, \varphi_N(\Psi^l))\}_{l \leq N_l}$. We then construct the surrogate model $\hat{\varphi}$ by training another neural network to approximate the mapping $\varphi(\Psi)$ using the simulated data. We obtain the estimated parameters by solving the optimization problem (2.14) with $\varphi_i(\Psi)$ replaced by $\hat{\varphi}_i(\Psi; \Theta^\varphi)$ for all $i \leq N$. We summarize all steps in Algorithm 2.

Algorithm 2: Generic Solution and Estimation Algorithm

1. Train the model across the state and structural parameter spaces:
 - (a) Approximate the surplus function by a neural network that includes the economic parameters Ψ as inputs $S(x, y, z, g, \Psi) \approx \hat{S}(x, y, z, g, \Psi; \Theta)$.
 - (b) Train the neural network, as in Algorithm 1 but with the expanded sampling scheme described in the text.
 2. Compute a surrogate model mapping structural parameters to model moments $\hat{\varphi}(\cdot) \approx \varphi(\cdot)$ and optimize over $\hat{\varphi}$ to solve problem (2.14).
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3 Labor Market With Two-Sided Heterogeneity

In this section we solve a concrete version of the simple labor search model we set up in Section 2. Even though this setup is a widely used building block in the literature, to our knowledge, we are the first to solve this model without simplifying assumptions. Our goals are three-fold: first, we illustrate our DeepSAM solution method; second, we evaluate its numerical performance, highlighting its accuracy and computational efficiency in addressing high-dimensional problems even with highly non-linear crises dynamics; lastly, we use the model to study the impact of the COVID-19 shock and show the importance of distributional feedback in aggregate dynamics. We leave the demonstration of our deep learning estimation approach to the setup in Section 4.

3.1 Model Details and Parameters

Our environment is a special case of Section 2.1 with the following features. The exogenous aggregate state z follows a three-state Markov chain, corresponding to expansion (z_H), normal recession (z_L), and major crisis (z_D), the latter of which we calibrate to the COVID-19 crisis but could also be interpreted as a depression shock. The match output $F(z, x, y) = A_z f(x, y)$ and the separation rate $\delta_z(x, y)$ depend upon the state z , where we use the notation that A and δ are indexed by a z subscript.

Economic parameters: The calibrated economic parameters are presented in Table 7 in Appendix B. All parameters are calibrated at the annual frequency. For non-crisis states, where possible, we take standard parameters from the literature. We calibrate the matching efficiency κ to target an average unemployment rate of 6%. For the cri-

sis state, we calibrate the separation rate $\delta_D(x, y)$ across workers and firms to match the observed peak declines in employment levels during the COVID-19 recession, as calculated in [Cajner et al. \(2020\)](#). [Cajner et al. \(2020\)](#) uses detailed data from a major US payroll company to estimate the employment drop for workers in different skill groups (corresponding to five groups in our model) and firms in different two-digit NAICS industries (corresponding to 11 groups in our model) during the COVID-19 recession. These drops peaked in April 2020, about 0.2 years after the onset of the pandemic in the US. As shown in Figure 1, we calibrate $\delta_D(x, y)$ such that the model’s simulated declines in employment across worker and firm groups following a change to the crisis state z_D for $t = 0.2$ years match the corresponding empirical moments. The detailed values of $\delta_D(x, y)$ are provided in Appendix B.1.

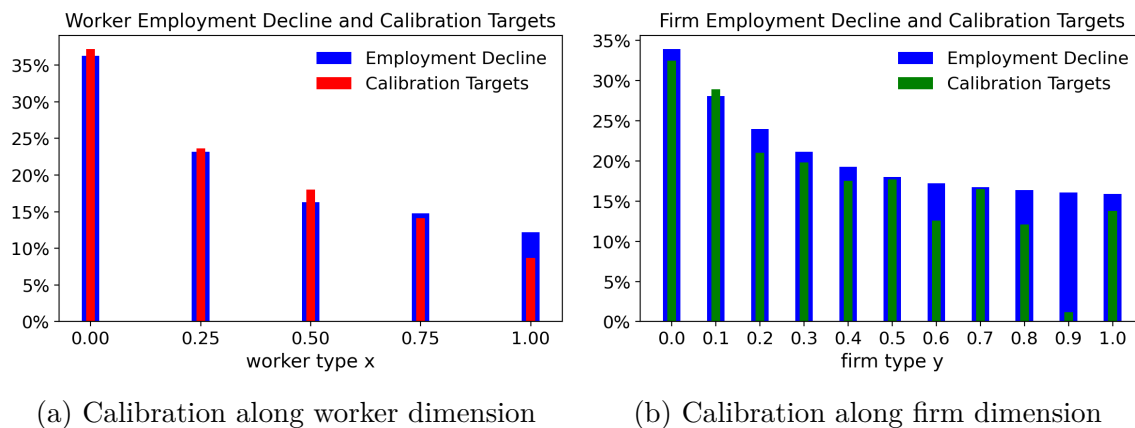


Figure 1: Calibrating $\delta_D(x, y)$ to match effect of COVID-19 shock on heterogeneous workers and firms in [Cajner et al. \(2020\)](#).

Neural network parameters: We use a fully connected feed-forward network with 4 layers, 50 neurons per layer, and a $\tanh(\cdot)$ activation function. We describe additional details of the neural network and the sampling in Table 8 in Appendix B.

3.2 Numerical Performance

Table 3 summarizes the numerical performance of the DeepSAM method. The numerical loss for solving the model is reported in the first column. We achieve small errors, on the order of 10^{-3} root mean square error (RMSE), across our sample. Notably, no existing method has been able to solve our 58-dimensional PDE. We discuss

the detailed numerical scheme and the numerical stability of the results in Appendix B.2. We also depict the numerical accuracy visually in Figure 8 in Appendix B.2, which shows the RMSE is on the order of magnitude of 10^{-3} across the (x, y) grid and not highly biased to a particular part of the state space.

We further validate the DeepSAM method by solving a simpler, non-aggregate shock version of the model for which there are conventional solution techniques that can be used for comparison. We set $(z_t, \delta_t) \equiv (\bar{z}, \bar{\delta})$ and use the fixed point approach in Shimer and Smith (2000) and Hagedorn et al. (2017) to solve for the deterministic steady state (DSS) solution $S^{\text{DSS}}(x, y)$. We then use DeepSAM to solve the 57-dimensional PDE for $S(x, y, \underline{g})$ and compute the DeepSAM solution at the DSS by: $S_{\text{DeepSAM}}^{\text{DSS}}(x, y) = S(x, y, \underline{g} = \underline{g}^{\text{DSS}})$. We define the squared difference of the two methods for each (x, y) pair as $\|S_{\text{DeepSAM}}^{\text{DSS}}(x, y) - S_{\text{Conventional}}^{\text{DSS}}(x, y)\|^2$, where the mean squared difference takes the average of the squared difference across all (x, y) pairs. We report the average numerical performance in the second column of Table 3 and show the root mean squared difference is on the order of 10^{-3} . Overall, we interpret our results in Table 3 and Appendix B.2 as evidence that our solution method has high accuracy.

	Model with agg. shock	Model without agg. shock
PDE Training Loss (RMSE)	9.2×10^{-4}	1.2×10^{-3}
PDE Evaluation Loss (RMSE)	1.7×10^{-3}	—
RMSE to The Existing Solution	No existing solution	2.4×10^{-3}

Table 3: Numerical performance of DeepSAM for models with and without aggregate shocks. Computations are performed on the A100 GPU at Google Colab. Evaluation loss denotes the loss computed on model simulation paths that are not used during neural-network training. We explain this term in detail in Appendix B.2.

3.3 Distributional Feedback to Aggregate Dynamics

An advantage of our DeepSAM method is that it can be used to study non-linear distributional dynamics in response to large aggregate shocks. This is because, even with aggregate risk, we can explicitly solve for the agent acceptance policy, α , as a function of the distribution and so can fully capture the feedback between distribution changes and aggregate dynamics. This is particularly valuable when aggregate shocks have asymmetric effects across the population, a phenomenon well-documented in a

large empirical literature, such as by [Guvenen et al. \(2017\)](#). In this subsection, we illustrate these points by studying the recent COVID-19 crisis and disentangling the role of distributional feedback.

As discussed above, we model the COVID-19 shock as a change to the disaster state z_D that lasts for $t = 0.2$ years, followed by the recovery phase with stochastic aggregate shocks $z_t \in \{z_H, z_L, z_D\}$. Using our DeepSAM solution, we can simulate the dynamics of the distribution g_t with the Kolmogorov forward equation (2.12) and compute aggregate dynamics for unemployment, employment, average wage, and other variables. The average path of employment change is plotted as the blue line of Figure 2a. As targeted in the data, we see a sharp initial drop in employment reflecting the impact of the shutdowns. Employment then continues to decline until April 2020 when restrictions gradually start to relax and employment recovers.

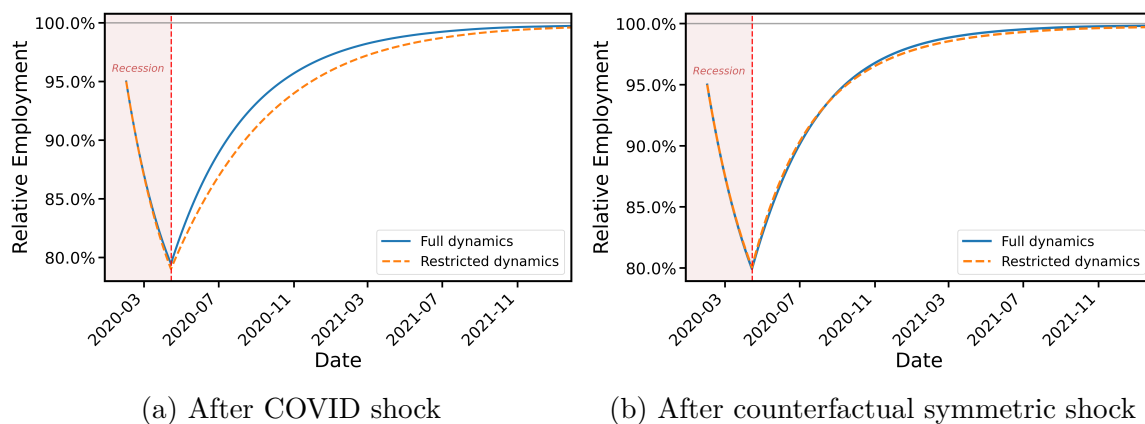


Figure 2: Relative change of employment compared to the initial level after the COVID-19 shock: full dynamics calculated with the KFE (2.12) vs. “restricted” dynamics in which the decision of workers and firms is myopic to change in distribution g_t as in Equation (3.1). The left panel shows the dynamics after the true asymmetric COVID-19 shock, while the right one corresponds to the counterfactual symmetric shock.

To understand the contribution of distributional feedback to aggregate dynamics, we decompose the impulse responses of employment into two channels:

- (i) the change of employment when agents’ acceptance decision is always evaluated at the ergodic mean of the match distribution which corresponds to the long-run

average of $g_t(x, y)$ over time t of the stochastic process (z_t, g_t) :¹²

$$g^{\text{ergodic}}(x, y) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T g_t(x, y) dt,$$

- (ii) the additional change of employment when the acceptance function reacts to the changing distribution of matches.

We refer to the former as the “restricted” dynamics for the experiment without the additional feedback from the distribution to the acceptance function. Mathematically, under the restricted dynamics, the distribution g_t^R evolves with the same jumps but with different drift:

$$\begin{aligned} \mu^g(x, y, z_t, \underline{g}_t^R) = & -(\delta(x, y, z_t) + \varsigma(z_t))g_t^R(x, y) \\ & + \frac{m_t(z_t, \underline{g}_t^R)}{\mathcal{U}_t(\underline{g}_t^R)\mathcal{V}_t(\underline{g}_t^R)}\alpha(x, y, z_t, \underline{g}^{\text{ergodic}})g_t^{u,R}(x)g_t^{v,R}(y) \end{aligned} \quad (3.1)$$

where $\underline{g}^{\text{ergodic}}$ denotes the vector of $g^{\text{ergodic}}(x, y)$ evaluated on the $(\mathcal{X}, \mathcal{Y})$ grid. We plot the restricted dynamics with the dashed orange line of Figure 2a. To focus on dynamic feedback, we impose the same initial jump as the full model. Compared to the full dynamics calculated with the KFE (2.12), the difference in Equation (3.1) is that agents make decisions assuming the distribution is always at the ergodic mean. Thus the gap between the full and restricted dynamics can be interpreted as the contribution of distributional feedback to aggregate impulse responses. From Figure 2a, we can see that distributional feedback accounts for approximately 25% of the cumulative change in employment dynamics during the first half year of the recovery. When agents make decisions based on the true distribution change, the aggregate employment recovers faster than in the economy where agents ignore distribution changes.

To understand why distributional feedback is significant, in Figure 3 we plot the agents’ decision functions before and after the COVID-19 shock. The left panels of Figure 3 depict the ergodic mean of the match distribution and the acceptance functions at the ergodic mean. Evidently, the matching function exhibits positive assortative matching (PAM) in the sense that workers and firms tend to match with counterparts with similar productivity levels. The right panels of Figure 3 depict the

¹²Note that the ergodic mean differs from the stationary (ergodic) distribution of the process. Also note that the limit is being computed separately for each (x, y) .

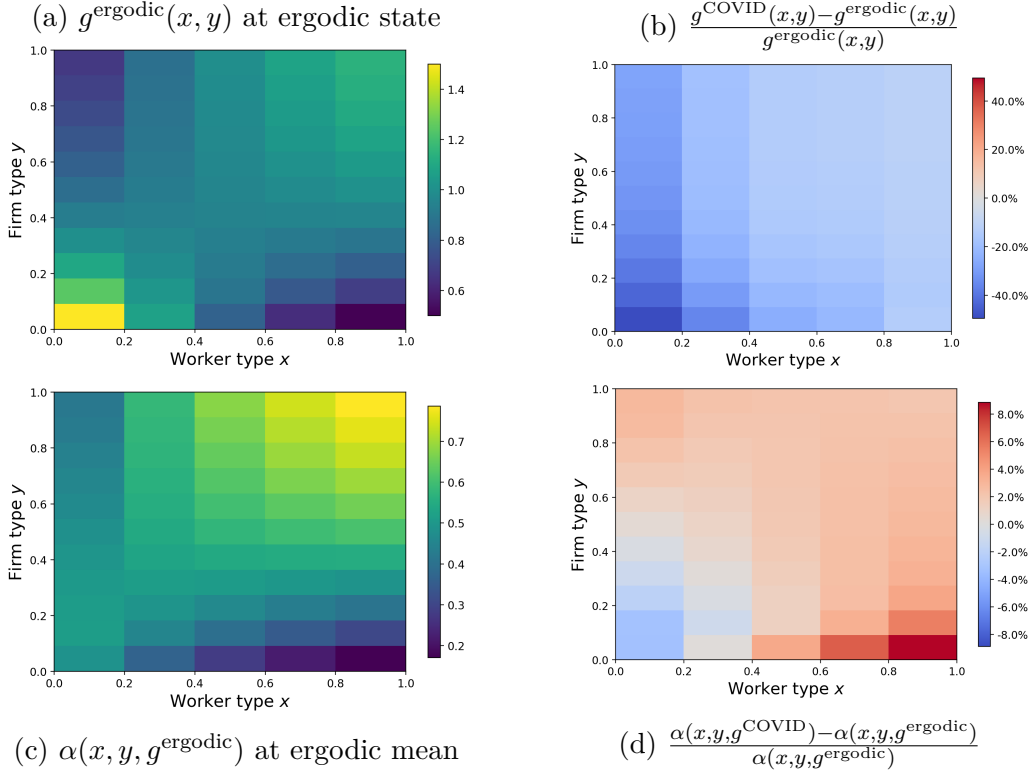


Figure 3: Distribution of matches and associated acceptance decision. Panels (a) and (c): ergodic mean of the distribution and acceptance at the ergodic mean. Panels (b) and (d): relative difference of distribution and acceptance at the end of the asymmetric COVID-19 shock, compared to the ergodic.

change in the distribution and the matching function at the end of the COVID-19 recession. Matches are disproportionately destroyed at the lower ends of both the worker and firm distributions. For high-type workers and firms, this decreases the relative probability of finding a match with a high-type counterpart, namely $\frac{g^v(y)}{v}$ and $\frac{g^u(x)}{u}$ decrease for large x and y . As a result, high-type workers and firms become less “picky” and are more willing to accept matches with low types. This is reflected in the increase in α for high x to low y pairs and (to a lesser extent) low x to high y pairs. For low-type workers and firms, the relative probability of finding a high-type match also decreases, but the relative probability that a high-type will accept the match significantly increases (particularly for low-type firms and high-type workers). The net result is that α decreases for low x to low y matches. Collectively, these changes show that the COVID-19 shock weakens positive assortative matching and speeds up the recovery.

To further explore the role of asymmetric shocks, we conduct a counterfactual analysis with a disaster shock that has a symmetric impact across workers and firms through the separation rate, namely $\tilde{\delta}_D(x, y) \equiv \tilde{\delta}_D$ for all x, y . We calibrate $\tilde{\delta}_D$ such that the shock generates the same aggregate employment rate as our calibrated $\delta_D(x, y)$ at the peak of the COVID-19 recession. We plot the full and restricted dynamics of the economy after this counterfactual “symmetric” shock in Figure 2b. In this case, the trajectories of the “restricted” dynamics and the full solution are closely aligned, implying that the distributional feedback is small when aggregate shocks affect workers and firms symmetrically.

These findings underscore the significance of distributional feedback on aggregate dynamics in the labor market, particularly when workers and firms have heterogeneous exposure to large aggregate shocks. They also suggest that policy design should account for the asymmetric nature of shocks and the heterogeneous effects of the policies themselves. For example, the design of unemployment insurance should consider feedback from the distribution changes to fully understand its implications and efficacy.

4 On-The-Job Search (OJS) and Business Cycles

In this section, we study business cycle shocks in a richer labor market with endogenous job-to-job and job-to-unemployment transitions. Conceptually, this model extends [Lise and Robin \(2017\)](#) by giving workers non-zero bargaining power, allowing for on-the-job bargaining beyond Bertrand competition, and generalizing the free entry condition, all of which are changes that break “block-recursivity”. We estimate the model using the deep learning based simulated method of moments from Section 2.5.2. We use the estimated model to study how the business cycle impacts wages and employment for workers and firms with different skill levels, the full analysis of which has previously been unattainable. We find support for the [Okun \(1973\)](#) hypothesis that low-skill workers benefit disproportionately from longer expansions.

4.1 Model Details

Our environment is an extension of Section 2.1 with the following features. The exogenous aggregate state z follows a two-state Markov chain, corresponding to ex-

pansion (z_H) and recession (z_L). The match output $F(z, x, y) = A_z f(x, y)$ and the separation rate $\delta_z(x, y)$ depend upon the state z , where we use the notation that A and δ are indexed by a z subscript. We introduce on-the-job search and endogenous separation to enable a richer calibration.

Search and matching: All workers (unemployed and employed) now engage in random search. The matching function becomes $m(\mathcal{W}_t, \mathcal{V}_t)$ where $\mathcal{W}_t := \mathcal{U}_t + \phi \mathcal{E}_t$ denotes the total mass of searchers and ϕ is the exogenous relative search intensity of employed workers. The probabilities that an unemployed or an employed worker meets a potential employer are given by $\mathcal{M}_t^u = \frac{m(\mathcal{W}_t, \mathcal{V}_t)}{\mathcal{W}_t}$ and $\mathcal{M}_t^e = \phi \frac{m(\mathcal{W}_t, \mathcal{V}_t)}{\mathcal{W}_t}$, while the probability that a vacant firm meets a potential hire is $\mathcal{M}_t^v = \frac{m(\mathcal{W}_t, \mathcal{V}_t)}{\mathcal{V}_t}$. Conditional on a firm meeting a worker, the probabilities that the worker is unemployed or employed are defined as $\mathcal{C}_t^u = \frac{\mathcal{U}_t}{\mathcal{U}_t + \phi \mathcal{E}_t}$ and $\mathcal{C}_t^e = \frac{\phi \mathcal{E}_t}{\mathcal{U}_t + \phi \mathcal{E}_t}$ respectively. So, the probability for a firm to meet an unemployed worker $x \in X$ is $\mathcal{M}_t^v \mathcal{C}_t^u \int_X \frac{g_t^u(x)}{\mathcal{U}_t} dx$ and the probability for a firm to meet an employed worker $x \in X$ is $\mathcal{M}_t^v \mathcal{C}_t^e \int_X \frac{g_t^e(x)}{\mathcal{E}_t} dx$.

Bargaining with unemployed workers: Let $S_t(x, y)$, $V_t^u(x)$, and $V_t^v(y)$ again denote the match surplus, unemployed worker value, and vacant firm value at time t . When an x -type unemployed worker meets a y -type vacant firm at time t , the surplus is again divided by generalized Nash bargaining. Following the dynamic contracting literature, we now write this by specifying that the firm promises the worker a value q_t above their outside option $V_t^u(x)$. The values of the match to the employed worker and the producing firm are then denoted by $V_t^e(x, y, q_t) = V_t^u(x) + q_t$ and $V_t^p(x, y, q_t)$ respectively. Under Nash bargaining, $q_t = \beta S_t(x, y)$ at the start of the contract. In the previous sections, we assumed continuous contract renegotiation so the surplus division stayed at $q_{t'} = \beta S_{t'}(x, y)$ and $V_{t'}^p(x, y, q_{t'}) - V_{t'}^v(y) = (1 - \beta) S_{t'}(x, y)$ for all times $t' \geq t$ for all (x, y) matches. In this section, we relax this assumption to nest a wider collection of on-the-job bargaining protocols. We impose that the contract is implemented by providing a flow wage $w_t(x, y, q_t, z_t, g_t)$ to the worker and flow profit $f(x, y, z) - w_t(x, y, q_t, z_t, g_t)$ to the firm. Since the wage implementation is not allocative, we first solve for equilibrium allocations and then wages in Section 4.6.

Bargaining on-the-job: Suppose an x -type worker is in a match with an incumbent y -type firm that is currently promising value q . If the worker meets a new firm of type

$\tilde{y}$ at time t , then the worker, incumbent firm, and poaching firm follow a parametrized protocol that nests the outcomes from different bargaining models in the literature. We restrict to models where ultimately workers move to the poaching firm if and only if the surplus is higher $S_t(x, \tilde{y}) \geq S_t(x, y) \geq 0$.¹³ If $S_t(x, \tilde{y}) \geq S_t(x, y)$ and the worker moves to the poaching firm, then there are the following divisions for:

1. Incremental surplus created by the match: $\Delta S_t := S_t(x, \tilde{y}) - S_t(x, y)$. The worker captures a share β of ΔS_t and the poaching firm captures a share $1 - \beta$.
2. Incumbent's firm surplus before the worker's move: $W_t^p := V_t^p(x, y, q_t) - V_t^v(y)$. The incumbent firm retains a share $1 - \chi$ of their surplus while the worker gets $\chi \in [0, 1]$.

For clarity, Table 4 summarizes the division of value before and after the move. If $S_t(x, \tilde{y}) < S_t(x, y)$ and the worker does not move, then the incumbent firm offers them a new promised value $q'_t = (1 - \chi)q_t + \chi \max\{\beta S_t(x, y) + (1 - \beta)S_t(x, \tilde{y}), q_t\}$. So, conceptually, χ governs to what extent the worker can freely leave and use outside offers to renegotiate while β governs their strength in renegotiation.

	Worker	Incumbent	Poacher
Initial value	$V_t^u + q_t$	V_t^p	V_t^v
+ Share of ΔS_t	$\beta \Delta S_t$	0	$(1 - \beta) \Delta S_t$
+ Share of W_t^p	χW_t^p	$-\chi W_t^p$	0
Value after move	$V_t^u + q_t + \beta \Delta S_t + \chi W_t^p$	$V_t^p - \chi W_t^p$	$V_t^v + (1 - \beta) \Delta S_t$

Table 4: Distribution of value at a successful poaching event. For convenience, we have dropped the arguments of the functions.

In Appendix C.2, we show that different protocol parameters recover well-known bargaining models. When $\chi = 1$, the worker can freely leave their job in the sense that they can capture the incumbent's full surplus. When combined with $\beta = 0$, the worker cannot also capture incremental surplus and so arrives at the new match with value $V_t^u(x) + q_t + W_t^p$. This is the outcome that emerges when the poaching and incumbent firm compete for the worker under Bertrand competition (as in [Postel-Vinay and Robin, 2002](#)). When $\chi = 1$ is combined with $\beta > 0$, the worker also captures the

¹³Or equivalently $\tilde{y} > y$ and $S_t(x, \tilde{y}) > 0$ as in [Postel-Vinay and Robin \(2002\)](#); [Cahuc et al. \(2006\)](#).

Nash share of the incremental surplus and so arrives at the new match with value $V_t^u(x) + q_t + \beta\Delta S_t + W_t^p$. This is the outcome that emerges when the firms compete for the right to bargain with the worker and then the worker uses the losing firm's best offer as their outside option when bargaining with the winning firm (as in [Cahuc et al., 2006](#)). When $\chi = 0$, the worker must compensate the firm to renegotiate their wage contract and so can only capture the Nash share of the incremental surplus. In this case, the worker arrives at the new match with value $V_t^u(x) + q_t + \beta\Delta S_t$. If we also assume that away from the on-the-job transitions workers and firms continuously renegotiate their surplus allocation to the current split in new matches between unemployed workers and firms, then workers enter the on-the-job search bargaining with $q_t = \beta S_t(x, y)$ and so get value $V_t^u(x) + \beta S_t(x, \tilde{y})$ when they move on-the-job (the outcome in [Pissarides, 1994](#), which assumes that workers can only ever use unemployment as their outside option when bargaining with poaching firms).

Match separation: In addition to exogenous match separation at rate $\delta(x, y, z)$, we now also allow workers and firms to break up matches endogenously, in which case they return to unemployment at the rate η .

Entry and vacancy: We assume firms may choose to enter at any time. If a firm enters, then they get a draw of type y from the density $\bar{h}(y)$ and create a vacancy.¹⁴ To be able to compare more directly to [Lise and Robin \(2017\)](#), we start by assuming that firm vacancies are “short-lived” in the sense that at the end of each period all unfilled vacancies are destroyed and all unmatched firms leave. We no longer impose common exogenous exit rates (across matched and unmatched firms), so $\varsigma(z) = 0$ and $\sigma(z, \tilde{z}) = 0$ under the Section 2 notation. We also no longer compensate agents when firms exit. In Appendix E, we solve and compare to a model with long-lived vacancies and slow firm entry and exit.

4.2 Recursive Characterization of Equilibrium

We now define and characterize a recursive equilibrium under the general bargaining protocol described above. The aggregate states are once again (z, g) , where z is the aggregate productivity and g is the measure function over matches. The value func-

¹⁴[Lise and Robin \(2017\)](#) allows firms to know their y and then create vacancies, whereas our firms have to choose to enter before knowing their draw of y .

tions (V^u, V^v) are defined as in Section 2. We let $V^p(x, y, q, z, g)$ denote the value to a type- y firm that is matched with a type- x worker and has promised q . As before, we let $\alpha(x, y, z, g)$ denote an indicator function for whether a type- x worker and a type- y firm choose to accept a match. In addition, we let $\alpha^b(x, y, z, g)$ denote an indicator for when an (x, y) match endogenously breaks up and $\alpha^e(x, y, \tilde{y}, z, g)$ denote an indicator for when a worker moves from match (x, y) to match $(x, \tilde{y})$. As suggested by the notation, α^b and α^e do not depend upon the promise q under our protocol.

Kolmogorov Forward Equation: The measure of matches now evolves according to:

$$\begin{aligned} dg_t(x, y) = & \left[-(\delta(x, y, z) + \eta\alpha^b(x, y, z_t, g_t))g_t(x, y) \right. \\ & - \mathcal{M}_t^e \int \alpha^e(x, y, \tilde{y}, z_t, g_t) \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} g_t(x, y) d\tilde{y} + \mathcal{M}_t^u \alpha(x, y, z_t, g_t) \frac{g_t^v(y)}{\mathcal{V}_t} g_t^u(x) \\ & \left. + \mathcal{M}_t^e \int \alpha^e(x, \tilde{y}, y, z_t, g_t) \frac{g_t^v(y)}{\mathcal{V}_t} g_t(x, \tilde{y}) d\tilde{y} \right] dt =: \mu^g(x, y, z_t, g_t) dt. \end{aligned} \quad (4.1)$$

This KF equation has been written from the perspective of the workers (it can be written equivalently from the point of view of the firms) and so the terms are written as: (Prob. worker meets) $\times$ (Prob. acceptance) $\times$ (prob. y) $\times$ (mass of workers). The first term on the RHS is the exit rate due to exogenous and endogenous separations, the second term is the exit rate due to workers finding better matches, the third term is the entry rate from unemployed workers finding jobs, and the final term is the entry rate from employed workers moving on-the-job to (x, y) .

Unemployed workers and vacant firms: Unemployed worker value functions $V^u(x, z, g)$ have the same HJB equation (2.5) as the model without on-the-job search. Vacant firms are short-lived and so have continuation value $V^v(y, z, g) = 0$ for all y .

Employed matches: To characterize on-the-job search behavior, we follow [Postel-Vinay and Robin \(2002\)](#) and work with the value of an employed match rather than the individual value functions. We denote the combined value of a match between a type x worker and a type y firm that has promised q by:

$$P(x, y, q, z, g) := V^p(x, y, q, z, g) + V^u(x, z, g) + q.$$

We denote by $\tilde{P}(x, y, \tilde{y}, q, \tilde{q}, z, g)$ the combined continuation value of the worker and the incumbent firm after the worker moves to a poaching firm of type $\tilde{y}$ to get $\tilde{q}$.

From our general bargaining protocol, we can derive what happens when an employed worker meets a firm. If the worker does not move, then $P(x, y, q, z, g)$ does not change because the exchange of value is between the worker and incumbent firm. If the worker does move, then the worker arrives at the new firm with value $V^u(x, z, g) + q + \beta(S(x, \tilde{y}, z, g) - S(x, y, z, g)) + \chi(V^p(x, y, q, z, g) - V^v(y, z, g))$, while the incumbent firm retains $V^p(x, y, q, z, g) - \chi(V^p(x, y, q, z, g) - V^v(y, z, g))$. Summing these values implies that the change in joint value is independent of q :

$$\tilde{P}(x, y, \tilde{y}, q, \tilde{q}, z, g) - P(x, y, q, z, g) = \beta(S(x, \tilde{y}, z, g) - S(x, y, z, g)). \quad (4.2)$$

Given agent beliefs about the drift of the match density $\check{\mu}^g$, $P(x, y, q, z, g)$ solves the HJB equation:

$$\begin{aligned} \rho P(x, y, q, z, g) = & F(x, y, z) \\ & + (\delta(x, y, z) + \eta \alpha^b(x, y, z, g))(V^u(x, z, g) + V^v(y, z, g) - P(x, y, q, z, g)) \\ & + \mathcal{M}^e \int \alpha^e(x, y, \tilde{y}, z, g) \beta(S(x, \tilde{y}, z, g) - S(x, y, z, g)) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\ & + \sum_{\check{z} \neq z} \lambda(z, \check{z}) (P(x, y, q, \check{z}, g) - P(x, y, q, z, g)) + \langle D_g P, \check{\mu}^g \rangle, \end{aligned}$$

where the third line uses the change in joint value equation (4.2), the endogenous match breakup indicator is $\alpha^b = 1 - \alpha(x, y, z, g)$, and the on-the-job acceptance indicator is:

$$\alpha^e(x, y, \tilde{y}, z, g) := \begin{cases} 1, & \text{if } S(x, \tilde{y}, z, g) \geq S(x, y, z, g) \text{ and } S(x, \tilde{y}, z, g) \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

Since the right-hand side of the HJB equation for P does not depend upon q , we can conjecture and verify that $P(x, y, q, z, g) = P(x, y, z, g)$.

Master equation for surplus: The surplus of an existing match now becomes $S(x, y, z, g) = P(x, y, z, g) - V^u(x, z, g)$. We can again characterize the master equation for S by

imposing belief consistency $\check{\mu}^g = \mu^g$ and combining the HJB equations for P and V^u :

$$\begin{aligned} \rho S(x, y, z, g) = & F(x, y, z) - b - (\delta(x, y, z) + \eta \alpha^b(x, y, z, g)) S(x, y, z, g) \\ & + \mathcal{M}^e \int \alpha^e(x, y, \tilde{y}, z, g) \beta [S(x, \tilde{y}, z, g) - S(x, y, z, g)] \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\ & - \mathcal{M}^u \int \alpha(x, \tilde{y}, z, g) \beta S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\ & + \sum_{\check{z} \neq z} \lambda(z, \check{z}) (S(x, y, \check{z}, g) - S(x, y, z, g)) + \langle D_g S, \mu^g \rangle. \end{aligned}$$

Relative to the master equation (2.11) in Section 2, this equation is simplified because vacancies are short-lived so $V^v = 0$ and there is no firm opportunity cost of matching. When we reintroduce long-lived vacancies in Appendix E, the terms from the HJB equation for V^v reappear. In addition, the master equation gains the second line, which captures surplus creation through on-the-job search matches.

The worker's bargaining power β enters the surplus equation directly as the coefficient on the on-the-job search term and the unemployed-hire term. These terms connect the Master equation to the distribution because we need g to solve for $\mathcal{V}$, $\mathcal{M}^e$, and $\mathcal{M}^u$.¹⁵ Setting $\beta = 0$ (the worker has no bargaining power) eliminates both terms, decoupling the surplus equation from the distribution of vacancies g^v and recovering block-recursivity in g as in [Lise and Robin \(2017\)](#). For $\beta > 0$, the surplus depends on the distribution g through these integral terms, so the surplus equation is not block-recursive and must be solved jointly with g via the master equation.

Observe that χ does not appear in the HJB equation for P or the master equation for S . The bargaining protocol divides the incumbent firm's surplus between the worker and the incumbent according to χ , but this division cancels in the joint value P of the worker–incumbent pair and so does not enter $S = P - V^u$. As a result, the surplus master equation is identical across all $\chi \in [0, 1]$ and so the same surplus function $S(x, y, z, g)$ governs the matching dynamics whether the model nests the firm competition model of [Cahuc et al. \(2006\)](#), the on-the-job bargaining protocol of [Pissarides \(1994\)](#), or any interpolation between them.

Free entry, vacancy creation, and other variables: Since $\mathcal{W}_t$, $\mathcal{U}_t$, $\mathcal{E}_t$, $\mathcal{C}^u$, $\mathcal{C}^e$, and g^u are all determined by g , the mass of vacancies $\mathcal{V}_t$ can again be determined as a function

¹⁵Note that, unlike in the Master equation in Section 2, we no longer need to compute $g^u/\mathcal{U}$ because it no longer appears in the Master equation and $g^v/\mathcal{V}$ is now the firm entry density.

of (z, g) and the surplus function $S(\cdot, \cdot, z, g)$ through a free-entry condition. We make this explicit in Appendix C.1.

4.3 Deep Learning-Based Estimation

We estimate the parameters $\{\beta, \kappa, c, \delta, \phi, \eta, b\}$ (respectively the worker bargaining power, scale factor in the meeting function, entry cost, exogenous separation rate, on-the-job meeting rate, endogenous separation rate, and unemployment benefit) to match the ergodic mean of the unemployment rate, vacancy rate, employment-to-employment transition rate, unemployment-to-employment transition rate, and employment-to-unemployment transition rate, denoted by $\mathbb{E}[U]$, $\mathbb{E}[V]$, $\mathbb{E}[EE]$, $\mathbb{E}[UE]$, and $\mathbb{E}[EU]$ respectively, in the US data. The other parameters are taken from the literature and outlined in Table 10 in Appendix C. We discretize workers into seven productivity types ($n_x = 7$) and firms into eight productivity types ($n_y = 8$), which is sufficiently fine to match the empirical regularities in the data.

We undertake the deep learning based estimation outlined in Section 2.5.2. The estimated parameters, the data moments, and the (targeted and untargeted) model moments are shown in Table 5. The target moments are taken from [Lise and Robin \(2017\)](#) and the transition flow moments (including expected rate of employment-to-employment, employment-to-unemployment, and unemployment-to-employment transitions) are presented as monthly values to match with the original paper.¹⁶ Evidently, we are able to get a close match to all five targeted moments and reasonable match to the untargeted moments. In Table 11 in Appendix C, we show that the targeted model moments stay the same even when the model is retrained using simulated sampling at our estimated parameters. An attractive feature of our estimation approach is that we can use our surrogate model mapping parameters to moments to undertake parameter sensitivity analysis. We do this (quantitatively and visually) in Appendix C.5, which shows that the target moments are particularly sensitive to β .

The computational times for each step of the estimation, as well as the associated numerical losses, are shown in Table 6. The entire solution and estimation process takes 3 hours and 10 minutes, where the model is solved over the economic parameter space and simulated across 10,000 parameter combinations to build the surrogate model deployed for the simulated method of moments. To our knowledge, it's infeasible

¹⁶[Lise and Robin \(2017\)](#) uses the BLS data 1951:I–2012:IV to construct most of the moments, except for the EE transition, which is constructed with the CPS data 1994:I–2011:III.

Panel A: Estimated parameters and targeted moments

Parameter	Interpretation	Value	Moment	Data	Model
β	Surplus division factor	0.836	$\mathbb{E}[U]$	0.058	0.057
κ	Scale for meeting function	20.0	$\mathbb{E}[V]$	0.037	0.036
c	Firm entry cost	0.698	$\mathbb{E}[EE]$	0.025	0.026
δ	Match separation rate	0.159	$\mathbb{E}[UE]$	0.421	0.448
ϕ	Employed worker search rate	0.146	$\mathbb{E}[EU]$	0.025	0.027
η	Endogenous separation intensity	0.786			
b	Unemployment benefit	0.019			

Panel B: Untargeted moments

Moment	Data	Model	Moment	Data	Model
autocorr[VA]	0.932	0.915	corr[UE, VA]	0.878	0.957
corr[V, U]	-0.846	-0.782	corr[EU, VA]	-0.716	-0.946
corr[UE, EE]	0.695	0.893	corr[U, VA]	-0.860	-0.961
corr[V, VA]	0.721	0.791			

Table 5: Estimated parameters and model fit. Moments in Panel A are targeted in estimation; those in Panel B are not. $VA_t = \sum_{x,y} F(x, y, z) g_t(x, y)$ is aggregate value added.

ble to solve and estimate such a high-dimensional problem within such a time frame using other approaches.

	Solving Model For a Parameter Set	Solving Model With Structural Parameters as Pseudo-states	Training Surrogate Model	Simulated Method of Moments	Entire Estimation
RMSE Loss	9.97×10^{-4}	2.9×10^{-3}	7.82×10^{-4}	1.24×10^{-4}	-
Time	40min	2h 47min	23min	0.4min	3h 10min

Table 6: Evaluation loss and computational time for solving vs estimating the model. Computations are performed on the G4 GPU at Google Colab.

4.4 Alternative Bargaining Parameters

With the DeepSAM method, we can solve heterogeneous agent random search models with on-the-job search and endogenous separation without imposing the assumptions needed to get block recursivity. A key assumption in Lise and Robin (2017) for getting block recursivity is setting $\beta = 0$ so unemployed workers get zero surplus when

bargaining with a firm. We now study how this restriction impacts the calibration and impulse responses.

For alternative values of β , we recalibrate the other parameters to match the same moments as in our baseline calibration in Section 4.3 but without allowing the β to vary.¹⁷ In Figure 4, for the calibrations at different β , we plot the time paths for changes in the unemployment level (U_t), the vacancy level (V_t), the total quantity of poaching, and the total hires from unemployment in response to a 1.5% negative productivity shock that lasts for one year and then subsequently reverts to the stochastic process for aggregate TFP.¹⁸ We find that aggregate unemployment, vacancies, poaching, and hiring from unemployment all become more sensitive to aggregate shocks as $\beta \rightarrow 0$. Under our estimated value of $\beta = 0.84$ from Table 5, a 1.5% productivity shock leads to an increase in the unemployment rate of about 0.9% over the first year of the recession. By contrast, a block recursive setup predicts a more substantial 5.0% increase in the unemployment rate over the same time. This is because firm posting is more elastic when firms get the entire surplus from matches with unemployed workers, as shown in the upper right panel in Figure 4. We interpret these observations as evidence that the assumptions required to establish block recursivity are quantitatively important for estimation.

4.5 Welfare Dynamics Over The Business Cycle

We use our estimated model to study how the business cycle impacts employment and welfare for different workers and firms. Figure 5 shows the impulse responses for an economy that goes into recession for a year and then recovers to the high state. The left panel shows the average surplus of low-type (solid orange) and high-type (dashed blue) workers during the subsequent expansion. Although low-type workers start the recovery with less surplus growth, they gain surplus more quickly and end up with higher surplus growth approximately 2 years into the expansion. This is consistent with the hypothesis proposed by [Okun \(1973\)](#) that longer expansions disproportionately improve labor market outcomes for low-wage workers.

In our model, Okun's hypothesis is generated through the impact of distribution

¹⁷The calibration results, shown in Table 12 in Appendix C.5, reveal that we need a large increase in c in order to match the same moments with a lower β .

¹⁸The total quantity of *poaching* is defined as $\phi \frac{m(\mathcal{W}_t, \mathcal{V}_t)}{\mathcal{W}_t \mathcal{V}_t} \int \int \int \alpha_t^e(x, \tilde{y}, y) g_t^v(y) g_t(x, \tilde{y}) d\tilde{y} dx dy$. The total quantity of *hires from unemployment* is defined as $\frac{m(\mathcal{W}_t, \mathcal{V}_t)}{\mathcal{W}_t \mathcal{V}_t} \int \int \alpha_t(x, y) g_t^u(x) g_t^v(y) dx dy$.

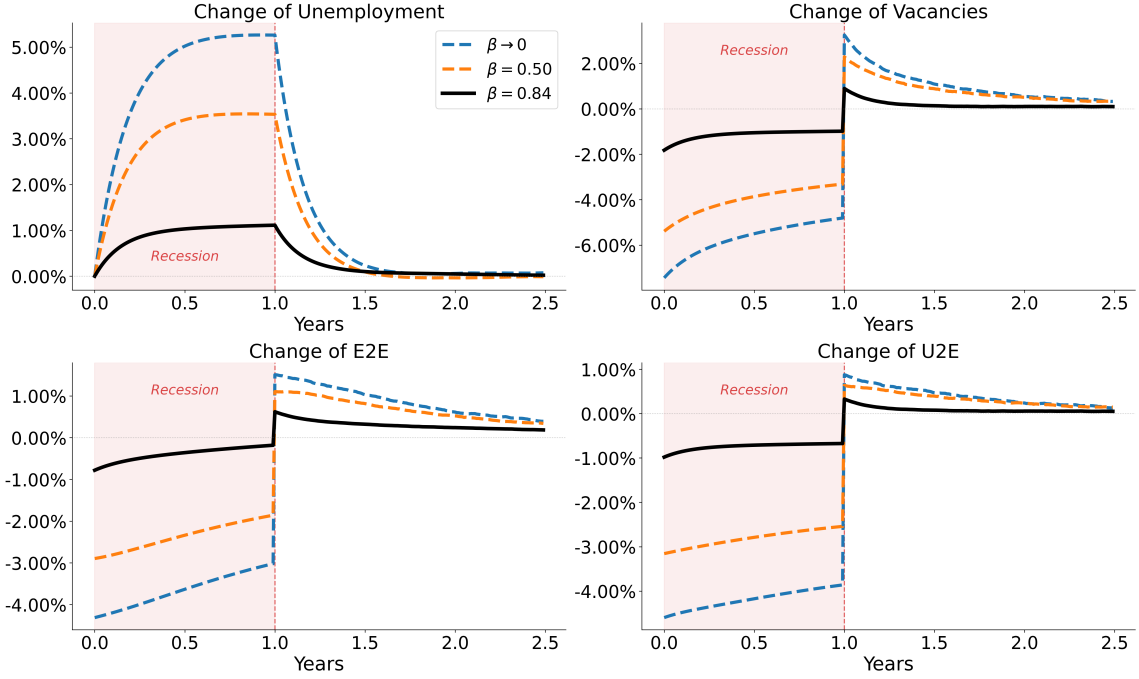


Figure 4: IRF following negative TFP Shock with different β 's. After one year, the plots show the average time path across 1000 simulations. E2E and U2E are the employment-to-employment and unemployment-to-employment transition rates.

dynamics on positive assortative matching (PAM, or positive sorting) during expansions of varying durations. To see this, consider the top right panel of Figure 5, which depicts the time path for the ratio of “non-PAM pairs” to “PAM pairs”. A matched pair (x, y) is called a “PAM pair” if $|x - y| \leq \frac{1}{2}$, and otherwise called a “non-PAM” pair. PAM pairs can be interpreted as worker-firm combinations with similar relative levels of idiosyncratic types. Evidently, the ratio of non-PAM to PAM pairs decreases during recessions and increases during expansions. This indicates that sorting is countercyclical, which aligns with the literature (e.g. [Lise and Robin, 2017](#)). To investigate how this affects workers specifically, the bottom right panel of Figure 5 depicts the change in the worker employment rate at high type firms (defined as firms with $y > 1/2$) for high and low type workers. This shows that low-type workers suffer larger declines in high type matches when PAM increases during the recession and enjoy larger increases in high type matches when PAM decreases during the subsequent expansion. Moreover, in both right panels, worker assortative matching weakens further as expansions lengthen. Conceptually, this is because high-type workers become increasingly scarce during expansions so high-type firms

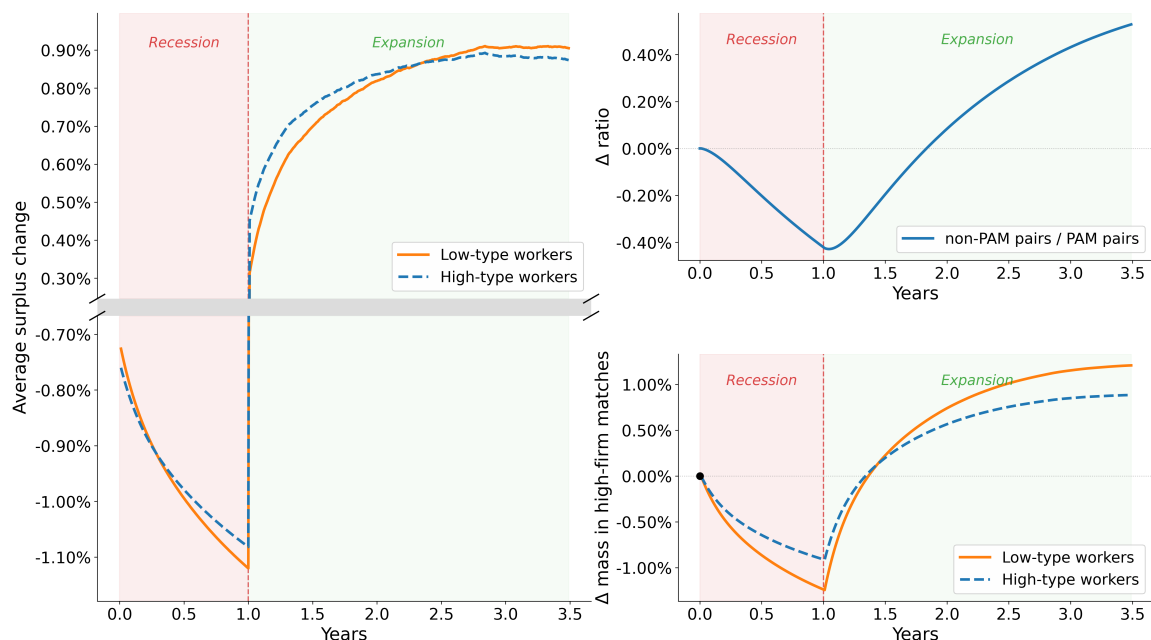


Figure 5: Impulse Responses. In all panels, low and high types refer to the lowest and highest types in our $n_x = 7$ -type discretization. Left panel: Change in average surplus for different types of workers (broken y -axis). Top-right panel: Change in the ratio of non-PAM to PAM matching mass. Bottom-right panel: Change of employment at firms with $y > 1/2$ for different types of workers.

become less picky. In other words, the distributional changes, which increase the relative employment of high and low type workers, determine the worker types who will benefit most from an additional period of economic expansion. Therefore, our model captures Okun’s hypothesis solely through the feedback between distribution changes and agent matching.

Due to its policy relevance, Okun’s hypothesis has attracted recent attention. Existing literature (e.g. [Alves and Violante, 2023](#)) introduces complex model elements such as human capital and skill accumulation to explain Okun’s observation. Interestingly, we can generate similar model predictions with a simpler model setup that drives dynamics through the interaction between distribution changes and acceptance decisions. This offers a complementary story that potentially has different policy implications. This is because, in our model, the decrease in low-skilled unemployment during long booms results from the formation of “non-PAM” low- x /high- y pairs, which are more likely to dissolve during subsequent recessions (as in Figure 5).

4.6 Dynamics of the Wage Distribution

We close the paper by returning to the wage implementation of the firm-worker contract. A well-known difficulty in the heterogeneous agent random search literature is that, even in “block-recursive” models, solving for wage dynamics has been intractable. This is because surplus division does not inherit the block recursive property of total surplus.¹⁹ By contrast, our DeepSAM method can solve for wage dynamics because it solves for the surplus function explicitly as a function of the match distribution. However, in order to do this, we must impose additional structure by specifying a bargaining protocol. We consider two extremes that are consistent with $\chi = 0$ and $\chi = 1$ in our parametrized surplus division protocol.

Case $\chi = 0$: We impose that $\chi = 0$ in on-the-job transitions and at other times worker-firm matches continuously renegotiate to the surplus division in new matches. In this case, at all times we have that in match (x, y) :

$$q = \beta S(x, y, z, g), \quad V^p(x, y, q, z, g) = (1 - \beta)S(x, y, z, g) \quad (4.3)$$

where we have used that $V^v(y, z, g) = 0$. This is because (i) away from on-the-job transitions, workers and firms continuously renegotiate to these values and (ii) during on-the-job transitions workers arrive with value $q_t = \beta S(x, y, z, g)$ and then leave with $q'_t = q_t + \beta(S(x, \tilde{y}, z, g) - S(x, y, z, g)) = \beta S(x, \tilde{y}, z, g)$ after bargaining. This is the same outcome as [Pissarides \(1994\)](#) and other papers with continuous wage renegotiation. It follows that $V^e(x, y, q, z, g) = V^u(x, z, g) + q = V^u(x, z, g) + \beta S(x, y, z, g)$ and $V^p(x, y, q, z, g) = V^p(x, y, z, g)$ are independent of q and satisfy:

$$\begin{aligned} \rho V^e(x, y, z, g) &= w(x, y, z, g) + \eta^b(x, y, z_t, g_t)(V^u(x, z, g) - V^e(x, y, z, g)) \\ &\quad + \mathcal{M}^e \int \alpha^e(x, y, \tilde{y}, z, g) \beta(S(x, \tilde{y}, z, g) - S(x, y, z, g)) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\ &\quad + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z})(V^e(x, y, \tilde{z}, g) - V^e(x, y, z, g)) + \langle D_g V^e, \mu^g \rangle \\ \rho V^p(x, y, z, g) &= F(x, y, z) - w(x, y, z, g) - \eta^b(x, y, z_t, g_t)V^p(x, y, z, g) \\ &\quad + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z})(V^p(x, y, \tilde{z}, g) - V^p(x, y, z, g)) + \langle D_g V^p, \mu^g \rangle \end{aligned} \quad (4.4)$$

¹⁹As [Lentz et al. \(2017\)](#) write: “wages cannot be solved for exactly, indeed one needs to solve for a fixed point in worker values where the distribution of workers across jobs is a state variable.”

where we have used the short-hand notation $\eta^b(x, y, z_t, g_t) := \delta(x, y, z) + \eta\alpha^b(x, y, z_t, g_t)$, used $V^v(y, z, g) = 0$, and imposed consistency $\check{\mu}^g = \mu^g$. These are the on-the-job search analogues of (2.6) and (2.8) in Section 2.3.2. Substituting (4.3) into (4.4) and rearranging gives the following equation for the unique wage in (x, y) matches in terms of $S(x, y, z, g)$:

$$w(x, y, z, g) = F(x, y, z) - \rho(1 - \beta)S(x, y, z, g) - \eta^b(x, y, z_t, g_t)(1 - \beta)S(x, y, z, g) + \sum_{\check{z} \neq z} \lambda(z, \check{z})(1 - \beta)(S(x, y, \check{z}, g) - S(x, y, z, g)) + \langle D_g((1 - \beta)S), \mu^g \rangle.$$

Case: $\chi = 1$: We also consider the other extreme in which workers can move freely to new firms ($\chi = 1$ in the protocol) and firms Bertrand compete for employed workers using the protocol in Cahuc et al. (2006). For the $\chi = 0$ case above, there is one wage function for each (x, y, z, g) . By contrast, under the Cahuc et al. (2006) protocol, the wage at any point in time reflects the history of offers the worker has received while employed. Consequently, the static Nash split (4.3) cannot hold at all times within a match like it does for $\chi = 0$. Instead, we need to keep track of the distribution of promised values q in different matches. Doing this mathematically is involved so we relegate the KF equation for q to Appendix C.3. However, conceptually, it involves accounting for the rate at which workers find poaching firms and move or renegotiate.

Business Cycle Wage Dynamics: We solve for the evolution of wages under the two model extremes. The left panel of Figure 6 shows the wage change across the worker distribution for the $\chi = 0$ case following a positive aggregate shock that lasts for 5 years. The right panel shows the same impulse response but for the $\chi = 1$ and $\beta \rightarrow 0$ case. In both cases, the wages of low-type workers are more procyclical. This reflects the observation from the previous section that booms lead to more favorable matching for low-type workers. The magnitude is very different because the labor market cyclicalities are very sensitive to β , as we saw in Figure 4. This indicates that by varying χ , our model can capture a wide range of wage outcomes.

5 Conclusion

In this paper, we developed a new method for globally solving and estimating search and matching models with aggregate shocks and heterogeneous agents. This allows

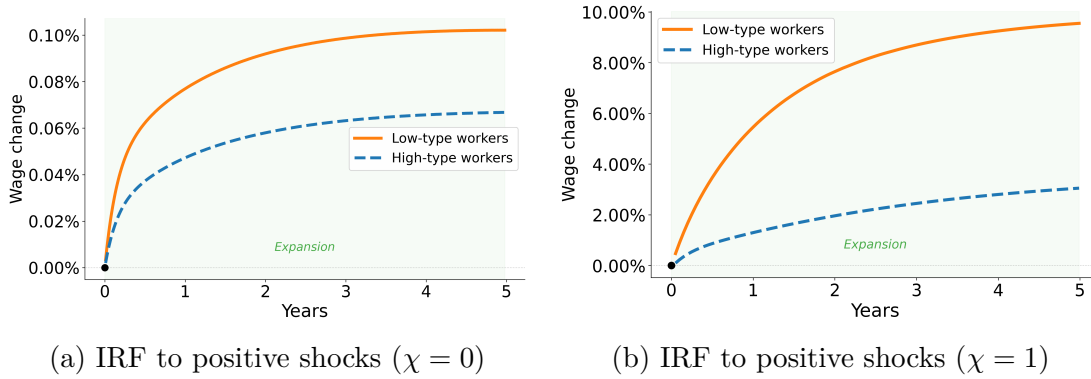


Figure 6: Wage change after z shocks for $\chi = 0$ and $\chi = 1$. Low, middle, and high types refer to the lowest, middle, and highest types in our $n_x = 7$ -type discretization for workers.

us to study dynamics in models where agent decisions depend upon the distribution so the model is not “block-recursive”. We believe our methodology is a major breakthrough in the literature of search and matching models and will open up many potential applications in the labor, finance, and spatial literature.

References

- ALVAREZ, F., F. LIPPI, AND P. SOUGANIDIS (2023): “Price setting with strategic complementarities as a mean field game,” *Econometrica*, 91, 2005–2039.
- ALVES, F. AND G. L. VIOLANTE (2023): “Some Like It Hot: Monetary Policy Under Okun’s Hypothesis,” .
- AZINOVIC, M., L. GAEGAUF, AND S. SCHEIDEGGER (2022): “Deep equilibrium nets,” *International Economic Review*, 63, 1471–1525.
- AZINOVIC, M. AND J. ŽEMLIČKA (2023): “Economics-inspired neural networks with stabilizing homotopies,” *arXiv preprint arXiv:2303.14802*.
- BALEY, I., A. FIGUEIREDO, AND R. ULBRICHT (2022): “Mismatch cycles,” *Journal of Political Economy*, 130, 2943–2984.
- BILAL, A. (2023): “Solving heterogeneous agent models with the master equation,” Tech. rep., National Bureau of Economic Research.
- BURDETT, K. AND D. T. MORTENSEN (1998): “Wage differentials, employer size, and unemployment,” *International Economic Review*, 257–273.
- CAHUC, P., F. POSTEL-VINAY, AND J.-M. ROBIN (2006): “Wage bargaining with on-the-job search: Theory and evidence,” *Econometrica*, 74, 323–364.
- CAJNER, T., L. D. CRANE, R. A. DECKER, J. GRIGSBY, A. HAMINS-PUERTOLAS, E. HURST, C. KURZ, AND A. YILDIRMAZ (2020): “The US Labor

- Market during the Beginning of the Pandemic Recession,” *Brookings Papers on Economic Activity*, 3–33.
- CALIENDO, L., M. DVORKIN, AND F. PARRO (2019): “Trade and labor market dynamics: General equilibrium analysis of the china trade shock,” *Econometrica*, 87, 741–835.
- CARDALIAGUET, P., F. DELARUE, J.-M. LASRY, AND P.-L. LIONS (2019): *The master equation and the convergence problem in mean field games*, Princeton University Press.
- CHEN, H., R. CUI, Z. HE, AND K. MILBRADT (2017): “Quantifying Liquidity and Default Risks of Corporate Bonds over the Business Cycle,” *The Review of Financial Studies*.
- CHEN, H., A. DIDISHEIM, AND S. SCHEIDEGGER (2023): “Deep surrogates for finance: With an application to option pricing,” *Available at SSRN 3782722*.
- DAVIS, M. AND R. ELLIOTT (1977): “Optimal control of a jump process,” *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete*, 40, 183–202.
- DUARTE, V., D. DUARTE, AND D. H. SILVA (2024): “Machine learning for continuous-time finance,” *The Review of Financial Studies*, 37, 3217–3271.
- DUARTE, V. AND J. FONSECA (2024): “Global identification with gradient-based structural estimation,” .
- DUFFIE, D., N. GÂRLEANU, AND L. H. PEDERSEN (2005): “Over-the-counter markets,” *Econometrica*, 73, 1815–1847.
- FERNÁNDEZ-VILLAYERDE, J. (2025): “Deep learning for solving economic models,” Tech. rep., National Bureau of Economic Research.
- FERNÁNDEZ-VILLAYERDE, J., S. HURTADO, AND G. NUNO (2023): “Financial frictions and the wealth distribution,” *Econometrica*, 91, 869–901.
- FERNÁNDEZ-VILLAYERDE, J., G. NUNO, AND J. PERLA (2024): “Taming the Curse of Dimensionality: Quantitative Economics with Deep Learning,” Tech. rep.
- FRIEDL, A., F. KÜBLER, S. SCHEIDEGGER, AND T. USUI (2023): “Deep uncertainty quantification: with an application to integrated assessment models,” .
- FUKUI, M. (2020): “A theory of wage rigidity and unemployment fluctuations with on-the-job search,” .
- GOPALAKRISHNA, G. (2021): “Aliens and continuous time economies,” Tech. rep.
- GREGORY, V., G. MENZIO, AND D. WICZER (2024): “The alpha beta gamma of the labor market,” *Journal of Monetary Economics*, 103695.
- GU, Z., M. LAURIERE, S. MERKEL, AND J. PAYNE (2023): “Deep Learning Solutions to Master Equations for Continuous Time Heterogeneous Agent Macroeconomic Models,” Tech. rep.
- GUVENEN, F., S. SCHULHOFER-WOHL, J. SONG, AND M. YOGO (2017): “Worker betas: Five facts about systematic earnings risk,” *American Economic Review*, 107,

398–403.

- HAGEDORN, M., T. H. LAW, AND I. MANOVSKII (2017): “Identifying equilibrium models of labor market sorting,” *Econometrica*, 85, 29–65.
- HAN, J., Y. YANG, AND W. E (2021): “DeepHAM: A Global Solution Method for Heterogeneous Agent Models with Aggregate Shocks,” .
- HUANG, J. (2022): “A Probabilistic Solution to High-Dimensional Continuous-Time Macro-Finance Models,” *Available at SSRN 4122454*.
- KAHOU, M. E., J. FERNÁNDEZ-VILLAYERDE, S. GÓMEZ-CARDONA, J. PERLA, AND J. ROSA (2025): “Spooky Boundaries at a Distance: Inductive Bias and Dynamic Macroeconomic Models,” .
- KAHOU, M. E., J. FERNÁNDEZ-VILLAYERDE, J. PERLA, AND A. SOOD (2021): “Exploiting symmetry in high-dimensional dynamic programming,” Tech. rep.
- KASE, H., L. MELOSI, AND M. ROTTNER (2024): “Estimating Nonlinear Heterogeneous Agent Models with Neural Networks,” .
- KRUSELL, P., T. MUKOYAMA, R. ROGERSON, AND A. ŞAHİN (2017): “Gross worker flows over the business cycle,” *American Economic Review*, 107, 3447–3476.
- KRUSELL, P. AND A. A. SMITH (1998): “Income and Wealth Heterogeneity in the Macroeconomy,” *Journal of Political Economy*, 106, 867–896.
- LAGOS, R. AND G. ROCHETEAU (2009): “Liquidity in asset markets with search frictions,” *Econometrica*, 77, 403–426.
- LAGOS, R. AND R. WRIGHT (2005): “A Unified Framework for Monetary Theory and Policy Analysis,” *Journal of Political Economy*, 113, 463–484.
- LENTZ, R., J. LISE, AND J.-M. ROBIN (2017): “The Macro Dynamics of the Wage Distribution,” in *2017 Meeting Papers*, vol. 1220.
- LIONS, P.-L. (2007-2011): “Lectures at College de France,” .
- LISE, J. AND J.-M. ROBIN (2017): “The macrodynamics of sorting between workers and firms,” *American Economic Review*, 107, 1104–1135.
- MALIAR, L., S. MALIAR, AND P. WINANT (2021): “Deep learning for solving dynamic economic models,” *Journal of Monetary Economics*, 122, 76–101.
- MENZIO, G. AND S. SHI (2010): “Block recursive equilibria for stochastic models of search on the job,” *Journal of Economic Theory*, 1.
- (2011): “Efficient Search on the Job and the Business Cycle,” *Journal of Political Economy*, 119, 468–510.
- MOEN, E. R. (1997): “Competitive Search Equilibrium,” *Journal of Political Economy*, 105, 385–411.
- MORTENSEN, D. T. AND C. A. PISSARIDES (1994): “Job creation and job destruction in the theory of unemployment,” *The review of economic studies*, 61, 397–415.
- MOSCARINI, G. AND F. POSTEL-VINAY (2018): “The cyclical job ladder,” *Annual*

- Review of Economics*, 10, 165–188.
- (2023): “The job ladder: Inflation vs. reallocation,” Tech. rep., National Bureau of Economic Research.
- MUELLER, A. I. (2017): “Separations, sorting, and cyclical unemployment,” *American Economic Review*, 107, 2081–2107.
- NORETS, A. (2012): “Estimation of dynamic discrete choice models using artificial neural network approximations,” *Econometric Reviews*, 31, 84–106.
- OKUN, A. M. (1973): “Upward mobility in a high-pressure economy,” *Brookings Papers on Economic Activity*, 1973, 207–261.
- PISSARIDES, C. A. (1994): “Search unemployment with on-the-job search,” *The Review of Economic Studies*, 61, 457–475.
- POSTEL-VINAY, F. AND J.-M. ROBIN (2002): “Equilibrium wage dispersion with worker and employer heterogeneity,” *Econometrica*, 70, 2295–2350.
- QIU, X. (2023): “Vacant Jobs,” .
- RAISSI, M., P. PERDIKARIS, AND G. E. KARNIADAKIS (2019): “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,” *Journal of Computational physics*, 378, 686–707.
- REITER, M. (2009): “Solving heterogeneous-agent models by projection and perturbation,” *Journal of Economic Dynamics and Control*, 33, 649–665.
- SANDMANN, C. AND N. BONNETON (2025): “Existence of a non-stationary equilibrium in search-and-matching models: TU and NTU,” *Theoretical Economics*, 20, 1411–1460.
- SCHAAL, E. (2017): “Uncertainty, Productivity and Unemployment in the Great Recession,” *Econometrica*, 85, 250–274.
- SHIMER, R. (2005): “The cyclical behavior of equilibrium unemployment and vacancies,” *American economic review*, 95, 25–49.
- SHIMER, R. AND L. SMITH (2000): “Assortative matching and search,” *Econometrica*, 68, 343–369.
- SMITH, L. (2006): “The marriage model with search frictions,” *Journal of political Economy*, 114, 1124–1144.
- STOKEY, N. L. AND R. E. LUCAS JR (1989): *Recursive methods in economic dynamics*, Harvard University Press.
- WEILL, P.-O. (2008): “Liquidity premia in dynamic bargaining markets,” *Journal of Economic Theory*, 140, 66–96.

Online Appendix

A Additional Details For The Setup in Section 2

In this Appendix, we complete the description of the environment from Section 2. In Subsection A.1 we characterize the full general equilibrium equations incorporating endogenous exit. In Subsection A.2, we prove terminal transversality conditions and provide a verification result that the solution to the recursive equilibrium is also a solution to the sequential equilibrium. In Subsection A.3 we construct a discrete time version of the model and show that the master equation is similar. In Subsection A.4 we comment on the possibility of multiple equilibria.

A.1 Equilibrium Characterization With Endogenous Exit

In the main text, we described the master equation under the assumption that there is no endogenous exit from the free entry condition and both firms and workers are fully compensated when they exit the economy. In this Appendix we describe the full equilibrium characterization in an environment with endogenous exit from the free entry condition and parametrized compensation for exogenous firm exit.

A.1.1 Parameterizing Compensation

We now parametrize the compensation for firm exit to nest different specifications of the economy. If a firm exits the economy, then we impose that they receive a transfer for a fraction τ of their lost surplus upon exit. We also impose that the worker receives a transfer for a fraction τ of their lost surplus if the match is broken up because the firm exits. Under this notation, the main text imposes that $\tau = 1$. These transfers could be interpreted as firm asset recovery, worker severance, or a government financial incentive, but we do not put an interpretation in the paper or explicitly account for the transfers in an aggregate resource constraint (similar to how the unemployment benefit and entry cost are not accounted for in an aggregate resource constraint). Instead, we parametrize the options and show how they change the master equation.

A.1.2 Endogenous Exit Accounting

We now need to account for both endogenous firm entry and endogenous firm exit. Denote the process for the total measure of firms, $\mathcal{F}_t := \mathcal{F}(z_t, g_t)$, by:

$$d\mathcal{F}_t = \mu^{\mathcal{F}}(z_t, g_t) dt + \sum_{\check{z}_t \neq z_t} \gamma^{\mathcal{F}}(z_t, \check{z}_t, g_t) dN_t(z_t; \check{z}_t).$$

So, $\mu^{\mathcal{F}}(z_t, g_t)$ is the drift and $\gamma^{\mathcal{F}}(z_t, \check{z}_t, g_t)$ is the jump in $\mathcal{F}_t$ when the exogenous aggregate state jumps from z_t to $\check{z}_t$.

Drift exit: Away from jumps in z_t , the free entry condition leads to endogenous firm entry if $\mu^{\mathcal{F}}(z_t, g_t) \geq -\varsigma(z_t)\mathcal{F}_t$ and endogenous firm exit if $\mu^{\mathcal{F}}(z_t, g_t) < -\varsigma(z_t)\mathcal{F}_t$. Let:

$$\mathbf{1}_{\mu^{\mathcal{F}}}(z_t, g_t) := \mathbf{1} \left(\mu^{\mathcal{F}}(z_t, g_t) < -\varsigma(z_t)\mathcal{F}_t \right) \quad (\text{A.1})$$

denote an indicator for endogenous drift exit. If endogenous exit occurs, $\mathbf{1}_{\mu^{\mathcal{F}}}(z_t, g_t) = 1$, we assume that each individual firm (vacant or producing) is randomly forced to exit at idiosyncratic rate $\alpha_{\mu}^g(z_t, g_t)$ to satisfy the free entry condition.

Jump exit: At a Poisson jump from z to $\check{z}$, the respective conditions are $\gamma^{\mathcal{F}}(z_t, \check{z}_t, g_t) \geq -\sigma(z_t; \check{z}_t)\mathcal{F}_t$ for endogenous firm entry and $\gamma^{\mathcal{F}}(z_t, \check{z}_t, g_t) < -\sigma(z_t; \check{z}_t)\mathcal{F}_t$ for endogenous firm exit. We let:

$$\mathbf{1}_{\sigma^{\mathcal{F}}}(z_t, \check{z}_t, g_t) := \mathbf{1} \left(\gamma^{\mathcal{F}}(z_t, \check{z}_t, g_t) < -\sigma(z_t; \check{z}_t)\mathcal{F}_t \right) \quad (\text{A.2})$$

denote an indicator for endogenous jump exit. If endogenous exit occurs, $\mathbf{1}_{\sigma^{\mathcal{F}}}(z_t, \check{z}_t, g_t) = 1$, we assume that each individual firm (vacant or producing) is randomly forced to exit with probability $\alpha_{\sigma}^g(z_t, \check{z}_t, g_t)$ to satisfy the free entry condition.

Reconciliation with main text: In the main text, we assumed $\mathbf{1}_{\sigma^{\mathcal{F}}}(z_t, \check{z}_t, g_t) = \mathbf{1}_{\mu^{\mathcal{F}}}(z_t, g_t) = 0$ for all (z_t, g_t) in our equations and verified it was true on equilibrium path under our calibration. For this Appendix, we derive α_{μ}^g and α_{σ}^g explicitly in Section A.1.7 below.

A.1.3 Kolmogorov Forward Equation For Measure of Matches

With both endogenous entry and exit, the match distribution evolves according to:

$$dg_t(x, y) = \mu^g(x, y, z_t, g_t) dt + \sum_{\check{z}_t \neq z_t} \gamma^g(x, y, z_t, \check{z}_t, g_t) dN_t(z_t; \check{z}_t)$$

where the drift and jump coefficients are:

$$\begin{aligned} \mu^g(x, y, z, g) := & -\left(\delta(x, y, z) + \varsigma(z) + \mathbf{1}_{\mu^F}(z, g)\alpha_\mu^g(z, g)\right)g(x, y) \\ & + \mathcal{M}^u\alpha(x, y, z, g)\frac{g^v(y)}{\mathcal{V}}g^u(x) \end{aligned} \quad (\text{A.3})$$

$$\gamma^g(x, y, z, \check{z}, g) := -\left(\sigma(z; \check{z}) + \mathbf{1}_{\sigma^F}(z, \check{z}, g)\alpha_\sigma^g(z, \check{z}, g)\right)g(x, y). \quad (\text{A.4})$$

The key difference compared to the main text is that now both the drift and the jump of g are endogenous. Setting $\mathbf{1}_{\mu^F}(z_t, g_t) = \mathbf{1}_{\sigma^F}(z_t, \check{z}_t, g_t) = 0$ recovers the baseline KF equation (2.2) in the main text.

The drift and jump of aggregate firm employment $\mathcal{P}_t = \int \int g_t(x, y) dx dy$ follow directly from integrating (A.3)–(A.4):

$$\begin{aligned} \mu^{\mathcal{P}}(z_t, g_t) &= \int \int \mu^g(x, y, z_t, g_t) dx dy \\ \gamma^{\mathcal{P}}(z_t, \check{z}_t, g_t) &= \int \int \gamma^g(x, y, z_t, \check{z}_t, g_t) dx dy. \end{aligned}$$

A.1.4 HJB Equations

We now specify the HJB equations with endogenous exit and parametrized compensation for exit. Agents now have beliefs about both the endogenous drift of the distribution, $\check{\mu}^g$, and the endogenous jump in the distribution, $\check{\gamma}^g$. In all cases below, the equations reduce to the equations in the text for $\tau = 1$, $\mathbf{1}_{\mu^F}(z, g) = 0$, and $\mathbf{1}_{\sigma^F}(z, \check{z}, g) = 0$.

Workers: The HJB equation for the value function $V^u(x, z, g)$ for an unemployed worker is unchanged from equation (2.5) in the main text. The HJB equation for an employed worker is modified to account for the possibility of match destruction by endogenous firm exit and that they may not be fully compensated for exit. Formally, given beliefs $(\check{\mu}^g, \check{\gamma}^g)$ and an optimal acceptance decision, the value function

$V^e(x, y, z, g)$ for a worker employed at a firm of type y is:

$$\begin{aligned}
\rho V^e(x, y, z, g) &= w(x, y, z, g) + \delta(x, y, z)(V^u(x, z, g) - V^e(x, y, z, g)) \\
&\quad + (\varsigma(z) + \mathbf{1}_{\mu^F}(z, g)\alpha_\mu^g(z, g))(1 - \tau)(V^u(x, z, g) - V^e(x, y, z, g)) \\
&\quad + \sum_{\check{z} \neq z} \lambda(z, \check{z}) \left[V^e(x, y, \check{z}, g + \check{\gamma}^g(z, \check{z}, g)) - V^e(x, y, z, g) \right. \\
&\quad + \left((1 - \sigma(z, \check{z}))\mathbf{1}_{\sigma^F}(z, \check{z}, g)\alpha_\sigma^g(z, \check{z}, g) + \sigma(z, \check{z}) \right) \\
&\quad \quad \times (1 - \tau)(V^u(x, \check{z}, g + \check{\gamma}^g(z, \check{z}, g)) - V^e(x, y, \check{z}, g + \check{\gamma}^g(z, \check{z}, g))) \left. \right] \\
&\quad + \langle D_g V^e, \check{\mu}^g \rangle
\end{aligned} \tag{A.5}$$

Firms: Given beliefs $(\check{\mu}^g, \check{\gamma}^g)$ and an optimal acceptance decision, the value function $V^v(y, z, g)$ for a vacant firm solves:

$$\begin{aligned}
\rho V^v(y, z, g) &= -c - (\varsigma(z) + \mathbf{1}_{\mu^F}(z, g)\alpha_\mu^g(z, g))(1 - \tau)V^v(y, z, g) \\
&\quad + \mathcal{M}^v \int \alpha(\tilde{x}, y, z, g)(V^p(\tilde{x}, y, z, g) - V^v(y, z, g)) \frac{g^u(\tilde{x})}{\mathcal{U}} d\tilde{x} \\
&\quad + \sum_{\check{z} \neq z} \lambda(z, \check{z}) \left[V^v(y, \check{z}, g + \check{\gamma}^g(z, \check{z}, g)) - V^v(y, z, g) \right. \\
&\quad \quad - \left((1 - \sigma(z, \check{z}))\mathbf{1}_{\sigma^F}(z, \check{z}, g)\alpha_\sigma^g(z, \check{z}, g) + \sigma(z, \check{z}) \right) (1 - \tau)V^v(y, \check{z}, g + \check{\gamma}^g(z, \check{z}, g)) \left. \right] \\
&\quad + \langle D_g V^v, \check{\mu}^g \rangle
\end{aligned} \tag{A.6}$$

and the value function $V^p(x, y, z, g)$ for a producing firm solves:

$$\begin{aligned}
\rho V^p(x, y, z, g) &= F(x, y, z) - w(x, y, z, g) \\
&\quad - (\varsigma(z) + \mathbf{1}_{\mu^F}(z, g)\alpha_\mu^g(z, g))(1 - \tau)V^p(x, y, z, g) \\
&\quad + \delta(x, y, z)(V^v(y, z, g) - V^p(x, y, z, g)) \\
&\quad + \sum_{\check{z} \neq z} \lambda(z, \check{z}) \left[V^p(x, y, \check{z}, g + \check{\gamma}^g(z, \check{z}, g)) - V^p(x, y, z, g) \right. \\
&\quad \quad - \left((1 - \sigma(z, \check{z}))\mathbf{1}_{\sigma^F}(z, \check{z}, g)\alpha_\sigma^g(z, \check{z}, g) + \sigma(z, \check{z}) \right) (1 - \tau)V^p(x, y, \check{z}, g + \check{\gamma}^g(z, \check{z}, g)) \left. \right]
\end{aligned}$$

$$+ \langle D_g V^p, \check{\mu}^g \rangle \quad (\text{A.7})$$

A.1.5 Master Equation for Surplus

Combining the HJB equations (A.5)–(A.7) with the surplus division (2.3) and imposing belief consistency $(\check{\mu}^g, \check{\gamma}^g) = (\mu^g, \gamma^g)$, the master equation for the equilibrium surplus function $S(x, y, z, g)$ is:

$$\begin{aligned} 0 = & -\rho S(x, y, z, g) + F(x, y, z) - \delta(x, y, z) S(x, y, z, g) \\ & - \left(\varsigma(z) + \mathbf{1}_{\mu^{\mathcal{F}}}(z, g) \alpha_{\mu}^g(z, g) \right) (1 - \tau) S(x, y, z, g) \\ & - b - \beta \frac{m(\mathcal{U}, \mathcal{V})}{\mathcal{U}} \int \alpha(x, \tilde{y}, z, g) S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\ & + c - (1 - \beta) \frac{m(\mathcal{U}, \mathcal{V})}{\mathcal{V}} \int \alpha(\tilde{x}, y, z, g) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}} d\tilde{x} \\ & + \sum_{\check{z} \neq z} \lambda(z, \check{z}) \left\{ S(x, y, \check{z}, g + \check{\gamma}^g(z, \check{z}, g)) - S(x, y, z, g) \right. \\ & \quad \left. - \left((1 - \sigma(z, \check{z})) \mathbf{1}_{\sigma^{\mathcal{F}}}(z, \check{z}, g) \alpha_{\sigma}^g(z, \check{z}, g) + \sigma(z, \check{z}) \right) (1 - \tau) S(x, y, \check{z}, g + \check{\gamma}^g(z, \check{z}, g)) \right\} \\ & + \langle D_g S, \mu^g \rangle. \end{aligned} \quad (\text{A.8})$$

Setting $\mathbf{1}_{\mu^{\mathcal{F}}}(z, g) = \mathbf{1}_{\sigma^{\mathcal{F}}}(z, \check{z}, g) = 0$ for all (z, g) recovers the main text master equation (2.11).

A.1.6 Free Entry Condition

For a homothetic matching function, $\mathcal{V}$ satisfies the free entry condition (2.9):

$$\mathcal{V} = \mathcal{U} \widehat{m}^{-1} \left(\frac{c}{\frac{1}{\mathcal{U}} \int \int \alpha(\tilde{x}, \tilde{y}, z, g) g^u(\tilde{x}) (1 - \beta) S(\tilde{x}, \tilde{y}, z, g) d\tilde{x} d\tilde{y}} \right) \quad (\text{A.9})$$

A.1.7 Law of Motion and Endogenous Exit Rates

To close the model we derive $(\mu^{\mathcal{F}}, \gamma^{\mathcal{F}})$ and $(\alpha_{\mu}^g, \alpha_{\sigma}^g)$. Define:

$$I(z, g) := \int \int \alpha(x, y, z, g) g^u(x) S(x, y, z, g) dx dy.$$

The drift of I is:

$$\begin{aligned}\mu^I(z, g) = & \int \int \alpha(x, y, z, g) g^u(x) \langle D_g S, \mu^g \rangle dx dy \\ & - \int \int \alpha(x, y, z, g) \left(\int \mu^g(x, \tilde{y}, z, g) d\tilde{y} \right) S(x, y, z, g) dx dy\end{aligned}$$

where the first term captures the effect of the drifting distribution on S and the second captures the changing mass of unemployed workers (which moves opposite to the change in matched workers, since $g^u(x) = g^w(x) - \int g(x, \tilde{y}) d\tilde{y}$ and g^w is time-invariant). The jump of I at a z -transition is:

$$\begin{aligned}\gamma^I(z, \check{z}, g) = & \int \int \alpha(x, y, \check{z}, g + \gamma^g) \left(g^u(x) - \int \gamma^g(x, \tilde{y}, z, \check{z}, g) d\tilde{y} \right) S(x, y, \check{z}, g + \gamma^g) dx dy \\ & - \int \int \alpha(x, y, z, g) g^u(x) S(x, y, z, g) dx dy\end{aligned}$$

where $g + \gamma^g$ denotes $g + \gamma^g(z, \check{z}, g)$ and the term $-\int \gamma^g(x, \tilde{y}, \cdot) d\tilde{y}$ accounts for the change in the mass of unemployed workers as matches dissolve at the jump.

The drift and jump of $\mathcal{V}$ follow from differentiating (A.9), using $\mu^{\mathcal{U}} = -\mu^{\mathcal{P}}$ (since g^w is time-invariant):

$$\begin{aligned}\mu^{\mathcal{V}}(z, g) = & -\mu^{\mathcal{P}}(z, g) \widehat{m}^{-1} \left(\frac{c\mathcal{U}}{(1-\beta)I} \right) \\ & - \mathcal{U} (\widehat{m}^{-1})' \left(\frac{c\mathcal{U}}{(1-\beta)I} \right) \frac{c}{1-\beta} \cdot \frac{\mu^{\mathcal{P}} I + \mathcal{U} \mu^I}{I^2} \\ \gamma^{\mathcal{V}}(z, \check{z}, g) = & \left(\mathcal{U} - \int \int \gamma^g(\tilde{x}, \tilde{y}, z, \check{z}, g) d\tilde{x} d\tilde{y} \right) \widehat{m}^{-1} \left(\frac{c(\mathcal{U} - \int \int \gamma^g(\tilde{x}, \tilde{y}, z, \check{z}, g) d\tilde{x} d\tilde{y})}{(1-\beta)(I + \gamma^I)} \right) \\ & - \mathcal{U} \widehat{m}^{-1} \left(\frac{c\mathcal{U}}{(1-\beta)I} \right)\end{aligned}$$

where the arguments (z, g) are suppressed for brevity. The drift and jump of $\mathcal{F} = \mathcal{V} + \mathcal{P}$ are:

$$\begin{aligned}\mu^{\mathcal{F}}(z, g) &= \mu^{\mathcal{V}}(z, g) + \mu^{\mathcal{P}}(z, g) \\ \gamma^{\mathcal{F}}(z, \check{z}, g) &= \gamma^{\mathcal{V}}(z, \check{z}, g) + \gamma^{\mathcal{P}}(z, \check{z}, g).\end{aligned}$$

The endogenous exit rates are then:

$$\begin{aligned}\alpha_\mu^g(z, g) &:= -\varsigma(z) - \frac{\mu^{\mathcal{F}}(z, g)}{\mathcal{F}} \\ \alpha_\sigma^g(z, \check{z}, g) &:= -\sigma(z; \check{z}) - \frac{\gamma^{\mathcal{F}}(z, \check{z}, g)}{\mathcal{F}}.\end{aligned}$$

A.2 Verification of Recursive Equilibrium

In this section, we verify that a solution to our recursive equilibrium is a solution to a sequential equilibrium. We start by defining a sequential equilibrium for our environment. We then prove that our recursive equilibrium has bounded value functions and so satisfies “terminal transversality” conditions. We use these results to prove our main verification result that the solution to the surplus master equation and HJB equations solves the sequential equilibrium. Finally, we comment on the Hamiltonian and the “costate” for our environment.

Verification, existence, and uniqueness proofs for continuous-time, heterogeneous agent economies are well known to be difficult because infinite-dimensional calculus is very challenging (e.g. see [Cardaliaguet et al., 2019](#)). To keep this Appendix tractable, we restrict attention to the version of our model from Section 2.4 in which (x, y) are discretized to a grid. To echo the setup in the main text, throughout this section we assume there is no endogenous exit ($\mathbf{1}_{\mu^{\mathcal{F}}}(z, \check{z}, g) = 0$, $\mathbf{1}_{\sigma^{\mathcal{F}}}(z, g) = 0$ in the notation from Section A.1) in equilibrium. The high-level logic of the argument would be similar for the case in which (x, y) is continuous, but the mathematics would be significantly more complicated and beyond the scope of this paper.

A.2.1 Sequential Equilibrium

In the main text, we defined equilibrium recursively. We now set up and define a sequential equilibrium.

Worker problem: Fix a worker of type x . At each point in time, the worker’s individual employment state is $s_t \in \{u, e(y)\}_{y \in \mathcal{Y}}$, where $s_t = u$ denotes unemployment and $s_t = e(y)$ denotes employment at a firm of type y . The worker takes as given the aggregate law of motion for $(z_t, \underline{g}_t)$, the process for the wage w_t , the vacancy distribution induced by the aggregate state, and the transfer process $H_t^w(y, s)$ that compensates workers

when matches are broken up because firms are forced to exit. The worker chooses an admissible acceptance policy process $\alpha_t(x, y) \in \{0, 1\}$ which determines whether to accept a meeting with a firm of type y at time t .

Given a policy process α , the individual state s_t evolves as a controlled Markov jump process. Conditional on the aggregate state $(z_t, \underline{g}_t)$, unemployed workers meet firms of type y according to the intensity kernel $\mathcal{M}^u(z_t, \underline{g}_t) \frac{\underline{g}_t^v(y)}{\mathcal{V}_t}$. Upon a meeting with a firm of type y , the worker transitions from u to $e(y)$ if $\alpha_t(x, y) = 1$. If the worker is employed at a firm of type y , the match is destroyed at intensity $\delta(x, y, z_t) + \varsigma(z_t)$ in which case the worker transitions from $e(y)$ back to unemployment.

The aggregate state z_t evolves as a continuous-time Markov chain with transition intensities $\lambda(z, \check{z})$. When z_t jumps from z to $\check{z}$, the match distribution shifts to $\underline{g}_t^+ = (1 - \sigma(z, \check{z}))\underline{g}_t$. If the worker is unemployed, their individual state remains unchanged. If the worker is employed at firm y , then the match survives with probability $1 - \sigma(z, \check{z})$ and the worker remains employed at $e(y)$. With probability $\sigma(z, \check{z})$, the match is destroyed and the worker transitions to unemployment. Between aggregate shocks, the match distribution evolves according to the drift equation (2.2).

For an initial state $(s_0, z_0, \underline{g}_0)$, taking as given the evolution of the aggregate states $(z_t, \underline{g}_t)$ and the process for the wage w_t , the worker chooses policy α to solve:

$$\hat{W}^w(x, s_0, z_0, \underline{g}_0) = \sup_{\alpha} \mathbb{E}_0^{\alpha} \left[\int_0^{\infty} e^{-\rho t} r_t(x, s_t) dt + \sum_{n \geq 1} e^{-\rho T_n} H_{T_n}^w(y, s_{T_n}) \right] \quad (\text{A.10})$$

where the realized flow payoff is:

$$r_t(x, s_t) = \begin{cases} b, & \text{if } s_t = u, \\ w_t(x, y_t), & \text{if } s_t = e(y_t). \end{cases}$$

and $\{T_n\}_{n \geq 1}$ are a series of stopping times for when the worker receives transfers because of firm exit. Define the augmented recursive worker value function analogously:

$$\hat{V}^w(x, s, z, \underline{g}) = \begin{cases} V^u(x, z, \underline{g}), & \text{if } s = u, \\ V^e(x, y, z, \underline{g}), & \text{if } s = e(y). \end{cases}$$

Firm problem: The firm problem is defined analogously. Fix a firm of type y . At each point in time, the firm's individual state is $s_t \in \{v, m(x)\}_{x \in \mathcal{X}}$, where $s_t = v$

denotes a vacant firm and $s_t = m(x)$ denotes a filled match with a worker of type x . The firm takes as given the law of motion for the aggregate state $(z_t, \underline{g}_t)$, the process for the wage w_t , the worker search behavior induced by the aggregate state, and the transfer process $H_t^f(y, s)$ that compensates firms when they exit. The firm chooses an admissible acceptance policy $\alpha_t(x, y) \in \{0, 1\}$, which determines whether to hire an unemployed worker of type x when vacant.

Given policy α , the individual state s_t evolves as a controlled Markov jump process analogous to the worker problem. Conditional on the aggregate state $(z_t, \underline{g}_t)$, a vacant firm meets unemployed workers of type x according to the intensity kernel $\mathcal{M}^v(z_t, \underline{g}_t) \frac{\underline{g}_t^u(x)}{u_t}$. Upon meeting a worker of type x , the firm transitions from v to $m(x)$ if $\alpha_t(x, y) = 1$. If the firm is matched with worker x , the match is destroyed at intensity $\delta(x, y, z_t)$, in which case the firm transitions back to vacancy. Regardless of the firm's individual state, it is forced to exit the economy at rate $\varsigma(z_t)$ and with probability $\sigma(z_t, \tilde{z}_t)$ when the aggregate state changes.

For an initial state $(s_0, z_0, \underline{g}_0)$, taking as given the law of motion for the aggregate state $(z_t, \underline{g}_t)$, the process for the wage w_t , and the transfer process $H_t^f(y, s)$, the firm's sequential problem is:

$$\hat{W}^f(y, s_0, z_0, \underline{g}_0) = \sup_{\alpha} \mathbb{E}_0^{\alpha} \left[\int_0^{T^f} e^{-\rho t} r(y, s_t, z_t, \underline{g}_t) dt + e^{-\rho T^f} H_{T^f}^f(y, s_{T^f-}) \right] \quad (\text{A.11})$$

where the realized flow payoff is:

$$r(y, s_t, z_t, \underline{g}_t) = \begin{cases} -c, & \text{if } s_t = v, \\ F(x_t, y, z_t) - w_t, & \text{if } s_t = m(x_t), \end{cases}$$

and T^f is the stopping time at which the firm exits the economy. Define the augmented recursive firm value function analogously:

$$\hat{V}^f(x, s, z, \underline{g}) = \begin{cases} V^v(y, z, \underline{g}), & \text{if } s = v, \\ V^p(y, s, z, \underline{g}), & \text{if } s = m(x). \end{cases}$$

Definition 2 (Sequential Equilibrium). A sequential equilibrium is a collection of stochastic processes $\{s_t^w, s_t^f, z_t, g_t, w_t, \alpha_t, \underline{g}_t^f, H_t^w, H_t^f\}_{t \geq 0}$ such that:

- (i) Given the aggregate processes $\{z_t, \underline{g}_t, \underline{g}_t^f, w_t, H_t^w\}_{t \geq 0}$, the worker acceptance pro-

cess α_t solves the worker sequential problem (A.10).

- (ii) Given the aggregate processes $\{z_t, \underline{\mathbf{g}}_t, \underline{\mathbf{g}}_t^f, w_t, H_t^f\}_{t \geq 0}$, the firm acceptance process α_t solves the firm sequential problem (A.11).
- (iii) The match distribution process g_t evolves according to the law of motion induced by equilibrium matching and separation decisions (the non-recursive formulation of the KFE (2.1) in the main text).
- (iv) The firm distribution process $\underline{\mathbf{g}}_t^f$ satisfies the free-entry condition at all times.
- (v) Wages are determined according to the bargaining protocol for the model.

The sequential equilibrium is defined for potentially non-recursive acceptance policies and bargaining protocols. When we reconcile to the recursive equilibrium, we will focus on Nash bargaining and recursive acceptance decisions.

A.2.2 Boundedness and Verification Transversality Conditions

In this section, we show that a sufficiently regular collection of functions

$$\{V^u, V^e, V^v, V^p, S, w, \alpha, \underline{\mathbf{g}}_t^f\}$$

that solve the recursive equilibrium in Definition 1 in the main text (for the state space discretization in Section 2.4) satisfy the terminal value conditions required for the verification argument in the following Subsubsection A.2.3:

$$\begin{aligned} \lim_{T \rightarrow \infty} \mathbb{E}_0 \left[e^{-\rho T} S(x, y, z_T, \underline{\mathbf{g}}_T) \right] &= 0, \\ \lim_{T \rightarrow \infty} \mathbb{E}_0 \left[e^{-\rho T} \hat{V}^w(x, s_T, z_T, \underline{\mathbf{g}}_T) \right] &= 0, \\ \lim_{T \rightarrow \infty} \mathbb{E}_0 \left[e^{-\rho T} \hat{V}^f(y, s_T, z_T, \underline{\mathbf{g}}_T) \right] &= 0. \end{aligned}$$

We prove results under the following set of assumptions. In particular, we work a finite type space like our model from Section 2.4 in which x and y are discretized to a grid.

Assumption 1 (Finite-Dimensional Boundedness and Compactness Conditions). The following hold:

- (i) The type spaces $\mathcal{X}$ and $\mathcal{Y}$ are finite and the aggregate state space $\mathcal{Z}$ is finite.
- (ii) The measure function of workers is $\underline{g}_t^w(x) = 1$ for all x and new firms draw types from a uniform distribution across $\mathcal{Y}$. The feasible set of match distributions

$$\mathcal{G} = \left\{ \underline{g} \in \mathbb{R}_+^{|\mathcal{X}| \times |\mathcal{Y}|} : \sum_{y \in \mathcal{Y}} \underline{g}(x, y) \leq g^w(x) \text{ for all } x \in \mathcal{X} \right\}$$

is compact, which in our environment follows from the previous assumptions but we state explicitly for clarity.

- (iii) The production function F is uniformly bounded:

$$\underline{F} \leq F(x, y, z) \leq \bar{F} \quad \forall (x, y, z).$$

- (iv) The exogenous transition intensities $\delta(x, y, z)$, $\varsigma(z)$, and $\lambda(z, \check{z})$ are uniformly bounded.
- (v) The matching function leads to uniformly bounded matching rates (e.g. for a truncated Cobb-Douglas matching function):

$$\mathcal{M}^u = \frac{m(\mathcal{U}, \mathcal{V})}{\mathcal{U}} \leq \bar{\mathcal{M}}^u < \infty, \quad \mathcal{M}^v = \frac{m(\mathcal{U}, \mathcal{V})}{\mathcal{V}} \leq \bar{\mathcal{M}}^v < \infty.$$

- (vi) The functions S, V^u, V^e, V^v, V^p are continuously differentiable with respect to $\underline{g}$ on $\mathcal{G}$. The KFE drift $\mu^g(z, \underline{g})$ is continuous in $\underline{g}$ and maps feasible states into feasible paths.

Assumption (i) specializes the verification argument to the finite-dimensional economy used in the computational implementation. Assumptions (ii), (iii), and (iv) are parametric restrictions that we impose throughout the paper. Assumption (v) replaces the standard Cobb-Douglas matching function with a truncated version that caps meeting rates (which is typically what is done in numerical implementations). This is a minor perturbation that has little effect on the quantitative results for reasonable parameter values, but ensures that meeting rates do not explode when the mass of vacancies or unemployed workers is small.

Since $\mathcal{X}$, $\mathcal{Y}$, and $\mathcal{Z}$ are finite and $\mathcal{G}$ is compact, any continuous recursive equilibrium object is uniformly bounded on the feasible state space. We exploit this to

bound S and the derivative operators.

Lemma 1 (Uniform Bound on Surplus and Value Functions). *Suppose Assumption 1 holds. Then there exists a finite constant $\bar{S} < \infty$ and $\{\bar{V}^j\}_{j \in \{u,e,v,p\}}$ such that $\forall (x, y, z, \underline{g}) \in \mathcal{X} \times \mathcal{Y} \times \mathcal{Z} \times \mathcal{G}$:*

$$\begin{aligned} |S(x, y, z, \underline{g})| &\leq \bar{S}, \\ |V^u(x, z, \underline{g})| &\leq \bar{V}^u, \\ |V^e(x, y, z, \underline{g})| &\leq \bar{V}^e, \\ |V^v(y, z, \underline{g})| &\leq \bar{V}^v, \\ |V^p(x, y, z, \underline{g})| &\leq \bar{V}^p. \end{aligned}$$

Moreover, the aggregate generator term $\mathcal{A}^{z,g}S$ for the surplus process is uniformly bounded.

Proof. Since $\mathcal{X}$, $\mathcal{Y}$, and $\mathcal{Z}$ are finite and $\mathcal{G}$ is compact, the feasible state space $\mathcal{X} \times \mathcal{Y} \times \mathcal{Z} \times \mathcal{G}$ is compact. By Assumption 1(vi), $S(x, y, z, \underline{g})$ is continuous in $\underline{g}$ for each finite tuple (x, y, z) . Therefore S is continuous on the compact feasible state space. By the extreme value theorem, there exists $\bar{S} < \infty$ such that:

$$|S(x, y, z, \underline{g})| \leq \bar{S} \quad \forall (x, y, z, \underline{g}).$$

An analogous argument applies to the value functions $\{V^j\}_{j \in \{u,e,v,p\}}$.

It remains to note that the aggregate generator term is also uniformly bounded. With exogenous jump exit and compensation parameter τ , from the general Master equation (A.8) the relevant aggregate jump operator for surplus is:

$$\begin{aligned} \mathcal{A}^{z,g,\tau}S(x, y, z, \underline{g}) &= \sum_{\check{z} \neq z} \lambda(z, \check{z}) \left[S(x, y, \check{z}, (1 - \sigma(z, \check{z}))\underline{g}) - S(x, y, z, \underline{g}) \right. \\ &\quad \left. - \sigma(z, \check{z})(1 - \tau)S(x, y, \check{z}, (1 - \sigma(z, \check{z}))\underline{g}) \right] \\ &\quad + \langle D_g S(x, y, z, \underline{g}), \mu^g(z, \underline{g}) \rangle. \end{aligned}$$

The jump part is uniformly bounded because $\mathcal{Z}$ is finite, $\lambda(z, \check{z})$, $\sigma(z, \check{z})$, and τ are

bounded, and S is uniformly bounded:

$$\begin{aligned} & \left| \sum_{\check{z} \neq z} \lambda(z, \check{z}) \left[S(x, y, \check{z}, (1 - \sigma)\underline{\mathbf{g}}) - S(x, y, z, \underline{\mathbf{g}}) - \sigma(1 - \tau)S(x, y, \check{z}, (1 - \sigma)\underline{\mathbf{g}}) \right] \right| \\ & \leq \bar{S} \sum_{\check{z} \neq z} \lambda(z, \check{z}) (2 + \sigma(z, \check{z})|1 - \tau|) < \infty. \end{aligned}$$

Since $D_g S$ and μ^g are continuous and $\mathcal{G}$ is compact, both are uniformly bounded. Hence

$$\left| \left\langle D_g S(x, y, z, \underline{\mathbf{g}}), \mu^g(z, \underline{\mathbf{g}}) \right\rangle \right| < \infty.$$

□

Corollary 1 (Limiting Terminal Value Conditions). *Under Assumption 1,*

$$\begin{aligned} \lim_{T \rightarrow \infty} \mathbb{E}_0 \left[e^{-\rho T} S(x, y, z_T, \underline{\mathbf{g}}_T) \right] &= 0, \\ \lim_{T \rightarrow \infty} \mathbb{E}_0 \left[e^{-\rho T} \hat{V}^w(x, s_T, z_T, \underline{\mathbf{g}}_T) \right] &= 0, \\ \lim_{T \rightarrow \infty} \mathbb{E}_0 \left[e^{-\rho T} \hat{V}^f(y, s_T, z_T, \underline{\mathbf{g}}_T) \right] &= 0. \end{aligned}$$

Proof. By Lemma 1 $|S(x, y, z, \underline{\mathbf{g}})| \leq \bar{S}$ uniformly over $(x, y, z, \underline{\mathbf{g}})$. Hence:

$$\left| \mathbb{E}_0 \left[e^{-\rho T} S(x, y, z_T, \underline{\mathbf{g}}_T) \right] \right| \leq e^{-\rho T} \bar{S} \xrightarrow{T \rightarrow \infty} 0.$$

An analogous argument applies for the value functions. □

Lemma 2 (Uniform Bound on Wages). *Under Lemma 1 and Assumption 1(ii)-(v), there exists a finite constant $\bar{w} < \infty$ such that:*

$$|w(x, y, z, \underline{\mathbf{g}})| \leq \bar{w} \quad \forall (x, y, z, \underline{\mathbf{g}}).$$

Proof. By Nash bargaining $V^e(x, y, z, \underline{\mathbf{g}}) - V^u(x, z, \underline{\mathbf{g}}) = \beta S(x, y, z, \underline{\mathbf{g}})$. Subtracting the unemployed worker HJB equation (2.5) from the employed worker HJB equation

(A.5) at general τ with $\mathbf{1}_{\mu^F}(z, g) = \mathbf{1}_{\sigma^F}(z, \check{z}, g) = 0$ gives:

$$\begin{aligned} w(x, y, z, \underline{g}) - b &= \rho \beta S(x, y, z, \underline{g}) \\ &\quad + \left(\delta(x, y, z) + (1 - \tau) \varsigma(z) \right) \beta S(x, y, z, \underline{g}) \\ &\quad + \mathcal{M}^u \sum_{\tilde{y} \in \mathcal{Y}} \alpha(x, \tilde{y}, z, \underline{g}) \beta S(x, \tilde{y}, z, \underline{g}) \frac{\underline{g}^v(\tilde{y})}{\mathcal{V}} \\ &\quad - \mathcal{A}^{z, g, \tau}(\beta S)(x, y, z, \underline{g}). \end{aligned}$$

Each term on the right-hand side is uniformly bounded by Lemma 1 and Assumption 1. Therefore w is uniformly bounded. $\square$

Comparison to other economic environments: The boundedness result here contrasts sharply with other economic environments, such as recursive asset pricing or economic growth models. In standard asset pricing models, endogenous variables such as wealth, capital gains, or asset prices may evolve on unbounded state spaces, and continuation values may inherit explosive dynamics from forward-looking pricing equations. As a result, recursive equilibrium objects need not be uniformly bounded and transversality conditions may need to be imposed explicitly.

By contrast, our environment has strong mean-reverting forces. First, the endogenous aggregate state, the match distribution $\underline{g}$, evolves on a bounded feasible set because total worker mass is finite and type spaces are compact. Second, flow payoffs are uniformly bounded by Assumption 1(iii). Third, discounting, exogenous match destruction, and free entry jointly prevent market tightness and meeting rates from exploding. Consequently, the recursive equilibrium objects are uniformly bounded.

A.2.3 Verification

We now verify that the recursive equilibrium (Definition 1) induces a sequential equilibrium (Definition 2). The argument is the continuous-time analogue of the standard Bellman verification argument from [Stokey and Lucas Jr \(1989\)](#). That is, the HJB equations imply a supermartingale for every admissible policy and a martingale under the candidate optimal policy. The terminal value conditions from Corollary 1 then allow us to pass to the infinite horizon.

Proposition 1 (Verification of Recursive Equilibrium). *Suppose the recursive objects*

$$\{V^u, V^e, V^v, V^p, S, w, \alpha, \underline{\mathbf{g}}^f\}$$

solve the recursive equilibrium in Definition 1 with $\mathbf{1}_{\mu^f}(z, g) = \mathbf{1}_{\sigma^f}(z, \check{z}, g) = 0$ and general τ , and satisfy Assumption 1. Define the transfer processes by:

$$\begin{aligned} H_t^f(y, s) &:= \tau \hat{V}^f(y, s, z_t, \underline{\mathbf{g}}_t), \\ H_t^w(x, e(y)) &:= \tau \left[\hat{V}^w(x, e(y), z_t, \underline{\mathbf{g}}_t) - \hat{V}^w(x, u, z_t, \underline{\mathbf{g}}_t) \right], \quad H_t^w(x, u) := 0. \end{aligned}$$

Then the induced Markov processes

$$w_t = w(x_t, y_t, z_t, \underline{\mathbf{g}}_t), \quad \alpha_t = \alpha(x_t, y_t, z_t, \underline{\mathbf{g}}_t), \quad g_t^f = g^f(z_t, \underline{\mathbf{g}}_t), \quad H_t^w, \quad H_t^f$$

form a sequential equilibrium in Definition 2. Moreover,

$$\hat{W}^w(x, s, z, \underline{\mathbf{g}}) = \hat{V}^w(x, s, z, \underline{\mathbf{g}}), \quad \hat{W}^f(y, s, z, \underline{\mathbf{g}}) = \hat{V}^f(y, s, z, \underline{\mathbf{g}}).$$

Proof. We first prove the result for firms and then for workers.

Firm problem: Fix firm type y and initial state $(s_0, z_0, \underline{\mathbf{g}}_0)$. Let $\bar{T} < \infty$ denote a deterministic verification horizon and let T^f denote the stopping time at which the firm exits the economy. We use an augmented generator that incorporates the transfer paid when the firm exits. Formally, let $\mathcal{A}^{\tilde{\alpha}, f}$ denote the augmented infinitesimal generator of the controlled process $(s_t, z_t, \underline{\mathbf{g}}_t)$ under admissible policy $\tilde{\alpha}$. For a vacant firm,

$$\begin{aligned} &\mathcal{A}^{\tilde{\alpha}, f} \Phi^f(y, v, z, \underline{\mathbf{g}}) \\ &= \mathcal{M}^v(z, \underline{\mathbf{g}}) \sum_{x \in \mathcal{X}} \tilde{\alpha}(x, y, z, \underline{\mathbf{g}}) \left[\Phi^f(y, m(x), z, \underline{\mathbf{g}}) - \Phi^f(y, v, z, \underline{\mathbf{g}}) \right] \frac{\underline{\mathbf{g}}^u(x)}{\mathcal{U}} \\ &\quad + \varsigma(z) \left[\tau \Phi^f(y, v, z, \underline{\mathbf{g}}) - \Phi^f(y, v, z, \underline{\mathbf{g}}) \right] \\ &\quad + \sum_{\check{z} \neq z} \lambda(z, \check{z}) \left[\Phi^f(y, v, \check{z}, (1 - \sigma(z, \check{z})) \underline{\mathbf{g}}) - \Phi^f(y, v, z, \underline{\mathbf{g}}) \right. \\ &\quad \quad \left. - \sigma(z, \check{z})(1 - \tau) \Phi^f(y, v, \check{z}, (1 - \sigma(z, \check{z})) \underline{\mathbf{g}}) \right] \\ &\quad + \left\langle D_g \Phi^f(y, v, z, \underline{\mathbf{g}}), \mu^g(z, \underline{\mathbf{g}}) \right\rangle. \end{aligned}$$

For a producing firm,

$$\begin{aligned}
& \mathcal{A}^{\tilde{\alpha},f} \Phi^f(y, m(x), z, \underline{g}) \\
&= \delta(x, y, z) \left[\Phi^f(y, v, z, \underline{g}) - \Phi^f(y, m(x), z, \underline{g}) \right] \\
&+ \varsigma(z) \left[\tau \Phi^f(y, m(x), z, \underline{g}) - \Phi^f(y, m(x), z, \underline{g}) \right] \\
&+ \sum_{\check{z} \neq z} \lambda(z, \check{z}) \left[\Phi^f(y, m(x), \check{z}, (1 - \sigma(z, \check{z})) \underline{g}) - \Phi^f(y, m(x), z, \underline{g}) \right. \\
&\quad \left. - \sigma(z, \check{z})(1 - \tau) \Phi^f(y, m(x), \check{z}, (1 - \sigma(z, \check{z})) \underline{g}) \right] \\
&+ \left\langle D_g \Phi^f(y, m(x), z, \underline{g}), \mu^g(z, \underline{g}) \right\rangle.
\end{aligned}$$

We now show that the recursive value function $\hat{V}^f$ can be written sequentially to reconcile with $\hat{W}^f$. Applying Dynkin's formula to $\hat{V}^f$ over $[0, \bar{T}]$ gives

$$\begin{aligned}
\hat{V}^f(y, s_0, z_0, \underline{g}_0) &= \mathbb{E}_0^{\tilde{\alpha}} \left[e^{-\rho \bar{T}} \hat{V}^f(y, s_{\bar{T}}, z_{\bar{T}}, \underline{g}_{\bar{T}}) \right. \\
&\quad \left. - \int_0^{\bar{T}} e^{-\rho t} \left(\mathcal{A}^{\tilde{\alpha},f} \hat{V}^f(y, s_t, z_t, \underline{g}_t) - \rho \hat{V}^f(y, s_t, z_t, \underline{g}_t) \right) dt \right].
\end{aligned}$$

The firm HJB equations at general τ with $\mathbf{1}_{\mu^f}(z, g) = \mathbf{1}_{\sigma^f}(z, \check{z}, g) = 0$ imply the pointwise inequality

$$\mathcal{A}^{\tilde{\alpha},f} \hat{V}^f(y, s, z, \underline{g}) - \rho \hat{V}^f(y, s, z, \underline{g}) \leq -r(y, s, z, \underline{g})$$

for every admissible policy $\tilde{\alpha}$, with equality under the recursive acceptance rule $\alpha^*(x, y, z, \underline{g}) = \mathbf{1}\{S(x, y, z, \underline{g}) \geq 0\}$. Therefore,

$$\begin{aligned}
\hat{V}^f(y, s_0, z_0, \underline{g}_0) &\geq \mathbb{E}_0^{\tilde{\alpha}} \left[\int_0^{\bar{T}} e^{-\rho t} r(y, s_t, z_t, \underline{g}_t) dt + e^{-\rho \bar{T}} \hat{V}^f(y, s_{\bar{T}}, z_{\bar{T}}, \underline{g}_{\bar{T}}) \right] \\
&= \mathbb{E}_0^{\tilde{\alpha}} \left[\int_0^{\bar{T} \wedge T^f} e^{-\rho t} r(y, s_t, z_t, \underline{g}_t) dt + \mathbf{1}_{\{T^f \leq \bar{T}\}} e^{-\rho T^f} H_{T^f}^f(y, s_{T^f-}) \right. \\
&\quad \left. + \mathbf{1}_{\{T^f > \bar{T}\}} e^{-\rho \bar{T}} \hat{V}^f(y, s_{\bar{T}}, z_{\bar{T}}, \underline{g}_{\bar{T}}) \right].
\end{aligned}$$

Taking $\bar{T} \rightarrow \infty$, the terminal term vanishes by Corollary 1. Hence

$$\hat{V}^f(y, s_0, z_0, \underline{g}_0) \geq \mathbb{E}_0^{\tilde{\alpha}} \left[\int_0^{T^f} e^{-\rho t} r(y, s_t, z_t, \underline{g}_t) dt + e^{-\rho T^f} H_{T^f}^f(y, s_{T^f-}) \right].$$

Under α^* the HJB holds with equality, so the inequality is attained. Therefore α^* solves the firm sequential problem and

$$\hat{W}^f(y, s, z, \underline{g}) = \hat{V}^f(y, s, z, \underline{g}).$$

Worker problem: Fix worker type x and initial state $(s_0, z_0, \underline{g}_0)$. Workers do not exit the economy permanently. Instead, when an employed worker's firm exits, the worker receives a transfer and transitions to unemployment. Let $\{T_n^w\}_{n \geq 1}$ denote the stopping times at which the worker receives such transfers.

For any test function $\Phi^w(x, s, z, \underline{g})$, let $\mathcal{A}^{\tilde{\alpha}, w}$ denote the augmented generator of the worker's controlled process. For an unemployed worker,

$$\begin{aligned} \mathcal{A}^{\tilde{\alpha}, w} \Phi^w(x, u, z, \underline{g}) &= \mathcal{M}^u(z, \underline{g}) \sum_{y \in \mathcal{Y}} \tilde{\alpha}(x, y, z, \underline{g}) \left[\Phi^w(x, e(y), z, \underline{g}) - \Phi^w(x, u, z, \underline{g}) \right] \frac{\underline{g}^v(y)}{\nu} \\ &\quad + \sum_{\check{z} \neq z} \lambda(z, \check{z}) \left[\Phi^w(x, u, \check{z}, (1 - \sigma(z, \check{z}))\underline{g}) - \Phi^w(x, u, z, \underline{g}) \right] \\ &\quad + \left\langle D_g \Phi^w(x, u, z, \underline{g}), \mu^g(z, \underline{g}) \right\rangle. \end{aligned}$$

For an employed worker,

$$\begin{aligned} \mathcal{A}^{\tilde{\alpha}, w} \Phi^w(x, e(y), z, \underline{g}) &= \delta(x, y, z) \left[\Phi^w(x, u, z, \underline{g}) - \Phi^w(x, e(y), z, \underline{g}) \right] \\ &\quad + \varsigma(z) \left[\Phi^w(x, u, z, \underline{g}) + \tau \left(\Phi^w(x, e(y), z, \underline{g}) - \Phi^w(x, u, z, \underline{g}) \right) - \Phi^w(x, e(y), z, \underline{g}) \right] \\ &\quad + \sum_{\check{z} \neq z} \lambda(z, \check{z}) \left[\Phi^w(x, e(y), \check{z}, (1 - \sigma(z, \check{z}))\underline{g}) - \Phi^w(x, e(y), z, \underline{g}) \right. \\ &\quad \left. + \sigma(z, \check{z})(1 - \tau) \left(\Phi^w(x, u, \check{z}, (1 - \sigma(z, \check{z}))\underline{g}) - \Phi^w(x, e(y), \check{z}, (1 - \sigma(z, \check{z}))\underline{g}) \right) \right] \\ &\quad + \left\langle D_g \Phi^w(x, e(y), z, \underline{g}), \mu^g(z, \underline{g}) \right\rangle. \end{aligned}$$

We again show that the recursive value function $\hat{V}^w$ can be written sequentially to reconcile with $\hat{W}^w$. Applying Dynkin's formula to $\hat{V}^w$ over $[0, \bar{T}]$ gives:

$$\hat{V}^w(x, s_0, z_0, \underline{g}_0) = \mathbb{E}_0^{\tilde{\alpha}} \left[e^{-\rho \bar{T}} \hat{V}^w(x, s_{\bar{T}}, z_{\bar{T}}, \underline{g}_{\bar{T}}) \right]$$

$$- \int_0^{\bar{T}} e^{-\rho t} \left(\mathcal{A}^{\tilde{\alpha}, w} \hat{V}^w(x, s_t, z_t, \underline{\mathbf{g}}_t) - \rho \hat{V}^w(x, s_t, z_t, \underline{\mathbf{g}}_t) \right) dt \Big].$$

The worker HJB at general τ with $\mathbf{1}_{\mu^F}(z, g) = \mathbf{1}_{\sigma^F}(z, \check{z}, g) = 0$ implies the pointwise inequality

$$\mathcal{A}^{\tilde{\alpha}, w} \hat{V}^w(x, s, z, \underline{\mathbf{g}}) - \rho \hat{V}^w(x, s, z, \underline{\mathbf{g}}) \leq -r(x, s, z, \underline{\mathbf{g}})$$

for every admissible policy $\tilde{\alpha}$, with equality under the recursive acceptance rule α^* . Therefore,

$$\begin{aligned} \hat{V}^w(x, s_0, z_0, \underline{\mathbf{g}}_0) &\geq \mathbb{E}_0^{\tilde{\alpha}} \left[\int_0^{\bar{T}} e^{-\rho t} r(x, s_t, z_t, \underline{\mathbf{g}}_t) dt + e^{-\rho \bar{T}} \hat{V}^w(x, s_{\bar{T}}, z_{\bar{T}}, \underline{\mathbf{g}}_{\bar{T}}) \right] \\ &= \mathbb{E}_0^{\tilde{\alpha}} \left[\int_0^{\bar{T}} e^{-\rho t} r(x, s_t, z_t, \underline{\mathbf{g}}_t) dt + \sum_{n: T_n^w \leq \bar{T}} e^{-\rho T_n^w} H_{T_n^w}^w(x, s_{T_n^w-}) \right. \\ &\quad \left. + e^{-\rho \bar{T}} \hat{V}^w(x, s_{\bar{T}}, z_{\bar{T}}, \underline{\mathbf{g}}_{\bar{T}}) \right]. \end{aligned}$$

Taking $\bar{T} \rightarrow \infty$, the terminal term vanishes by Corollary 1, giving

$$\hat{V}^w(x, s_0, z_0, \underline{\mathbf{g}}_0) \geq \mathbb{E}_0^{\tilde{\alpha}} \left[\int_0^\infty e^{-\rho t} r(x, s_t, z_t, \underline{\mathbf{g}}_t) dt + \sum_{n \geq 1} e^{-\rho T_n^w} H_{T_n^w}^w(x, s_{T_n^w-}) \right].$$

Under α^* the HJB holds with equality, so the inequality is attained. Therefore α^* solves the worker sequential problem and

$$\hat{W}^w(x, s, z, \underline{\mathbf{g}}) = \hat{V}^w(x, s, z, \underline{\mathbf{g}}).$$

Other equilibrium objects: The KFE, free-entry condition, and bargaining protocol hold by the definition of recursive equilibrium. Therefore the recursive equilibrium induces a sequential equilibrium. $\square$

A.2.4 Comment on Transversality Conditions and Co-states

It is helpful to compare the verification argument above with the stochastic maximum principle for controlled jump processes. [Davis and Elliott \(1977\)](#) (Corollary 5.2) write

the Hamiltonian for a controlled jump process in the form:

$$N(t, \alpha) = \zeta^\alpha(t) \left(\int_X f^\alpha(t, x) \beta^\alpha(t, x) d\lambda(x) - \eta(t) \right).$$

Here α is the control, $\zeta^\alpha(t)$ is the controlled jump intensity, $\beta^\alpha(t, x) d\lambda(x)$ is the conditional distribution of the post-jump mark, $f^\alpha(t, x)$ is the continuation value after a jump to mark x , and $\eta(t)$ is the continuation value if no jump occurs. Thus the Hamiltonian is the controlled jump intensity multiplied by the expected gain from the jump.

We can map this into our environment. For an unemployed worker, the relevant controlled jump is the transition from unemployment to employment at a firm of type y . The controlled jump intensity is the matching rate under acceptance policy α :

$$\zeta^\alpha(t) = \mathcal{M}^u(z_t, \underline{\mathbf{g}}_t) \sum_{y \in \mathcal{Y}} \alpha(x, y, z_t, \underline{\mathbf{g}}_t) \frac{\underline{\mathbf{g}}_t^v(y)}{\mathcal{V}_t}.$$

The post-jump conditional density is the firm density weighted by the acceptance policy α :

$$\beta^\alpha(t, y) d\lambda(y) = \frac{\alpha(x, y, z_t, \underline{\mathbf{g}}_t) \frac{\underline{\mathbf{g}}_t^v(y)}{\mathcal{V}_t}}{\int \alpha(x, \tilde{y}, z_t, \underline{\mathbf{g}}_t) \frac{\underline{\mathbf{g}}_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y}}.$$

The post-jump continuation and no-jump continuation values are:

$$\begin{aligned} f^\alpha(t, y) &= V^e(x, y, z_t, \underline{\mathbf{g}}_t), \\ \eta(t) &= V^u(x, z_t, \underline{\mathbf{g}}_t). \end{aligned}$$

Therefore:

$$\begin{aligned} N(t, \alpha) &= \zeta^\alpha(t) \left(\int V^e(x, y, z_t, \underline{\mathbf{g}}_t) \beta^\alpha(t, y) d\lambda(y) - V^u(x, z_t, \underline{\mathbf{g}}_t) \right) \\ &= \mathcal{M}^u(z_t, \underline{\mathbf{g}}_t) \sum_{y \in \mathcal{Y}} \alpha(x, y, z_t, \underline{\mathbf{g}}_t) V^e(x, y, z_t, \underline{\mathbf{g}}_t) \frac{\underline{\mathbf{g}}_t^v(y)}{\mathcal{V}_t} \\ &\quad - \mathcal{M}^u(z_t, \underline{\mathbf{g}}_t) \sum_{y \in \mathcal{Y}} \alpha(x, y, z_t, \underline{\mathbf{g}}_t) V^u(x, z_t, \underline{\mathbf{g}}_t) \frac{\underline{\mathbf{g}}_t^v(y)}{\mathcal{V}_t} \\ &= \mathcal{M}^u(z_t, \underline{\mathbf{g}}_t) \sum_{y \in \mathcal{Y}} \alpha(x, y, z_t, \underline{\mathbf{g}}_t) \left[V^e(x, y, z_t, \underline{\mathbf{g}}_t) - V^u(x, z_t, \underline{\mathbf{g}}_t) \right] \frac{\underline{\mathbf{g}}_t^v(y)}{\mathcal{V}_t}. \end{aligned}$$

Thus the worker jump gain associated with a transition to firm type y is:

$$\Lambda_t^w(x, y) = V^e(x, y, z_t, \underline{\mathbf{g}}_t) - V^u(x, z_t, \underline{\mathbf{g}}_t),$$

while the corresponding integrated jump gain (or “co-state”) entering the Hamiltonian is:

$$\Lambda_t^w(x) = \sum_{y \in \mathcal{Y}} \alpha(x, y, z_t, \underline{\mathbf{g}}_t) \left[V^e(x, y, z_t, \underline{\mathbf{g}}_t) - V^u(x, z_t, \underline{\mathbf{g}}_t) \right] \frac{\underline{\mathbf{g}}_t^v(y)}{\mathcal{V}_t}.$$

The firm-side mapping is analogous. For a vacant firm meeting a worker of type x , the post-jump value is $V^p(x, y, z, \underline{\mathbf{g}})$ and the no-jump value is $V^v(y, z, \underline{\mathbf{g}})$. Thus the firm jump gain associated with a transition to worker type x is:

$$\Lambda_t^f(x, y) = V^p(x, y, z_t, \underline{\mathbf{g}}_t) - V^v(y, z_t, \underline{\mathbf{g}}_t),$$

while the corresponding integrated jump gain (or “co-state”) entering the Hamiltonian is:

$$\Lambda_t^f(y) = \sum_{x \in \mathcal{X}} \alpha(x, y, z_t, \underline{\mathbf{g}}_t) \left[V^p(x, y, z_t, \underline{\mathbf{g}}_t) - V^v(y, z_t, \underline{\mathbf{g}}_t) \right] \frac{\underline{\mathbf{g}}_t^u(x)}{\mathcal{U}_t}.$$

Using Nash bargaining, the jump gains are proportional to the surplus:

$$\begin{aligned} \Lambda_t^w(x, y) &= V^e(x, y, z_t, \underline{\mathbf{g}}_t) - V^u(x, z_t, \underline{\mathbf{g}}_t) = \beta S(x, y, z_t, \underline{\mathbf{g}}_t), \\ \Lambda_t^f(x, y) &= V^p(x, y, z_t, \underline{\mathbf{g}}_t) - V^v(y, z_t, \underline{\mathbf{g}}_t) = (1 - \beta) S(x, y, z_t, \underline{\mathbf{g}}_t). \end{aligned}$$

Hence the relevant jump co-state is essentially the integrated surplus function $S(x, y, z_t, \underline{\mathbf{g}}_t)$ and the surplus function is uniformly bounded by Lemma 1.

This differs from the usual co-state transversality condition in optimal control problems with continuous idiosyncratic state x and smooth choice c (e.g. a capital accumulation problem in economics). In such a problem, perturbing the control generates a first-order perturbation of the future state and so the co-state transversality condition then takes the familiar form:

$$\lim_{T \rightarrow \infty} e^{-\rho T} \lambda_T x_T = 0.$$

In the present environment, the individual state is categorical: a worker is either unemployed or employed at a particular firm type, and a firm is either vacant or matched with a particular worker type. The control changes the intensity of jumps across these states rather than perturbing a continuous state variable. Therefore the natural marginal object is not a derivative with respect to an individual state variable, but the value difference across jump states.

Because these value differences are uniformly bounded, discounting implies that their discounted tails vanish:

$$\lim_{T \rightarrow \infty} \mathbb{E}_0 \left[e^{-\rho T} \left(V^e(x, y, z_T, \underline{g}_T) - V^u(x, z_T, \underline{g}_T) \right) \right] = 0,$$

for all (x, y) and similarly for the firm-side co-state. This can be viewed as the jump-process analogue of the usual co-state tail condition.

A.3 Discrete Time Version

We consider a discrete time version of our model from Section 2 with time step $\Delta > 0$. As in the main text, for notational convenience, we write the model under the assumption that there is no endogenous firm exit and that exiting agents are fully compensated for their lost value ($\mathbf{1}_{\mu^F}(z, g) = 0$, $\mathbf{1}_{\sigma^F}(z, \tilde{z}, g) = 0$, $\tau = 1$). The continuous-time model in Section 2.1 is obtained as the limit as $\Delta \rightarrow 0$. Let $\beta_\Delta := e^{-\rho\Delta}$ denote the one-period discount factor.

The timing within a period is as follows. At the beginning of period t , the aggregate state is (z_t, g_t) . Agents receive flow payoffs over the interval $[t, t + \Delta)$. Unemployed workers and vacant firms meet during the period, and accepted meetings form new matches at $t + \Delta$. Existing matches separate exogenously during the period with probability $\delta(x, y, z_t)\Delta$. Firms exit exogenously during the period with probability $\varsigma(z_t)\Delta$, and exiting agents are fully compensated for their lost value. Finally, the aggregate state transitions from z to $\tilde{z}$ with probability $P_\Delta(z, \tilde{z})$, where:

$$P_\Delta(z, \tilde{z}) = \begin{cases} \lambda(z, \tilde{z})\Delta + o(\Delta), & \tilde{z} \neq z, \\ 1 - \sum_{\bar{z} \neq z} \lambda(z, \bar{z})\Delta + o(\Delta), & \tilde{z} = z. \end{cases}$$

where we use the notation $f(\Delta) = o(\Delta)$ to denote a function satisfying $\lim_{\Delta \rightarrow 0} f(\Delta)/\Delta = 0$, so that $f(\Delta)$ becomes negligible relative to Δ as $\Delta \rightarrow 0$. If the aggregate state

changes from z to $\check{z}$, a fraction $\sigma(z, \check{z})$ of firms exits and the match distribution is scaled by $1 - \sigma(z, \check{z})$; exiting agents are again fully compensated.

A.3.1 Discrete-Time Law of Motion for Matches

Given the acceptance rule $\alpha(x, y, z, g)$, the match distribution evolves according to

$$\begin{aligned}\Gamma_{t+\Delta}(x, y) &= g_t(x, y) - (\delta(x, y, z_t)\Delta + \varsigma(z_t)\Delta) g_t(x, y) \\ &\quad + \frac{m(\mathcal{U}_t, \mathcal{V}_t)}{\mathcal{U}_t \mathcal{V}_t} \alpha(x, y, z_t, g_t) g_t^u(x) g_t^v(y) \Delta,\end{aligned}$$

where $\Gamma_{t+\Delta}$ denotes the post-matching, pre-aggregate-shock match distribution. Here $(\mathcal{U}, \mathcal{V}, g^u, g^v)$ are computed from (z, g) using Table 2 and the free-entry condition. If the aggregate state then jumps from z_t to $\check{z}$, the beginning-of-next-period match distribution is:

$$g_{t+\Delta|\check{z}}(x, y) = (1 - \sigma(z_t, \check{z})) \Gamma_{t+\Delta}(x, y).$$

To echo the continuous time formulation in the main text, we write the transition recursively using the definitions:

$$\begin{aligned}\Gamma_{\Delta}^g(x, y, z, g) &:= [1 - \delta(x, y, z)\Delta - \varsigma(z)\Delta] g(x, y) \\ &\quad + \frac{m(\mathcal{U}, \mathcal{V})}{\mathcal{U}\mathcal{V}} \alpha(x, y, z, g) g^u(x) g^v(y) \Delta\end{aligned}$$

so the next-period distribution conditional on $z' = \check{z}$ is:

$$g'_{\check{z}} = (1 - \sigma(z, \check{z})) \Gamma_{\Delta}^g(z, g).$$

A.3.2 Surplus Division and Acceptance Decisions

Let $V^u(x, z, g)$ denote the value of an unemployed worker of type x at the beginning of the period, $V^e(x, y, z, g)$ denote the value of a worker of type x employed at a firm of type y at the beginning of the period, $V^v(y, z, g)$ denote the value of a vacancy for firm y at the beginning of the period, and $V^p(x, y, z, g)$ denote the value of a producing firm y employing a worker of type x at the beginning of the period.

The surplus of a match is defined as:

$$S(x, y, z, g) := V^p(x, y, z, g) - V^v(y, z, g) + V^e(x, y, z, g) - V^u(x, z, g)$$

The bargaining protocol implies that the division of surplus is given by:

$$\begin{aligned}\beta S(x, y, z, g) &= V^e(x, y, z, g) - V^u(x, z, g) \\ (1 - \beta) S(x, y, z, g) &= V^p(x, y, z, g) - V^v(y, z, g).\end{aligned}$$

Optimizing workers and firms accept the match when the surplus is positive. So the matching decision α is given by:

$$\alpha(x, y, z, g) := \begin{cases} 1, & \text{if } S(x, y, z, g) \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

A.3.3 Discrete-Time HJB Equations

Unemployed worker: Taking as given the next-period match distribution, under the optimal acceptance decision the unemployed worker value function satisfies:

$$\begin{aligned}V^u(x, z, g) &= b\Delta + \beta_\Delta \sum_{\check{z}} P_\Delta(z, \check{z}) \left[(1 - \mathcal{M}^u \Delta) V^u(x, \check{z}, g'_z) \right. \\ &\quad \left. + \mathcal{M}^u \Delta \int \left(\alpha(x, \tilde{y}, z, g) V^e(x, \tilde{y}, \check{z}, g'_z) + (1 - \alpha(x, \tilde{y}, z, g)) V^u(x, \check{z}, g'_z) \right) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \right].\end{aligned}$$

Employed worker: Taking as given the next-period match distribution, under the optimal acceptance decision the employed worker value function satisfies:

$$\begin{aligned}V^e(x, y, z, g) &= w(x, y, z, g)\Delta + \beta_\Delta \sum_{\check{z}} P_\Delta(z, \check{z}) \left[(1 - \delta(x, y, z)\Delta) V^e(x, y, \check{z}, g'_z) \right. \\ &\quad \left. + \delta(x, y, z)\Delta V^u(x, \check{z}, g'_z) \right].\end{aligned}$$

Vacant firm: Taking as given the next-period match distribution, under the optimal

acceptance decision the vacancy value function satisfies:

$$V^v(y, z, g) = -c\Delta + \beta_\Delta \sum_{\tilde{z}} P_\Delta(z, \tilde{z}) \left[(1 - \mathcal{M}^v \Delta) V^v(y, \tilde{z}, g'_z) + \mathcal{M}^v \Delta \int \left(\alpha(\tilde{x}, y, z, g) V^p(\tilde{x}, y, \tilde{z}, g'_z) + (1 - \alpha(\tilde{x}, y, z, g)) V^v(y, \tilde{z}, g'_z) \right) \frac{g^u(\tilde{x})}{\mathcal{U}} d\tilde{x} \right].$$

Producing firm: Taking as given the next-period match distribution, under the optimal acceptance decision the producing firm value function satisfies:

$$V^p(x, y, z, g) = (F(x, y, z) - w(x, y, z, g)) \Delta + \beta_\Delta \sum_{\tilde{z}} P_\Delta(z, \tilde{z}) \times \left((1 - \delta(x, y, z) \Delta) V^p(x, y, \tilde{z}, g'_z) + \delta(x, y, z) \Delta V^v(y, \tilde{z}, g'_z) \right).$$

A.3.4 Free Entry and the Firm Distribution

As in the continuous time model, firms freely enter until the expected value of a vacancy equals zero:

$$0 = \mathbb{E}_{\tilde{y}}[V^v(\tilde{y}, z, g)] = \int V^v(\tilde{y}, z, g) d\tilde{y}.$$

A.3.5 Discrete-Time Master Equation

After combining the Bellman equations and imposing belief consistency, the equilibrium can be characterized by the discrete-time master equation:

$$S(x, y, z, g) = (F(x, y, z) - b + c) \Delta + \beta_\Delta \sum_{\tilde{z}} P_\Delta(z, \tilde{z}) \times \left[(1 - \delta(x, y, z) \Delta) S(x, y, \tilde{z}, g'_z) - \beta \frac{m(\mathcal{U}, \mathcal{V})}{\mathcal{U}} \Delta \int \alpha(x, \tilde{y}, z, g) S(x, \tilde{y}, \tilde{z}, g'_z) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} - (1 - \beta) \frac{m(\mathcal{U}, \mathcal{V})}{\mathcal{V}} \Delta \int \alpha(\tilde{x}, y, z, g) S(\tilde{x}, y, \tilde{z}, g'_z) \frac{g^u(\tilde{x})}{\mathcal{U}} d\tilde{x} \right].$$

We can observe that the discrete time Master equation is actually very similar to the continuous time Master equation (2.11). This is because we only have continuous time Markov processes. If we also had continuous time Brownian motion processes, then

the Master equation would look more different because there would also be second derivative terms in the continuous time Master equation that do not appear in the discrete time Master equation.

A.3.6 Reconciliation With Continuous Time

We now provide a heuristic reconciliation between the discrete time formulation in this Appendix and the continuous time formulation in the main text (without considering the technical issues associated with infinite dimensional calculus). For convenience, define:

$$S := S(x, y, z, g), \quad F := F(x, y, z), \quad \delta := \delta(x, y, z), \quad \varsigma := \varsigma(z),$$

and, for each $\check{z}$,

$$S'_{\check{z}} := S(x, y, \check{z}, g'_{\check{z}}), \quad S^w_{\check{z}}(\tilde{y}) := S(x, \tilde{y}, \check{z}, g'_{\check{z}}), \quad S^f_{\check{z}}(\tilde{x}) := S(\tilde{x}, y, \check{z}, g'_{\check{z}}).$$

Similarly, the integral terms are evaluated at the relevant continuation state $(\check{z}, g'_{\check{z}})$. Using this condensed notation, the discrete-time master equation is:

$$S = (F - b + c)\Delta + \beta_{\Delta} \sum_{\check{z}} P_{\Delta}(z, \check{z}) \left[(1 - \delta\Delta) S'_{\check{z}} - \beta \frac{m}{\mathcal{U}} \Delta \int \alpha S^w_{\check{z}} \frac{g^v}{\mathcal{V}} d\tilde{y} - (1 - \beta) \frac{m}{\mathcal{V}} \Delta \int \alpha S^f_{\check{z}} \frac{g^u}{\mathcal{U}} d\tilde{x} \right].$$

where we use the approximation:

$$\beta_{\Delta} = 1 - \rho\Delta + o(\Delta),$$

$$P_{\Delta}(z, \check{z}) = \begin{cases} 1 - \sum_{\bar{z} \neq z} \lambda(z, \bar{z})\Delta + o(\Delta), & \check{z} = z, \\ \lambda(z, \check{z})\Delta + o(\Delta), & \check{z} \neq z, \end{cases}$$

We split the sum into the no-jump term $\check{z} = z$ and the jump terms $\check{z} \neq z$. For the no-jump term, $\sigma(z, z) = 0$ and:

$$g'_z = g + \mu^g(z, g)\Delta + o(\Delta).$$

Therefore:

$$S'_z = S(x, y, z, g'_z) = S(x, y, z, g) + \langle D_g S(x, y, z, g), \mu^g(z, g) \rangle \Delta + o(\Delta).$$

Hence the no-jump contribution is (where we have incorporated all terms with Δ^n for $n > 1$ into $o(\Delta)$):

$$\begin{aligned} & \beta_\Delta P_\Delta(z, z)(1 - \delta\Delta)S'_z \\ &= (1 - \rho\Delta) \left(1 - \sum_{\check{z} \neq z} \lambda(z, \check{z})\Delta \right) (1 - \delta\Delta) [S + \langle D_g S, \mu^g \rangle \Delta] + o(\Delta) \\ &= \left(1 - \left(\rho + \delta + \sum_{\check{z} \neq z} \lambda(z, \check{z}) \right) \Delta \right) [S + \langle D_g S, \mu^g \rangle \Delta] + o(\Delta) \\ &= S - \left(\rho + \delta + \sum_{\check{z} \neq z} \lambda(z, \check{z}) \right) S\Delta + \langle D_g S, \mu^g \rangle \Delta + o(\Delta). \end{aligned}$$

For each jump term $\check{z} \neq z$, we have:

$$P_\Delta(z, \check{z}) = \lambda(z, \check{z})\Delta + o(\Delta).$$

Since this term is already multiplied by Δ , only the zeroth-order continuation value matters. Thus:

$$\begin{aligned} g'_z &= (1 - \sigma(z, \check{z}))g + o(1), \\ S'_z &= S(x, y, \check{z}, g'_z) = S(x, y, \check{z}, (1 - \sigma(z, \check{z}))g) + o(1). \end{aligned}$$

So the jump term becomes:

$$\beta_\Delta P_\Delta(z, \check{z})(1 - \delta\Delta)S'_z = \lambda(z, \check{z})\Delta S(x, y, \check{z}, (1 - \sigma(z, \check{z}))g) + o(\Delta).$$

The two search-option terms are already multiplied by Δ , so their no-jump com-

ponents are the only first-order contributions:

$$\begin{aligned}
& -\beta_{\Delta} \sum_{\tilde{z}} P_{\Delta}(z, \tilde{z}) \beta \frac{m}{\mathcal{U}} \Delta \int \alpha S_{\tilde{z}}^w \frac{g^v}{\mathcal{V}} d\tilde{y} \\
& = -\beta \frac{m}{\mathcal{U}} \Delta \int \alpha(x, \tilde{y}, z, g) S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} + o(\Delta), \\
& -\beta_{\Delta} \sum_{\tilde{z}} P_{\Delta}(z, \tilde{z}) (1 - \beta) \frac{m}{\mathcal{V}} \Delta \int \alpha S_{\tilde{z}}^f \frac{g^u}{\mathcal{U}} d\tilde{x} \\
& = -(1 - \beta) \frac{m}{\mathcal{V}} \Delta \int \alpha(\tilde{x}, y, z, g) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}} d\tilde{x} + o(\Delta).
\end{aligned}$$

Substituting these expressions into the discrete-time master equation gives:

$$\begin{aligned}
S &= S + (F - b + c)\Delta - (\rho + \delta)S\Delta + \langle D_g S, \mu^g \rangle \Delta \\
& - \beta \frac{m}{\mathcal{U}} \Delta \int \alpha(x, \tilde{y}, z, g) S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\
& - (1 - \beta) \frac{m}{\mathcal{V}} \Delta \int \alpha(\tilde{x}, y, z, g) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}} d\tilde{x} \\
& + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) \Delta \left[S(x, y, \tilde{z}, (1 - \sigma(z, \tilde{z}))g) - S \right] + o(\Delta).
\end{aligned}$$

Subtracting S from both sides, dividing by Δ , and taking the limit as $\Delta \rightarrow 0$ yields:

$$\begin{aligned}
0 &= -\rho S(x, y, z, g) + F(x, y, z) - b + c - \delta(x, y, z) S(x, y, z, g) \\
& - \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \int \alpha(x, \tilde{y}, z, g) S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}(z, g)} d\tilde{y} \\
& - (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g)} \int \alpha(\tilde{x}, y, z, g) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}(z, g)} d\tilde{x} \\
& + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) \left[S(x, y, \tilde{z}, (1 - \sigma(z, \tilde{z}))g) - S(x, y, z, g) \right] \\
& + \langle D_g S(x, y, z, g), \mu^g(z, g) \rangle,
\end{aligned}$$

where $m(z, g) = m(\mathcal{U}(z, g), \mathcal{V}(z, g))$. This is exactly the continuous-time master equation (2.11) in the main text.

A.4 Comment on Equilibrium Multiplicity

In the baseline model of Section 2, the free entry condition

$$\mathbb{E}_{\tilde{y}} [V^v(\tilde{y}, z, g)] = 0 \quad (\text{A.12})$$

must hold at every reachable aggregate state (z, g) . This makes the firm distribution g^f an endogenous jump variable that is selected simultaneously with the surplus function S . Since S depends on g^f through the opportunity cost terms in the master equation (2.11), and g^f is pinned down by (A.12) which depends on S , the equilibrium is the solution to a simultaneous non-linear fixed point problem. In general, such fixed point problems can admit multiple solutions, so the possibility of equilibrium multiplicity is present. Attempting a general proof of existence and uniqueness in high dimensional models is well known to be difficult (e.g. [Shimer and Smith, 2000](#) and [Sandmann and Bonneton, 2025](#) in the search and matching context and [Cardaliaguet et al. \(2019\)](#) for mean field games) and beyond the scope of this paper.

However, in the slow-entry model of Appendix E, the firm mass $\mathcal{F}_t$ is a predetermined state variable governed by:

$$d\mathcal{F}_t = \psi \alpha^f(z_t, g_t, \mathcal{F}_t) dt - \varsigma \mathcal{F}_t dt$$

so the contemporaneous fixed-point problem that arises from free entry is absent by construction. Consequently, the source of multiplicity associated with instantaneous vacancy adjustment is removed. So equilibrium selection is replaced by equilibrium dynamics. What remains is the possibility of multiple steady states (in the model without aggregate shocks) or multiple regions of attraction (in the model with aggregate shocks). In the former case, this would take the form of multiple values $\mathcal{F}^{ss}$ satisfying $\psi \alpha^f(z, g^{ss}, \mathcal{F}^*) = \varsigma \mathcal{F}^{ss}$ for a given z .

Even when uniqueness cannot be guaranteed, [Kahou et al. \(2025\)](#) show that neural-network-based solvers may exhibit an endogenous equilibrium-selection mechanism due to their inductive bias toward low-complexity solutions. To the extent that the low-complexity solution coincides with the economically stable solution, this can act as a practical equilibrium selection device.

B Appendix For Section 3

In this Appendix, we provide additional details on the crisis model of the labor market in Section 3. In Subsection B.1, we outline the calibrated parameters. In Subsection B.3, we specify the master equation for the model without aggregate shocks.

B.1 Calibration Details

Parameter	Interpretation	Value	Target/Source
ρ	Discount rate	0.05	Interest rate
ξ	Extreme value for α choice	2.0	
$f(x, y)$	Production for match (x, y)	$0.6 + 0.4(\sqrt{x} + \sqrt{y})^2$	Hagedorn et al. (2017)
β	Surplus division factor	0.72	Shimer (2005)
$m(\mathcal{U}, \mathcal{V})$	Matching function	$\kappa \mathcal{U}^\nu \mathcal{V}^{1-\nu}$	Hagedorn et al. (2017)
ν	Elasticity in meeting function	0.5	Hagedorn et al. (2017)
κ	Scale for meeting function	6.4	Unemployment rate
b	Worker unemployment benefit	1.0	Unemployment rate
c	Entry cost	2.400	Steady state $\mathcal{V}/\mathcal{U} = 1$
Steady State:			
$\bar{z}$	Steady state TFP	1	Shimer (2005)
$\bar{\delta}$	Steady state separation rate	0.2	BLS job tenure 5 years
$\bar{\varsigma}$	Steady state exogenous exit rate	0.01	No endogenous exit
Exogenous Aggregate Shock Process:			
A	TFP level	1.0	
$\varsigma_D, \varsigma_L, \varsigma_H$	exogenous exit rates	0.63, 0.01, 0.01	No endogenous exit
δ_L, δ_H	Separation rates	0.22, 0.18	Shimer (2005)
$\delta_D(x, y)$	Separation at crisis state	See Appendix B.1	Match Cajner et al. (2020)
λ_z	Poisson transition probability	See Appendix B.1	Shimer (2005)
$\sigma(z, \check{z})$	exogenous exit rate at Poisson jumps	See Appendix B.1	No endogenous exit
n_x	Discretization of worker types	5	Match Cajner et al. (2020)
n_y	Discretization of firm types	11	Match Cajner et al. (2020)

Table 7: Economic Parameters of the Labor Search Model in Section 3.

The Poisson transition rates for aggregate shocks across high, low, and disaster states are:

$$\lambda_z = \begin{bmatrix} - & \lambda_{HL} & \lambda_{HD} \\ \lambda_{LH} & - & \lambda_{LD} \\ \lambda_{DH} & \lambda_{DL} & - \end{bmatrix} = \begin{bmatrix} - & 0.4 & 0.001 \\ 0.4 & - & 0.001 \\ 0.0995 & 0.0995 & - \end{bmatrix}$$

The calibrated separation rate across worker and firm types $\delta_D(x, y) =$

3.8844	3.2176	2.7549	2.4429	2.2388	2.1093	2.0292	1.9804	1.9505	1.9317	1.9195
2.4663	1.8736	1.4721	1.2102	1.0466	0.9493	0.8944	0.8650	0.8498	0.8417	0.8372
1.5824	1.1386	0.8430	0.6546	0.5408	0.4765	0.4431	0.4273	0.4207	0.4182	0.4172
1.4336	1.0136	0.7366	0.5624	0.4592	0.4026	0.3746	0.3626	0.3584	0.3573	0.3571
0.8883	0.6295	0.4596	0.3533	0.2909	0.2570	0.2406	0.2338	0.2315	0.2311	0.2310

We calibrate $\delta_D(x, y)$ to match the observed peak declines in employment levels during the COVID-19 recession calculated in [Cajner et al. \(2020\)](#). Using detailed data from a major US payroll company, [Cajner et al. \(2020, Figure 4\)](#) estimates the employment drop of workers in different wage quintiles, which corresponds to five worker groups in our calibration. They also estimate the employment drop for 15 two-digit NAICS industries ([Cajner et al., 2020, Table 1](#)). We use the industry-level output and hours worked data from the Bureau of Labor Statistics (BLS) to compute the labor productivity of each industry in 2020. We then consolidate smaller industries to get 11 “composite” sectors, ensuring that after merging, these sectors exhibit similar levels of sectoral output.²⁰ These sectors are mapped into the 11 groups of firms in our model. We calibrate $\delta_D(x, y)$ such that the model’s simulated declines in employment for these heterogeneous groups match these empirical moments after the disaster shock z_D hits the ergodic mean of the economy for $t = 0.2$ years.

The calibrated exogenous exit rates at jumps are:

$$\sigma = \begin{bmatrix} - & \sigma(z_H, z_L) & \sigma(z_H, z_D) \\ \sigma(z_L, z_H) & - & \sigma(z_L, z_D) \\ \sigma(z_D, z_H) & \sigma(z_D, z_L) & - \end{bmatrix} = \begin{bmatrix} - & 0.0025 & 0.05 \\ 0 & - & 0.05 \\ 0 & 0 & - \end{bmatrix}$$

B.2 Numerical method and performance

In this Subsection, we provide additional details on the training process and the numerical errors.

B.2.1 Sampling and Training Scheme for the Surplus Network

We start by describing the sampling and training procedure used to approximate the surplus function $S(x, y, z, g)$ by a neural network $\hat{S}(x, y, z, g; \Theta)$. We work with

²⁰Data link <https://www.bls.gov/productivity/tables/major-industry-total-factor-productivity-klems.xlsx>. Last accessed on September 23, 2024.

discrete worker types $x \in \{x_1, \dots, x_{n_x}\}$, firm types $y \in \{y_1, \dots, y_{n_y}\}$, and an aggregate state $z \in \{z_L, z_H, z_D\}$ evolving as a continuous-time Markov chain with generator $\Lambda = (\lambda_{ij})_{i,j \in \{L,H,D\}}$. The goal of the numerical procedure below is to construct a weight vector Θ such that $\hat{S}(x, y, z, g; \Theta)$ solves the surplus equation with small residuals under the relevant distributions of (x, y, z, g) . We describe the training ingredients in the following order: the mixed steady-state sampler used in phase 1 training, phase 1 training, the ergodic simulation sampler used in phase 2 training, phase 2 training, and phase 3 out-of-sample evaluation.

Mixed steady-state sampler (for phase 1 training). For each aggregate state $z \in \{z_L, z_H, z_D\}$, let $g_m^L(x, y)$, $g_m^H(x, y)$ and $g_m^D(x, y)$ denote the stationary distributions of matches under z_L , z_H and z_D , respectively. In the first training phase, each draw of network inputs (x, y, z, g) is constructed as follows. We draw (x, y, z) as in Section 2.5.1: values of (x, y, z) are obtained from a multinomial sampler on the discretized support of (x, y, z) . We then construct the match distribution as a random convex combination of the three steady-state objects,

$$g = w_L g_m^L + w_H g_m^H + w_D g_m^D, \quad w_s \geq 0, \quad \sum_{s \in \{L,H,D\}} w_s = 1.$$

The variables (w_L, w_H, w_D) are generated as follows. Let (π_L, π_H, π_D) denote the stationary probabilities of the three aggregate states. First, we draw an index $s^* \in \{L, H, D\}$ with $\Pr(s^* = s) = \pi_s$. Conditional on $s^* = s$, we draw

$$w_s \sim \text{Beta}(\alpha, 1), \quad \alpha \gg 1,$$

so that w_s is typically close to one but remains random. Let $\{r_1, r_2\} = \{L, H, D\} \setminus \{s\}$ denote the remaining two states. We allocate the residual mass $1 - w_s$ to these states in proportion to their stationary probabilities,

$$w_r = (1 - w_s) \frac{\pi_r}{\pi_{r_1} + \pi_{r_2}}, \quad r \in \{r_1, r_2\}.$$

This procedure yields nonnegative variables summing to one, emphasizes one state in each draw, and on average reflects the stationary probabilities (π_L, π_H, π_D) in the mixture of g_m^L , g_m^H , and g_m^D . Phase 1 training uses these mixed steady-state draws to bring the neural network surplus function approximation $\hat{S}$ close to the true surplus

function S .

Phase-1 training and stopping rule. Given the mixed steady-state sampler described above, we train the surplus network weights Θ by minimizing the average mean squared error of the surplus master equation (2.13) on sampled states generated by the mixed steady-state sampler described above. We do this by following Step 3 of Algorithm 1 in the main text. For iteration n in Step 3, we generate K sample points from the mixed steady state sampler:

$$Q^n = \{x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y}\}_{k \leq K}$$

and compute the empirical loss in the master equation:

$$L(\Theta^n, Q^n) := \frac{1}{K} \sum_{k \leq K} \left| \mathcal{L}^S \hat{S}(x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y}, \Theta^n) \right|^2,$$

where $\hat{S}$ denotes the neural-network approximation of the surplus. The weight vector Θ^n is updated using stochastic gradient descent (or a variant) to decrease $L(\Theta^n, Q^n)$. Phase 1 training consists of repeated iterations to decrease $L(\Theta^n, Q^n)$ until this loss falls below a prespecified threshold. This stopping rule ensures that $\hat{S}$ is close to the steady-state surplus before we move to the ergodic-simulation-based phase of the training scheme that uses the current best approximation of $\hat{S}$ to generate samples.

Ergodic simulation sampler (for phase 2 training). After Phase 1 has brought $\hat{S}$ close to the steady-state surplus, we switch to a sampling scheme based on the ergodic mean of the distribution implied by the current network. We first simulate the aggregate state path $\{z_t\}_{t \geq 0}$ as the three-state continuous-time Markov chain with generator Λ , using a small time step Δt . Conditional on $\{z_t\}$ and on the current best neural network approximation $\hat{S}$, we then simulate the cross-sectional distribution of matches $\{g_t(x, y)\}_{t \geq 0}$ forward over time using the Kolmogorov forward equation (2.12) with the state-dependent separation rates $\delta(x, y, z_t)$ and the primitives specified in Section 3. We discard an initial burn-in period of this simulation and retain only the post-burn-in segment, which provides a large pool of draws $\{(g^n, z^n)\}_{n=1}^N$ approximating the ergodic mean of the distribution of (g, z) under Θ .

From this ergodic pool, we construct the data used in Phase 2 training. We

randomly split the set $\{(g^n, z^n)\}_{n=1}^N$ into a training set and a validation set, and for each pair (g^n, z^n) in either subset we augment with worker and firm types (x^n, y^n) drawn as in Section 2.5.1, using a multinomial sampler on the discretized support of (x, y, z) . This yields two datasets:

$$\mathcal{D}_{\text{train}} = \{(x^n, y^n, z^n, g^n)\}_{n=1}^{N_{\text{train}}} \quad \text{and} \quad \mathcal{D}_{\text{val}} = \{(x^n, y^n, z^n, g^n)\}_{n=1}^{N_{\text{val}}},$$

which are both drawn from the ergodic mean of the distribution induced by the current neural network approximation $\hat{S}$. In the subsequent phase, only $\mathcal{D}_{\text{train}}$ is used to compute stochastic gradients, while $\mathcal{D}_{\text{val}}$ is reserved for evaluating out-of-sample performance and selecting the best parameter vector.

Phase-2 inner-outer training loop and validation-based model selection.

Given the ergodic datasets $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{val}}$ constructed above, Phase 2 proceeds in repeated inner-outer iterations. In the inner loop, we fix $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{val}}$ and follow Step 3 in Algorithm 1 to use stochastic gradient descent (or a variant) to minimize the surplus loss in (2.13), now evaluated on draws from $\mathcal{D}_{\text{train}}$. At each iteration n we draw a minibatch of size $N_{\text{batch}} < N_{\text{train}}$ from $\mathcal{D}_{\text{train}}$ denoted by

$$Q^n(\mathcal{D}_{\text{train}}) = \{(x^n, y^n, z^n, g^n)\}_{n=1}^{N_{\text{batch}}},$$

evaluate the surplus master equation loss $L(\Theta^n, Q^n(\mathcal{D}_{\text{train}}))$ on the minibatch, and update the weights Θ^n to reduce the loss. At regular intervals, we compute a validation loss using the validation data set $\mathcal{D}_{\text{val}}$, defined analogously by $L(\Theta^n, \mathcal{D}_{\text{val}})$, and keep track of the weight vector Θ^* that attains the lowest validation loss within the current inner loop.

In the outer loop, we use this validation-based model selection to update the simulation. After a predetermined number of gradient steps (or epochs) in the inner loop, we set $\Theta \leftarrow \Theta^*$, so that the model used for resimulation is precisely the one with the best performance on the validation samples in that inner block. We then resimulate the economy under the updated surplus network $\hat{S}$, using the Kolmogorov forward equation (2.12) and the aggregate Markov chain to generate a new ergodic pool of (g, z) draws. From this pool we reconstruct fresh $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{val}}$ as in the previous paragraph, and then repeat the inner training loop. Thus, at each outer iteration we use a set of validation samples that is not involved in gradient updates

to select the best-performing model, and this model in turn generates the next round of ergodic samples for training. This resimulation-and-training cycle continues until the validation losses stabilizes at the desired scale.

Phase-3 out-of-sample evaluation. We stop training once the validation loss in phase 2 drops below a prespecified threshold. We then use the ergodic simulation sampler to generate a new “evaluation” data set:

$$\mathcal{D}_{\text{eval}} = \{(x^n, y^n, z^n, g^n)\}_{n=1}^{N_{\text{eval}}}$$

and compute the loss $L(\Theta^n, \mathcal{D}_{\text{eval}})$ on this dataset. We use this loss to evaluate the performance of the deep learning algorithm.

B.2.2 Hyperparameters for the neural networks

Table 8 provides the set of hyperparameters governing our feed-forward neural network and training procedure.

Hyperparameter	Value
Number of hidden layers	4
Neurons per hidden layer	50
Activation function	$\tanh(\cdot)$
Optimizer	Adam
Learning rate schedule	Time-varying
Phase-1 learning rate	From 10^{-4} to 10^{-5}
Phase-2 learning rate	From 5×10^{-5} to 10^{-5}
Batch size schedule	Time-varying
Phase-1 batch size	From 256 to 512
Phase-2 batch size	From 256 to 512
Phase-1 sample-regeneration frequency	—
Phase-2 sample-regeneration frequency	Every 10^5 epochs

Table 8: Neural-network and training hyperparameters for the labor search model in Section 3.

B.2.3 Results

Figure 7 reports the training and validation losses over 1.4 million epochs in the Phase 2 procedure. At each resimulation point (indicated by red vertical lines), we

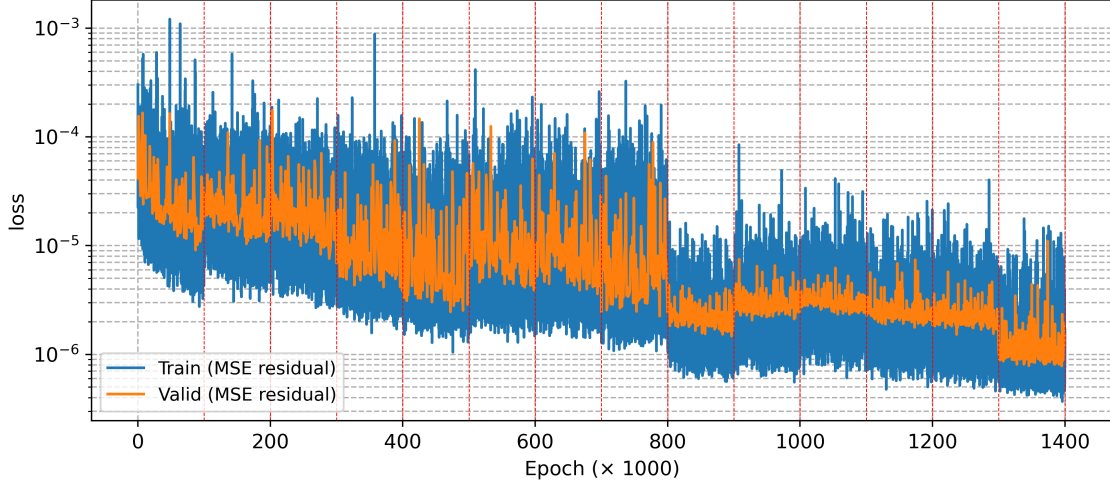


Figure 7: Training and validation losses over 1.4 million epochs in Phase 2. The solid curves show the mean squared residual of the surplus master equation on training minibatches (blue) and on the full validation dataset (orange). Red vertical lines indicate the epochs at which the economy is resimulated and new ergodic training and validation samples are generated from the updated surplus network.

use the currently selected surplus network to generate new ergodic samples and rebuild the training and validation datasets. Both training and validation losses exhibit a clear downward trend over the course of training. A key difference between the two curves is that, in each step, the training loss is computed on a single minibatch rather than on the full training dataset, whereas the validation loss is computed over the entire validation sample. As a result, the validation loss is smoother and provides a more reliable measure of overall performance. At the end of the procedure, we select the parameter vector corresponding to the minimum validation loss as our final network.

We assess the performance of this training process using three different simulations of the ergodic mean of the distribution: the training sample generated in the last inner loop training step, the validation sample generated in the last outer loop, and a completely new simulation of the economy after finishing the training process, which we refer to as the evaluation set. Figures 8a, 8b, and 8c show the root mean squared error (RMSE) of the surplus loss over the ergodic mean of the distribution, computed

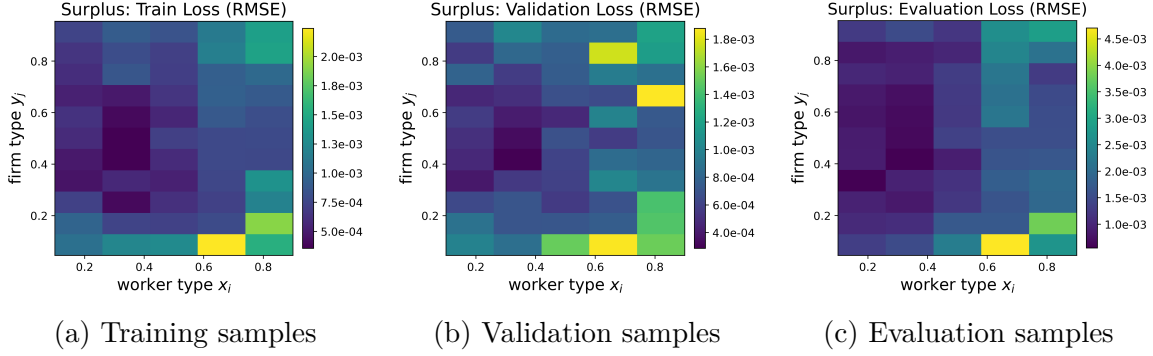


Figure 8: Root mean squared error (RMSE) of the surplus loss computed on (a) training samples from the last inner loop, (b) validation samples from the last outer loop, and (c) an independently simulated evaluation set generated after the training procedure has converged.

using the training, validation, and evaluation samples, respectively.²¹ The average RMSEs for the training, validation, and evaluation samples are approximately 9.2×10^{-4} , 9.1×10^{-4} , and 1.7×10^{-3} , respectively, which indicates a good approximation of the surplus function in the neighborhood of the ergodic mean of the distribution, the key region for simulating the economy and conducting policy analysis.

B.3 Comparison to Model Without Aggregate Shocks

To verify the accuracy of the DeepSAM method, we apply it to solve the labor search model in Section 3 without aggregate shocks (or any exogenous exit). This is a model which can also be solved with a conventional numerical method such as that in [Hagedorn et al. \(2017\)](#) so we have a clear benchmark to which to compare.

B.3.1 Master equation for the model without aggregate shocks

The master equation and loss function for the Surplus is given by:

$$0 = \mathcal{L}^S S = F(x, y) - (\rho + \delta(x, y))S(x, y, \underline{g})$$

²¹For a given sample $\{(x_k, y_k, z_k, g_k)\}_{k \leq K}$, we define the RMSE of the surplus loss as

$$\sqrt{\frac{1}{K} \sum_{k \leq K} \left| \mathcal{L}^S \hat{S}(x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y}) \right|^2}.$$

$$\begin{aligned}
& -b - \beta \frac{m(\underline{\mathbf{g}})}{\mathcal{U}(\underline{\mathbf{g}})\mathcal{V}(\underline{\mathbf{g}})} \frac{1}{n_y} \sum_{j=1}^{n_y} \alpha(x, y_j, \underline{\mathbf{g}}) S(x, y_j, \underline{\mathbf{g}}) \underline{\mathbf{g}}^v(y_j) \\
& + c - (1 - \beta) \frac{m(\underline{\mathbf{g}})}{\mathcal{U}(\underline{\mathbf{g}})\mathcal{V}(\underline{\mathbf{g}})} \frac{1}{n_x} \sum_{i=1}^{n_x} \alpha(x_i, y, \underline{\mathbf{g}}) S(x_i, y, \underline{\mathbf{g}}) \underline{\mathbf{g}}^u(x_i) \\
& + \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \partial_{g_{ij}} S(x, y, \underline{\mathbf{g}}) \mu^g(x_i, y_j, \underline{\mathbf{g}})
\end{aligned} \tag{B.1}$$

in which

$$\begin{aligned}
dg_t(x, y)/dt = \mu^g(x, y, \underline{\mathbf{g}}_t) = & -\delta(x, y) g_t(x, y) + \frac{m(\underline{\mathbf{g}}_t)}{\mathcal{U}_t \mathcal{V}_t} \alpha(x, y, \underline{\mathbf{g}}_t) \\
& \times \left(g^w(x) - \frac{1}{n_y} \sum_{j=1}^{n_y} \underline{\mathbf{g}}_t(x, y_j) \right) \left(g^f(y) - \frac{1}{n_x} \sum_{i=1}^{n_x} \underline{\mathbf{g}}_t(x_i, y) \right)
\end{aligned}$$

and $\alpha(x, y, \underline{\mathbf{g}})$ is given by:

$$\alpha(x, y, \underline{\mathbf{g}}) = \left(1 + e^{-\xi S(x, y, \underline{\mathbf{g}})} \right)^{-1}.$$

B.3.2 Numerical Comparison Between Methods

As described in the main text, we set $(z_t, \delta_t) \equiv (\bar{z}, \bar{\delta})$ and use the fixed point approach in [Shimer and Smith \(2000\)](#) and [Hagedorn et al. \(2017\)](#) to solve for the deterministic steady state (DSS) solution $S^{\text{DSS}}(x, y)$ defined by equation (B.1). We then use DeepSAM to solve the 57-dimensional PDE for $S(x, y, \underline{\mathbf{g}})$ and compute the DeepSAM solution at the DSS by: $S_{\text{DeepSAM}}^{\text{DSS}}(x, y) = S(x, y, \underline{\mathbf{g}} = \underline{\mathbf{g}}^{\text{DSS}})$. We define the squared difference of the two methods for each (x, y) pair as $\|S_{\text{DeepSAM}}^{\text{DSS}}(x, y) - S_{\text{Conventional}}^{\text{DSS}}(x, y)\|^2$. Figure 9 plots the absolute difference between the DeepSAM solution and the conventional solution at the DSS. The difference is in the order of magnitude of 10^{-3} across the state space.

B.4 Checking Endogenous Exit

In our numerical implementation, the neural network is trained under the maintained assumption that endogenous exit does not occur along the simulated paths. As outlined in Appendix A.1, under this assumption the terms in the master equation (A.8) that are associated with endogenous exit drop out and we get the simpler equation (2.11) outlined in the main text. This simplification is convenient computationally,

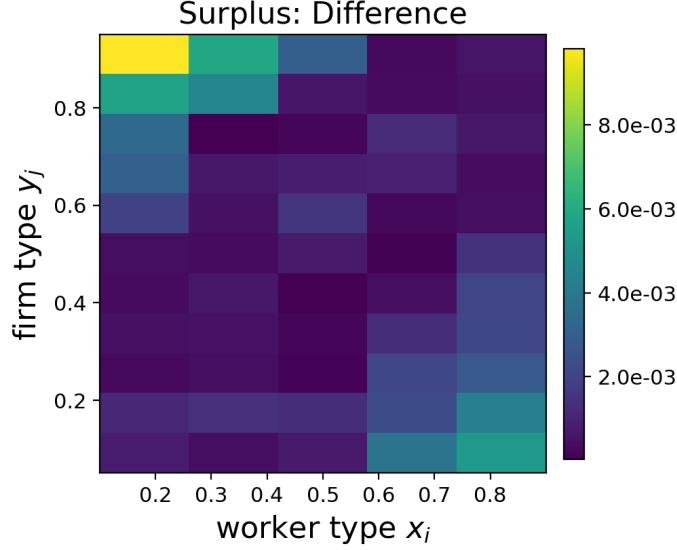


Figure 9: Model without aggregate shock: absolute difference from conventional solution. Since the maximum value of the surplus function $S(x, y)$ is approximately 0.65, expressing the error values relative to this scale multiplies them by $1/0.65 \simeq 1.54$. Hence, the scaled error values are about 54% larger than the unscaled error values.

since it allows us to evaluate the loss at each iteration without having to keep track of the endogenous-exit region and the corresponding additional terms in the law of motion.

For this numerical solution to be consistent with the model, the exogenous exit rates must be sufficiently large, both in the drift and at aggregate-state jumps, so that the decline in the total mass of firms never exceeds the decline implied by exogenous exit alone. In this section we verify that the calibrated exogenous exit rates are large enough to satisfy this requirement. In particular, we check that throughout the relevant transitions, both in continuous-time drift and at aggregate-state jumps, the observed decline in the total mass of firms is always weakly smaller than the decline generated by exogenous exit. This confirms that endogenous exit does not bind in the regions used for training and simulation, and hence that the implemented loss function is consistent with the model.

More precisely, we verify that the two endogenous-exit indicators $\mathbf{1}_{\mu^{\mathcal{F}}}(z_t, g_t)$ and $\mathbf{1}_{\sigma^{\mathcal{F}}}(z_t, \check{z}_t, g_t)$, defined in (A.1) and (A.2), respectively, are always equal to zero along

the simulated paths. To do this, we need to check that:

$$K_\mu(z_t, g_t) := \frac{\mu^{\mathcal{F}}(z_t, g_t)}{\mathcal{F}_t} + \varsigma(z_t) \geq 0,$$

$$K_\gamma(z_t, \check{z}_t, g_t) := \frac{\gamma^{\mathcal{F}}(z_t, \check{z}_t, g_t)}{\mathcal{F}_t} + \sigma(z_t; \check{z}_t) \geq 0.$$

at all times along simulated paths. The first inequality is the drift condition. It requires that, between aggregate-state jumps, the proportional drift of the total mass of firms is not more negative than the decline generated by exogenous flow exit. The second inequality is the jump condition. It requires that, at an aggregate-state transition from z_t to $\check{z}_t$, the realized proportional jump in the total mass of firms is not more negative than the calibrated exogenous jump-exit rate. We refer to $K_\mu(z_t, g_t)$ and $K_\gamma(z_t, \check{z}_t, g_t)$ as the slack in the drift and jump terms.

We check these conditions numerically for the simulations reported in Section 3, where both the “free-entry” condition and exogenous exit are included in the model. To make the check robust, we evaluate the inequalities over 200 simulations with independent realizations of the aggregate-state path after the COVID-19 shock. For illustration, Figure 10 reports the corresponding drift and jump diagnostics for one realized simulation.

Table 9 reports the numerical diagnostics for the no-endogenous-exit conditions along the simulated paths. The column “Jumps” summarizes the jump condition, while the remaining columns summarize the drift condition separately for each aggregate state. For jumps, the table reports the total number of aggregate-state transitions observed across the simulations, the number of transitions for which the jump condition is violated, the corresponding violation share, and the minimum value of the jump slack. For the drift condition, the table reports the total amount of valid simulated time spent in each aggregate state, the amount of time for which the drift condition is violated, the corresponding violation share, and the minimum drift slack in that state. All the diagnostics in Table 9 confirm that endogenous exit does not bind anywhere in the simulated region. The violation share is zero for aggregate-state jumps and also zero for the drift component in all three aggregate states. In addition, the minimum slack is nonnegative in every column of the table. These results confirm that the calibrated exogenous exit rates are sufficiently large for the maintained no-endogenous-exit approximation to be numerically consistent in the regions used

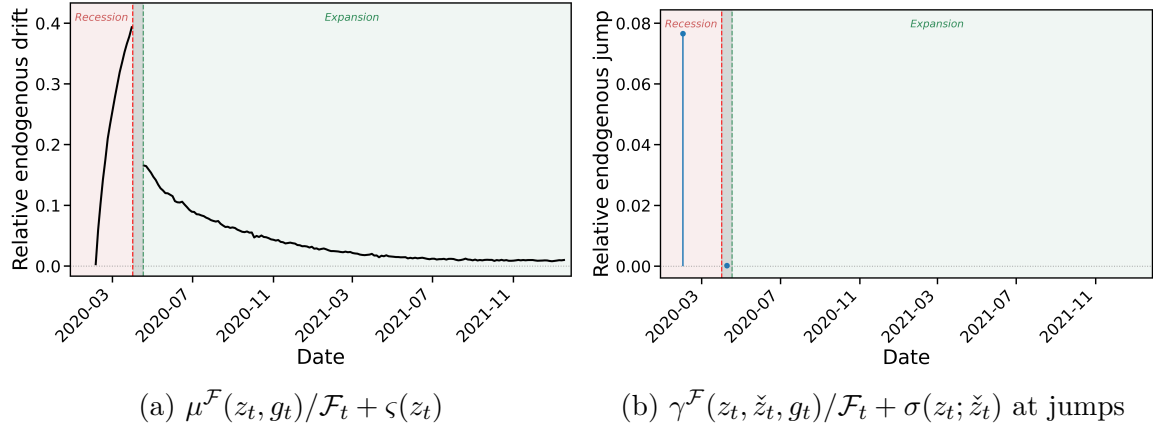


Figure 10: Checking endogenous exit after the COVID-19 shock. Panel (a) compares the realized proportional drift of the total mass of firms with the lower bound implied by exogenous flow exit. Panel (b) compares the realized proportional change in the total mass of firms at aggregate-state jumps with the lower bound implied by exogenous jump exit. In both panels, the realized values remain above the corresponding exogenous-exit bounds, confirming that endogenous exit does not bind along the simulated path.

for the quantitative exercises.

Table 9: Checking endogenous exit across simulated paths

	Jumps	Drift: L	Drift: H	Drift: D
Total observations/time	550	84.55	261.57	32.0
Violations	0	0.00	0.00	0.00
Violation share	0.00	0.00	0.00	0.00
Minimum slack	0.00	0.00	0.00	0.01

Notes. The column “Jumps” reports the number of aggregate-state jumps and the number and share of jumps at which $\gamma^{\mathcal{F}}(z_t, \check{z}_t, g_t)/\mathcal{F}_t + \sigma(z_t; \check{z}_t) < 0$. The drift columns report, for each aggregate state, the simulated time spent in that state and the amount of time for which $\mu^{\mathcal{F}}(z_t, g_t)/\mathcal{F}_t + \varsigma(z_t) < 0$. A nonnegative minimum slack implies that endogenous exit does not bind in the corresponding category.

B.5 Solution to the Problem with Discrete Choices

In the main text, we solve the master equation (2.13) by approximating the acceptance function $\alpha(x, y, z, \underline{\mathbf{g}})$ with a smooth function,

$$\alpha(x, y, z, \underline{\mathbf{g}}) = \left(1 + e^{-\xi S(x, y, z, \underline{\mathbf{g}})}\right)^{-1},$$

which ensures differentiability of value and policy functions and enables efficient gradient-based training. While this smooth formulation is computationally convenient, the model can also be solved under the exact discrete rule,

$$\alpha(x, y, z, \underline{\mathbf{g}}) = \mathbb{1}\{S(x, y, z, \underline{\mathbf{g}}) \geq 0\}, \quad (\text{B.2})$$

although the problem becomes more demanding. From the mathematical side, the combination of finite types and the exact binary choice rule means that the surplus function is potentially not smooth on a zero measure set and so the solution to the master equation should be interpreted as a viscosity solution. From the computational side, we need to approximate a non-smooth surplus function. In this section, we present how to solve the exact discrete rule numerically.

B.5.1 The Homotopy Approach

The main challenge arises from the discontinuity of α , which generates sharp decision boundaries in high-dimensional state space and renders gradient-based training less stable. To overcome this difficulty, we adopt a “homotopy” style approach: we solve a sequence of problems with increasing curvature parameters ξ , each time using the solution from the previous stage as the initial guess for the next. After convergence at a sufficiently large ξ , we take the resulting solution as the initial condition to solve the model under the exact discrete rule in Equation (B.2). This approach preserves numerical stability throughout training and ensures convergence even in the presence of discrete choices.

Under the discrete acceptance choices, we use this homotopy approach to solve both the steady state and the dynamic problem. Algorithm 3 outlines the procedure for computing the steady state of this problem, which provides the stationary distributions used in sampling for the dynamic training. Algorithm 4 extends this idea to the full dynamic model, gradually increasing the curvature parameter ξ and finally solving the model under the discrete rule using the high- ξ solution as initialization.

B.5.2 Solution to the Shimer-Smith model with aggregate shocks and discrete choice

We apply this method to the model in Section 3 without the crisis state, where the aggregate shock follows a two-state Markov chain. Figure 11 compares the steady-

Algorithm 3: Steady-state homotopy for discrete choice

Input: Schedule $(\xi_i)_{i=1}^N$ with $\xi_1 < \dots < \xi_N$ and ξ_N large.

Output: Steady-state solution $\mathcal{X}^* = (\underline{g}, \alpha, w, V^u, V^e, V^v, V^p, S, g^f)$ under the discrete rule.

Initialize: Choose an initial guess $\mathcal{X}^{(0)}$ for the steady state when $\xi = \xi_1$.

for $i = 1$ **to** N **do**

Stage i (smooth rule). Set $\alpha(x, y, z, \underline{g}) = (1 + e^{-\xi_i S(x, y, z, \underline{g})})^{-1}$.

 Using $\mathcal{X}^{(i-1)}$ as the initial guess, solve the *steady-state fixed point*; denote the converged solution by $\mathcal{X}^{(i)}$.

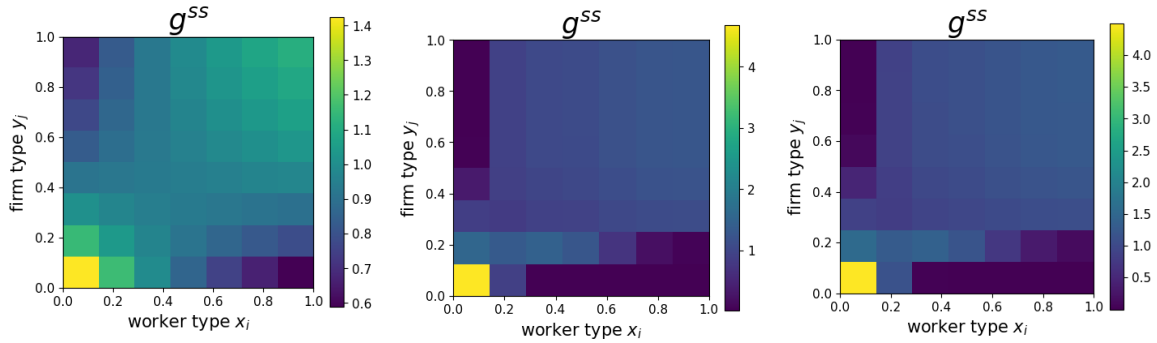
end

Discrete stage. Set $\alpha(x, y, z, \underline{g}) = \mathbb{1}\{S(x, y, z, \underline{g}) \geq 0\}$.

Solve the *steady-state fixed point* of this discrete choice problem using $\mathcal{X}^{(N)}$ as the initial guess; denote the converged solution by $\mathcal{X}^*$.

return $\mathcal{X}^*$.

state distributions $\underline{g}$ at $z = 1$ under three specifications: a smooth acceptance rule with low curvature ($\xi = 2$), a high-curvature specification ($\xi = 42$), and the exact discrete rule. Evidently, the high-curvature case provides a very close approximation to the discrete model.



(a) Continuous α with $\xi = 2$ (b) Continuous α with $\xi = 42$ (c) Discrete α

Figure 11: Comparison of steady-state distributions $\underline{g}$ at $z = 1$ under the continuous acceptance rule with low curvature ($\xi = 2$), high curvature ($\xi = 42$), and the discrete acceptance rule.

Having solved for the steady-state solutions, we then apply Algorithm 4 to solve the dynamic model. Figures 12 and 13 display, respectively, the acceptance rule $\alpha(x, y, z, \underline{g})$ and the surplus function $S(x, y, z, \underline{g})$ under the same three specifications. Again, the high-curvature model produces results that are very close to the discrete

Algorithm 4: Dynamic homotopy for discrete choice

Input: Schedule $(\xi_i)_{i=1}^N$ with $\xi_1 < \dots < \xi_N$ and ξ_N large; initial NN parameters $\Theta^{(0)}$.

Output: Trained neural networks $(\hat{S}, \hat{V}^u, \hat{V}^e, \hat{V}^v, \hat{V}^p, \hat{w})$ under the discrete rule.

Step 1. Approximate the surplus function by a neural network (NN),

$$S(x, y, z, \underline{g}) \approx \hat{S}(x, y, z, \underline{g}; \Theta^{(0)}).$$

for $i = 1$ **to** N **do**

Stage i (smooth rule). Set $\alpha(x, y, z, \underline{g}) = (1 + e^{-\xi_i S(x, y, z, \underline{g})})^{-1}$.

Using parameters $\Theta^{(i-1)}$ as initialization, train the NN by minimizing the residual loss of the surplus master equation on the sampled points; denote the converged parameters by $\Theta^{(i)}$.

end

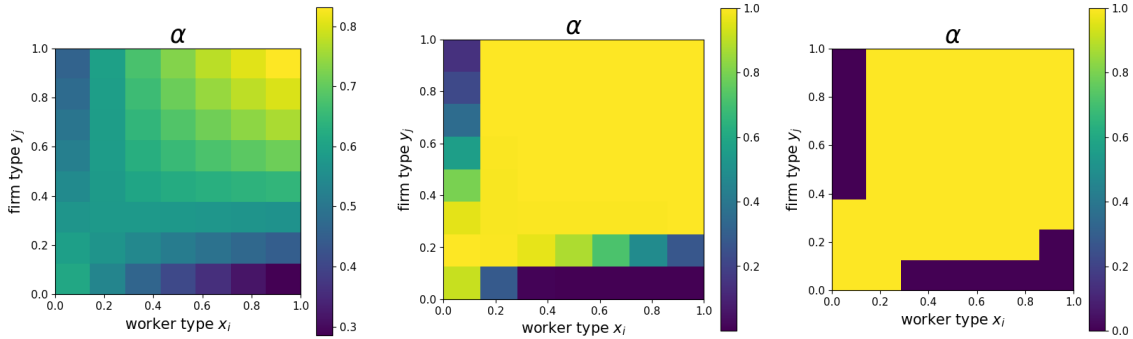
Discrete stage. Set $\alpha(x, y, z, \underline{g}) = \mathbf{1}\{S(x, y, z, \underline{g}) \geq 0\}$.

Train the NN starting from $\Theta^{(N)}$ to convergence under the discrete acceptance rule; denote the final parameters by Θ^* .

Step 2. Using $\hat{S}(\cdot; \Theta^*)$, train neural-network approximations to the value and wage functions $(\hat{V}^u, \hat{V}^e, \hat{V}^v, \hat{V}^p, \hat{w})$.

return $(\hat{S}, \hat{V}^u, \hat{V}^e, \hat{V}^v, \hat{V}^p, \hat{w})$.

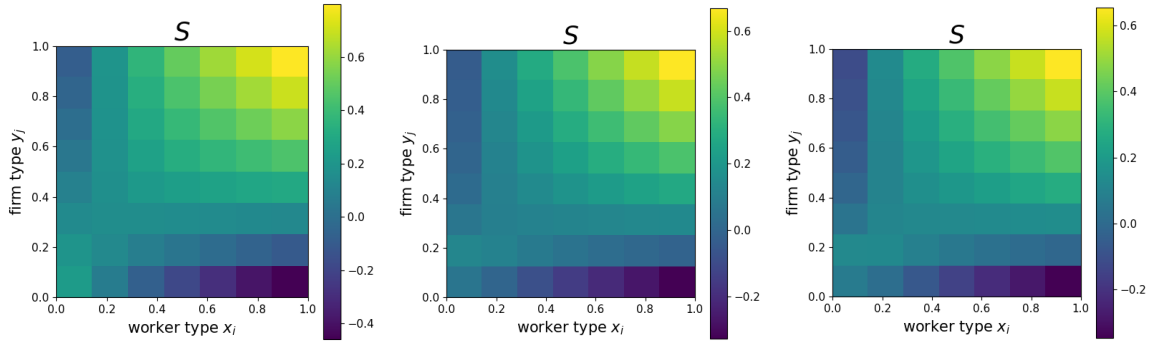
case.



(a) Continuous α with $\xi = 2$ (b) Continuous α with $\xi = 42$ (c) Discrete α

Figure 12: Acceptance rule $\alpha(x, y, z, \underline{g})$ at the steady state $\underline{g}$ and $z = 1$, comparing the continuous acceptance rule with low curvature ($\xi = 2$), the continuous acceptance rule with high curvature ($\xi = 42$), and the discrete acceptance rule.

Figure 14 illustrates the training dynamics of the homotopy algorithm for the dynamic model. Within each stage of ξ_i , the training loss declines steadily as the network converges. At the transition between stages ($\xi_i \rightarrow \xi_{i+1}$), the change in the



(a) Continuous α with $\xi = 2$ (b) Continuous α with $\xi = 42$ (c) Discrete α

Figure 13: Surplus function $S(x, y, z, \underline{g})$ at steady state $\underline{g}$ and $z = 1$: comparison of models with the continuous choice with low curvature, high curvature, and the discrete choice.

acceptance rule induces a temporary jump in the loss, followed by renewed convergence. The overall pattern is piecewise monotonic, with the final discrete stage converging smoothly due to the warm start from the high-curvature solution. Figure 15 further shows the distribution of the residual loss over the (x, y) grid after convergence at each stage: panels (a)–(c) correspond to the low-curvature, high-curvature, and discrete specifications, respectively. After the homotopy procedure, we obtain highly accurate solutions for the discrete choice problem.

Therefore, our DeepSAM approach can obtain highly accurate solutions with the discrete acceptance rule. This distinguishes our approach from the perturbation-based approach, such as [Reiter \(2009\)](#) and [Bilal \(2023\)](#), which rely on differentiability throughout. However, the computational burden is significant: for a given precision threshold, the required training time and number of epochs are roughly an order of magnitude greater than for the smooth specification. In models where a smooth approximation is theoretically acceptable, such as those micro-founded along the lines of [Caliendo et al. \(2019\)](#), it is possibly preferable to train with a continuous acceptance rule using a suitable curvature parameter ξ , which balances computational efficiency with accuracy.

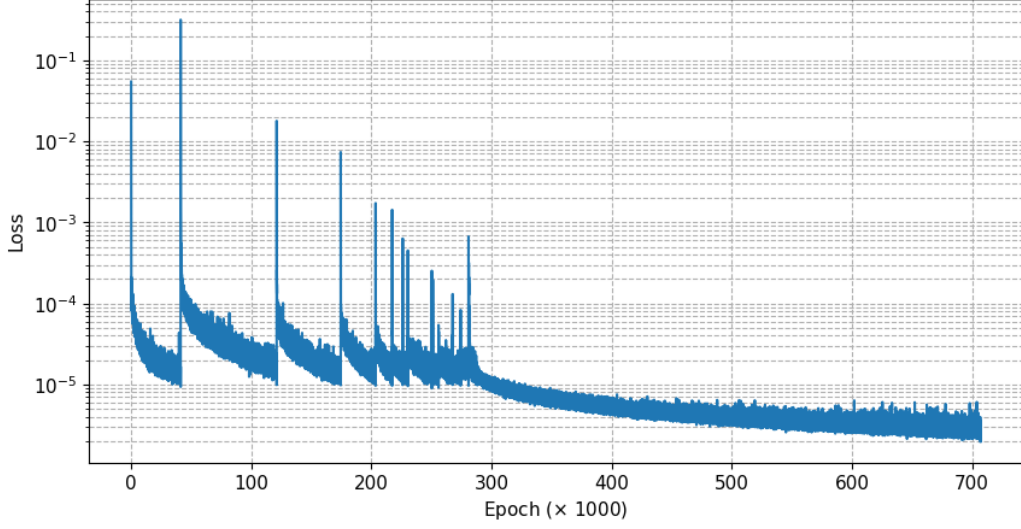


Figure 14: Training loss over epochs under the homotopy schedule. Each upward jump marks a transition to a higher curvature parameter ξ , where the acceptance rule becomes steeper or discrete. Within each stage, the loss declines monotonically until convergence.

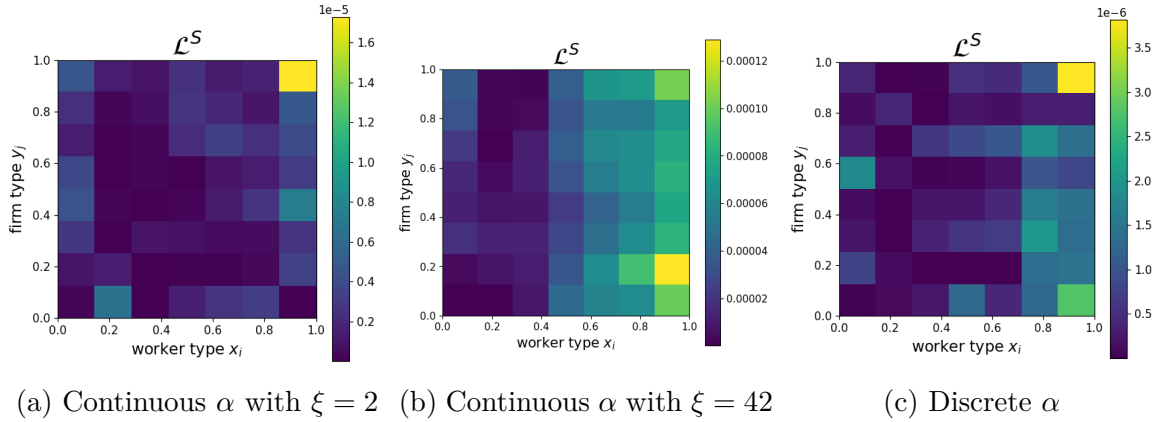


Figure 15: Residual loss over the (x, y) grid at the end of training under the homotopy schedule, for models with the continuous choice with low curvature, high curvature, and the discrete choice.

C Appendix For the OJS Model in Section 4

In this Appendix, we provide additional details for the on-the-job-search (OJS) model from Section 4. We start in Subsection C.1 by completing the recursive equilibrium characterization. In Subsection C.2, we then map our parameterized OJS allocations

into the outcomes from common models in the literature. In Subsection C.3, we derive the differential equation for the evolution of wages. In Subsections C.4 and C.5, we discuss the training and calibration of the model.

C.1 Completing the Recursive Equilibrium Characterization

In the main text, we outlined the agent HJB equations, the KF equation, and the master equation for surplus. The remaining part of the equilibrium is the free entry condition.

The flow value $J^v(y, z, g)$ of creating a vacancy with productivity y comes from the rents to matching with unemployed workers and to poaching employed workers:

$$\begin{aligned} J^v(y, z, g) = & \mathcal{M}^v \mathcal{C}^u \int \alpha(\tilde{x}, y, z, g)(1 - \beta) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}} d\tilde{x} \\ & + \mathcal{M}^v \mathcal{C}^e \int \int \alpha^p(y, \tilde{x}, \tilde{y}, z, g)(1 - \beta) [S(\tilde{x}, y, z, g) - S(\tilde{x}, \tilde{y}, z, g)] \frac{g(\tilde{x}, \tilde{y})}{\mathcal{E}} d\tilde{x} d\tilde{y}, \end{aligned}$$

where $\alpha^p(y, \tilde{x}, \tilde{y}, z, g) := \alpha^e(\tilde{x}, \tilde{y}, y, z, g)$ is the (smoothed) indicator that the poaching firm wins and hires the worker. The free entry condition is then:

$$0 = -c + \int_0^1 J^v(\tilde{y}, z, g) \bar{h}^v(\tilde{y}) d\tilde{y}.$$

where $\bar{h}^v(\tilde{y}) = 1$. Assuming that $m(\mathcal{W}, \mathcal{V})$ is homothetic in $\mathcal{W}$ and $\mathcal{V}$, we can invert this equation and solve for $\mathcal{V}_t$ in an analogous manner to Section 2. Since $\mathcal{W}_t, \mathcal{U}_t, \mathcal{E}_t, \mathcal{C}^u, \mathcal{C}^e$, and g^u are all determined by g , this gives $\mathcal{V}_t$ as a function of (z, g) and the surplus function $S(\cdot, \cdot, z, g)$.

Observe that we define the free-entry condition in terms of the flow value of creating a vacancy rather than the expected continuation value of holding a vacancy because vacancies are short-lived, and so the continuation value is $V^v(y, z, g) = 0$. This approach directly imposes that the model has short-lived vacancies. Alternatively, we could define the value of creating a vacancy in a model in which vacancies are destroyed at rate ς^v and then take the limit as $\varsigma^v \rightarrow \infty$.

C.2 Reconciliation With Bargaining Models

In this Appendix, we reconcile our model with a collection of bargaining protocols from the labor search literature.

C.2.1 Relationship to Cahuc et al. (2006)

We start by reconciling our model with that of Cahuc et al. (2006) and showing that their bargaining protocol (with some adjustments to account for aggregate shocks) is nested as a special case of our formulation when $\chi = 1$.

Notation: Cahuc et al. (2006) focus on a stationary environment with $z = 1$ and $F(x, y, z) = xy$. They define $V_0(x)$ as the value of unemployment for a worker of type x and $V(x, w, y)$ as the value of employment at a firm of productivity y paying wage w .²² If the worker receives the entire match value and the employer earns zero profit, then the worker gets wage $w = F(x, y)$. Thus, they write the match surplus as:

$$V(x, F(x, y), y) - V_0(x).$$

Reconciliation with our notation: Our environment has stochastic productivity so $F(x, y, z)$ depends on z and the value functions depend on the aggregate states (z, g) . Even though the value functions in Cahuc et al. (2006) are not expressed as explicit functions of the state, we make this dependence explicit in this section so we can have consistent notation. Mapping Cahuc et al. (2006)'s notation to our notation in Section 4 (both in Section 4.1 and Section 4.6) leads to the following relationships:

$$\begin{aligned} V_0(x, z, g) &\leftrightarrow V^u(x, z, g), \\ V(x, w, y, z, g) &\leftrightarrow V^e(x, y, q, z, g) = V^u(x, z, g) + q(w), \\ V(x, F(x, y, z), y, z, g) &\leftrightarrow P(x, y, z, g) \\ V(x, F(x, y, z), y, z, g) - V_0(x, z, g) &\leftrightarrow S(x, y, z, g) \end{aligned}$$

where w is the promised wage to the worker in Cahuc et al. (2006)'s notation and $q(w)$ is the corresponding promised value to the worker in our notation, chosen such that:

$$V^e(x, y, q, z, g) = V(x, w, y, z, g).$$

For convenience, throughout this reconciliation note, we assume that there is a one-

²²Technically, Cahuc et al. (2006) use ϵ for x and p for y but, for clarity, I have used variables corresponding to our setup.

to-one mapping between promised values and wages so $q(w)$ is well defined.

Bargaining in the Style of Cahuc et al. (2006): We fix $z = 1$ and $g = \bar{g}$, where $\bar{g}$ is the steady state, since there are no aggregate shocks in Cahuc et al. (2006). We consider the following interpretation of the bargaining protocol in Cahuc et al. (2006). Suppose an employed worker of type x currently in a match with a firm of type y promising wage w encounters an alternative firm with y' . Then, following Appendix A in Cahuc et al. (2006) (with some changes to the notation and descriptions) the agents play the following game:

- (i) The firms make simultaneous noncooperative job contracts,
- (ii) The worker chooses one of the two firms at which to work,
- (iii) If the worker chooses to stay at the incumbent firm and keep their current promise, then they keep q . Otherwise, the worker and the chosen firm use generalized Nash Bargaining (with worker bargaining parameter β) to divide surplus under the assumption that the worker uses the offer from Step (i) as the threat point in the bargaining.

Proposition 2 of Cahuc et al. (2006) shows that the bargaining protocol leads to the following equilibrium outcomes. In stage (i-ii), the firm with the higher productivity wins the contest by offering infinitesimally more than the maximum bid of the other firm. So, if $y' > y$, then firm y' wins by offering $V(x, F(x, y, z), y, 1, \bar{g})$, the worker moves to the new firm, and the outcome in stage (iii) bargaining is that the worker gets a wage $\phi(x, y, y')$ satisfying (equation (A.7) in their paper):

$$\begin{aligned}
& V(x, \phi(x, y, y'), y', 1, \bar{g}) \\
&= V(x, F(x, y, z), y, 1, \bar{g}) + \beta (V(x, F(x, y', z), y', 1, \bar{g}) - V(x, F(x, y, z), y, 1, \bar{g})) \\
&= \beta V(x, F(x, y', z), y', 1, \bar{g}) + (1 - \beta) V(x, F(x, y, z), y, 1, \bar{g}). \tag{C.1}
\end{aligned}$$

By contrast, if $y' \leq y$, then firm y wins by offering $V(x, F(x, y', z), y', 1, \bar{g})$ and the worker does not move to the new firm. If the worker chooses not to renegotiate their contract with the incumbent firm, then they keep w . Otherwise, if the worker chooses to renegotiate, then under the terms of the stage (iii) bargaining game, the worker

gets a wage $\phi(x, y, y')$ satisfying the following (equation (A.8) in [Cahuc et al. \(2006\)](#)):

$$\begin{aligned}
& V(x, \phi(x, y', y), y, 1, \bar{g}) \\
&= V(x, F(x, y', z), y', 1, \bar{g}) + \beta (V(x, F(x, y, z), y, 1, \bar{g}) - V(x, F(x, y', z), y', 1, \bar{g})) \\
&= (1 - \beta)V(x, F(x, y', z), y', 1, \bar{g}) + \beta V(x, F(x, y, z), y, 1, \bar{g}). \tag{C.2}
\end{aligned}$$

So, for the case in which the worker does not move, they renegotiate if $\phi(x, y, y') > w$, which using (C.2) is equivalent to:

$$(1 - \beta)V(x, F(x, y', z), y', 1, \bar{g}) + \beta V(x, F(x, y, z), y, 1, \bar{g}) > V(x, w, y, 1, \bar{g})$$

[Cahuc et al. \(2006\)](#) focuses on a stationary equilibrium because there are no aggregate shocks in their model so they can ensure that $q(w) \leq S(x, y, 1, \bar{g})$ when the poacher arrives. Our model has aggregate shocks, which requires us to make some adjustments to the [Cahuc et al. \(2006\)](#) protocol. In particular, guaranteeing that $q \leq S(x, y, z, g)$ when a poacher arrives requires more care. Because the surplus now moves with the aggregate states, as z and g evolve over time $S(x, y, z, g)$ may fall below a worker's standing promise. To remedy this, we enforce that at the instant this happens the incumbent renegotiates the promise down to the current match surplus. This downward renegotiation is permanent and a promise reduced this way is not restored if the surplus later recovers. We also allow the worker and firm to end the match if $S(x, y, z, g) < 0$. This preserves the moving rule from [Cahuc et al. \(2006\)](#) that the worker moves if $y' > y$ (so long as at least one match has positive surplus) in an environment with aggregate shocks.

Bargaining in our model: We show that with $\chi = 1$, our protocol leads to the same surplus allocation as [Cahuc et al. \(2006\)](#). Under our surplus division protocol, we have the following. If $y' > y$, then $S(x, y', z, g) > S(x, y, z, g)$ and the worker moves. After the move, the worker gets a new value of:

$$V^u(x, z, g) + q + \beta(S(x, y', z, g) - S(x, y, z, g)) + \chi(V^p(x, y, q, z, g) - V^v(y, z, g)).$$

Imposing $\chi = 1$, $V^v(y, z, g) = 0$ and $S(x, y, z, g) > 0$, we can translate into the

notation from [Cahuc et al. \(2006\)](#):

$$\begin{aligned}
& V^u(x, z, g) + q + \beta(S(x, y', z, g) - S(x, y, z, g)) + V^p(x, y, q, z, g) - V^v(y, z, g) \\
&= P(x, y, z, g) + \beta(S(x, y', z, g) - S(x, y, z, g)) \\
&= V(x, F(x, y, z), y, z, g) + \beta(V(x, F(x, y', z), y', z, g) - V(x, F(x, y, z), y, z, g)) \\
&= \beta V(x, F(x, y', z), y', z, g) + (1 - \beta)V(x, F(x, y, z), y, z, g).
\end{aligned}$$

which is the same as equation (C.1) in our description of [Cahuc et al. \(2006\)](#) above but without $z = 1$ and $g = \bar{g}$.

Now, suppose that $y' \leq y$. Then $S(x, y, z, g) \geq S(x, y', z, g)$ and so under our protocol the worker does not move and gets a new value of:

$$V^u(x, z, g) + q + \chi(\max\{q, \beta S(x, y, z, g) + (1 - \beta)S(x, y', z, g)\} - q)$$

Imposing $\chi = 1$, $S(x, y, z, g) > q > 0$, and $\beta S(x, y, z, g) + (1 - \beta)S(x, y', z, g) > q$ so the poacher is a credible threat, we can translate the previous equation into the notation from [Cahuc et al. \(2006\)](#):

$$\begin{aligned}
& V^u(x, z, g) + \beta S(x, y, z, g) + (1 - \beta)S(x, y', z, g) \\
&= (1 - \beta)V(x, F(x, y', z), y', z, g) + \beta V(x, F(x, y, z), y, z, g),
\end{aligned}$$

which is the same as equation (C.2) in our description of [Cahuc et al. \(2006\)](#) above. If $\beta S(x, y, z, g) + (1 - \beta)S(x, y', z, g) \leq q$, then the poacher is not a credible threat and the worker value stays at $V^u + q$.

C.2.2 Relationship to [Postel-Vinay and Robin \(2002\)](#)

Our formulation also nests the bargaining environment of [Postel-Vinay and Robin \(2002\)](#). As in [Cahuc et al. \(2006\)](#), workers engage in on-the-job search and move if and only if the poaching firm they encounter has a higher $y' > y$. However, unlike [Cahuc et al. \(2006\)](#), there is no stage (iii) and so workers do not capture any share of the incremental surplus created by the new match. That is, the worker simply accepts the best offer from stage (i).

In the notation of Section C.2.1, [Postel-Vinay and Robin \(2002\)](#) corresponds to the special case $\beta = 0$. Suppose that a worker currently employed at a firm of type y en-

counters a more productive firm $y' > y$, the incumbent firm's maximum offer in stage (i) is $V(x, F(x, y), y, z, g)$ and the poaching firm wins the worker by matching this value. The worker therefore arrives at the new match with value $V(x, F(x, y), y, z, g)$, while the entire incremental surplus $V(x, F(x, y'), y', z, g) - V(x, F(x, y), y, z, g)$ accrues to the poaching firm. Translating this to our notation and imposing $V^v(y, z, g) = 0$, the worker arrives at the new match with value $V^u(x, z, g) + S(x, y, z, g)$. Under our surplus division protocol, this corresponds to the case $\chi = 1$ and $\beta = 0$, since

$$\begin{aligned} V^u(x, z, g) + q' &= V^u(x, z, g) + q + \chi(V^p(x, y, q, z, g) - V^v(y, z, g)) \\ &= V^u(x, z, g) + S(x, y, z, g). \end{aligned}$$

The case for $y' \leq y$ is analogous. Thus, the environment of [Postel-Vinay and Robin \(2002\)](#) is obtained as the special case in which workers can fully carry incumbent rents across jobs ($\chi = 1$) but do not share in the incremental surplus created by a more productive match ($\beta = 0$).

C.2.3 Relationship to [Pissarides \(1994\)](#) and Continuous Renegotiation

Our formulation also nests the continuous renegotiation protocol sometimes assumed in the search-and-matching literature, as in [Pissarides \(1994\)](#). In [Pissarides \(1994\)](#), wages always split the prevailing match surplus, with unemployment serving as the worker's outside option in bargaining. Conceptually, this means that an employed worker's only credible outside option in bargaining on-the-job is quitting into unemployment rather than returning to the previous job after bargaining with a poaching firm.

Notation: [Pissarides \(1994\)](#) considers an environment with (g)ood and (b)ad jobs, tenure-specific productivity, and on-the-job search only from bad jobs. Output in a job with tenure τ is $a(\tau)f$ for good jobs and $a(\tau)\psi y$ for bad jobs with $\psi \in (0, 1)$. The value of unemployment is denoted by U , the worker's value in a good job at tenure τ is denoted by $E_g(\tau)$ and the worker's value in a bad job while searching on the job is denoted by $E_{bs}(\tau)$. The firm's value from a filled good job is denoted by $J_g(\tau)$ and the value from a filled bad job with a searching worker is denoted by $J_{bs}(\tau)$. The values of vacant good and bad jobs are denoted by V_g and V_b , and free entry implies $V_g = V_b = 0$ in equilibrium. Wages are denoted by $w_g(\tau)$, $w_{bs}(\tau)$, and $w_b(\tau)$. The

worker's surplus share is denoted by β .

Reconciliation with our notation: Our environment has worker type x , firm type y , aggregate state (z, g) , and promised worker value q . The mapping from [Pissarides \(1994\)](#)'s notation to ours is:

$$\begin{aligned} U(z, g) &\leftrightarrow V^u(x, z, g), \\ E_g(\tau, z, g); E_{bs}(\tau, z, g) &\leftrightarrow V^e(x, y, q, z, g) = V^u(x, z, g) + q, \quad y \in \{g, b\}, \\ J_g(\tau, z, g); J_{bs}(\tau) &\leftrightarrow V^p(x, y, q, z, g), \quad y \in \{g, b\} \end{aligned}$$

and

$$\begin{aligned} V_g(z, g), V_{bs}(z, g) &\leftrightarrow V^v(y, z, g), \quad y \in \{g, b\} \\ E_{bs}(\tau, z, g) - U, E_g(\tau, z, g) - U &\leftrightarrow q, \quad y \in \{g, b\} \\ J_g(\tau, z, g) - V_g(z, g) + E_g(\tau, z, g) - U(z, g) &\leftrightarrow S(x, y_g, z, g) \\ J_{bs}(\tau, z, g) - V_{bs}(z, g) + E_{bs}(\tau, z, g) - U(z, g) &\leftrightarrow S(x, y_b, z, g). \end{aligned}$$

Bargaining in the style of [Pissarides \(1994\)](#): The key assumption on page 464-465 of [Pissarides \(1994\)](#) is that “wages always split the prevailing match surplus (in bargaining terms there is continuous wage renegotiation) so a job holder's only credible alternative to the current job is quitting into unemployment”. This means that for good and bad jobs before and after on-the-job transitions we have (equations (12) and (18) in [Pissarides, 1994](#)):

$$\begin{aligned} E_g(\tau, z, g) - U(z, g) &= \beta[J_g(\tau, z, g) - V_g(z, g) + E_g(\tau, z, g) - U(z, g)] \\ E_{bs}(\tau, z, g) - U(z, g) &= \beta[J_{bs}(\tau, z, g) - V_b(z, g) + E_{bs}(\tau, z, g) - U(z, g)] \end{aligned} \tag{C.3}$$

Reconciliation with our model: Suppose in our model we have $\chi = 0$ during on-the-job search meetings, and away from on-the-job search meetings, workers and firms renegotiate to the surplus division in new matches. Then, away from on-the-job search meetings, we have:

$$q = V^e(x, y, q, z, g) - V^u(x, z, g) = \beta S(x, y, z, g).$$

which is the same as (C.3) from [Pissarides \(1994\)](#) but with our notation. Now,

suppose the worker in a match (x, y) meets a new firm with y' . If the worker moves, then their value above replacement after the move becomes (with $\chi = 0$):

$$\begin{aligned} q' &= q + \beta(S(x, y', z, g) - S(x, y, z, g)) + \chi(V^p(x, y, q, z, g) - V^v(y, z, g)) \\ &= \beta S(x, y, z, g) + \beta(S(x, y', z, g) - S(x, y, z, g)) \\ &= \beta S(x, y', z, g) \end{aligned}$$

which again is the same as (C.3) from [Pissarides \(1994\)](#). If the worker does not move, then their value above replacement after the meeting becomes (with $\chi = 0$):

$$\begin{aligned} q' &= q + \chi(\max\{q, \beta S(x, y, z, g) + (1 - \beta)S(x, \tilde{y}, z, g)\} - q) \\ &= q \\ &= \beta S(x, y, z, g) \end{aligned}$$

So, for $\chi = 0$, the on-the-job search outcomes during on-the-job search transitions are consistent with continuous renegotiation.

Observe that for $\chi \neq 0$, the on-the-job search outcomes are inconsistent with continuous contract renegotiation because after moving, the worker gets:

$$\begin{aligned} q' &= q + \beta(S(x, y', z, g) - S(x, y, z, g)) + \chi(V^p(x, y, q, z, g) - V^v(y, z, g)) \\ &\neq \beta S(x, y', z, g). \end{aligned}$$

So, we see that continuous renegotiation is consistent with a special case of the division of the incumbent's surplus satisfying $\chi = 0$.

C.3 Wages For On-the-Job Search With $\chi = 1$

Deriving the wages for the $\chi = 0$ case is sufficiently simple that we could do it in the main text. Deriving the distribution of wages for $\chi > 0$ is significantly more complicated. In this Appendix we derive the Kolmogorov Forward (KF) equation for the distribution of promised worker surplus across employed worker–firm matches when on-the-job search bargaining follows the [Cahuc et al. \(2006\)](#) protocol described in the previous Appendix C.2.1. This corresponds to the $\chi = 1$ case in the On-The-Job-Search (OJS) model from Section 4.2. We then discuss how to solve the KFE numerically after the surplus function S has been solved for.

C.3.1 Contract Terms and Dynamics

High level, after contract formation, the firm's promise to the worker q does not change except when the worker meets a poaching firm and when match surplus S_t falls below q forcing renegotiation back down to S_t . We explain these changes in detail below.

Promised value changes. Suppose a worker currently employed at firm y with promised surplus q meets a poaching firm with $\tilde{y}$ at time t . Let $S_t(x, y) := S(x, y, z_t, g_t)$ denote the surplus at time t . Because promises are bounded at $S_t(x, y)$ through renegotiation, we can assume that $q \leq S_t(x, y)$ when the worker meets a poaching firm. We follow the [Cahuc et al. \(2006\)](#) protocol described in the previous Appendix C.2.1 where (i) the firms make simultaneous noncooperative job contracts, (ii) the firm with the lower productivity loses the competition with a maximum bid of $S_t(x, \min\{y, \tilde{y}\})$, and (iii) the firm with higher productivity then Nash Bargains with the worker to divide the surplus, with the worker using $S_t(x, \min\{y, \tilde{y}\})$ as their outside option.

The promised surplus q is used to determine the outcome of meetings. Using the results from the previous Appendix C.2.1, there are three possible outcomes:

1. If $\beta S_t(x, y) + (1 - \beta)S_t(x, \tilde{y}) \leq q$, then the poaching firm's threat is not credible and the worker retains promised surplus q .
2. If $q < \beta S_t(x, y) + (1 - \beta)S_t(x, \tilde{y}) \leq S_t(x, y)$, then the worker remains with the incumbent, but the outside offer improves the worker's reference bargaining position to $V^u(x, z, g) + S_t(x, \tilde{y})$. Under Nash bargaining, the worker's value becomes:

$$\begin{aligned} V_t^u(x) + S_t(x, \tilde{y}) + \beta(S_t(x, y) - S_t(x, \tilde{y})) \\ = V_t^u(x) + \beta S_t(x, y) + (1 - \beta)S_t(x, \tilde{y}) \end{aligned}$$

Hence the worker's new promised value above their outside option becomes:

$$Q_t^s(x, y, \tilde{y}) := \beta S_t(x, y) + (1 - \beta)S_t(x, \tilde{y}).$$

3. If $S_t(x, \tilde{y}) > S_t(x, y)$, then the worker moves to the new firm, their reference bargaining position improves to $V_t^u(x) + S_t(x, y)$, and under Nash bargaining

the worker's value improves to:

$$\begin{aligned} V_t^u(x) + S_t(x, y) + \beta(S_t(x, \tilde{y}) - S_t(x, y)) \\ = V_t^u(x) + (1 - \beta)S_t(x, y) + \beta S_t(x, \tilde{y}) \end{aligned}$$

So, the worker's new promised value above their outside option $V_t^u(x)$ is:

$$Q_t^m(x, y, \tilde{y}) := (1 - \beta)S_t(x, y) + \beta S_t(x, \tilde{y}).$$

Throughout this section we maintain the convention that the second argument is the firm the worker leaves and the third argument is the firm they move to, so the worker retains a share of $1 - \beta$ of the surplus at the old firm and captures a share of β of the surplus at the new firm.

The distribution of promised surplus. For each match type (x, y) , let $L_t(q; x, y)$ denote the cumulative mass of employed workers whose promised surplus is less than or equal to q and let $\ell_t(\cdot; x, y)$ denote the measure over promised surplus levels. When the measure $\ell_t(\cdot; x, y)$ is absolutely continuous, there exists a density, which we also denote by $\ell_t(q; x, y)$, such that

$$L_t(q; x, y) = \int_{-\infty}^q \ell_t(q'; x, y) dq'.$$

The evolution of promised values: For exposition we write the acceptance decisions as sharp indicators. In the solved model these are replaced by the smoothed (logit) acceptance probabilities α^e , as in the main text, which also ensures the surplus is differentiable in the match measure g_t .

Recall that a promise is renegotiated down whenever it would exceed the current match surplus $S_t(x, y) = S(x, y, z_t, g_t)$. This ceiling moves in two ways. First, at a fixed z_t it drifts smoothly as g_t evolves, and second, at a jump of z_t it moves discontinuously. The cumulative mass $L_t(q; x, y)$ therefore evolves by an endogenous drift between jumps of z_t and an exogenous jump at each change in z_t :

$$dL_t(q; x, y) = \mu_t^L(q; x, y) dt + \sum_{\check{z}_t \neq z_t} \left[\mathcal{R}_{\check{z}_t} L_t(q; x, y) - L_t(q; x, y) \right] dN_t(z_t; \check{z}_t),$$

subject at all times to the boundary condition

$$L_t(q; x, y) = g_t(x, y) \quad \text{for all } q \geq S_t(x, y).$$

Here $\mathcal{R}_{\check{z}_t}$ is the renegotiation map

$$(\mathcal{R}_{\check{z}_t} L_t)(q; x, y) = \begin{cases} L_t(q; x, y), & q < S(x, y, \check{z}, g_t), \\ g_t(x, y), & q \geq S(x, y, \check{z}, g_t), \end{cases}$$

$N_t(z_t; \check{z}_t)$ is a Poisson random variable such that $dN_t(z_t; \check{z}_t) = 1$ when z_t changes to $\check{z}_t$, and the endogenous drift $\mu_t^L(q; x, y) = \mu_t^L(q; x, y, z_t, g_t)$ is

$$\begin{aligned} \mu_t^L(q; x, y, z_t, g_t) = & -(\delta(x, y, z_t) + \eta \alpha^b(x, y, z_t, g_t)) L_t(q; x, y) \\ & - \mathcal{M}_t^e \int \mathbf{1}[S_t(x, \tilde{y}) > S_t(x, y)] \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y} L_t(q; x, y) \\ & - \mathcal{M}_t^e \int \mathbf{1}[q < Q_t^s(x, y, \tilde{y}) \leq S_t(x, y)] \mathbf{1}[q < Q_t^s(x, y, \tilde{y})] \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y} L_t(q; x, y) \\ & + \mathcal{M}_t^e \int \mathbf{1}[S_t(x, y) > S_t(x, \tilde{y})] \mathbf{1}[q \geq Q_t^m(x, \tilde{y}, y)] \frac{g_t^v(y)}{\mathcal{V}_t} g_t(x, \tilde{y}) d\tilde{y} \\ & + \mathcal{M}_t^u \alpha(x, y, z_t, g_t) \frac{g_t^v(y)}{\mathcal{V}_t} g_t^u(x) \mathbf{1}[q \geq \beta S_t(x, y)]. \end{aligned}$$

where we have left redundancy on line three to show the pattern in the equations and help build the intuition discussed below.

The boundary condition holds at the upper edge of the support of $L_t(\cdot; x, y)$, forcing all mass to lie weakly below $S_t(x, y)$, so the cumulative mass equals its total $g_t(x, y)$ for every $q \geq S_t(x, y)$. The renegotiation map $\mathcal{R}_{\check{z}_t}$ applies the downward renegotiation at each jump of z_t , leaving the mass below the new surplus unchanged and moving all mass above it onto the new ceiling, so that the boundary condition holds at the post-jump surplus.

We now describe the drift μ_t^L term by term. The first term is the standard separation outflow due to exogenous separation or endogenous breakup. The second term captures the impact of poaching outflow. Whenever a worker meets a firm with surplus exceeding that of the incumbent match, the worker leaves the match entirely and so exits the set of workers employed at (x, y) with promised surplus less than or equal to q . The third term captures the impact of incumbent matching.

Suppose a worker currently employed in match (x, y) receives an outside offer from a firm $\tilde{y}$ that is not sufficiently attractive to induce a move. If $Q_t^s(x, y, \tilde{y}) > q$, then the incumbent raises the worker's offer to $Q_t^s(x, y, \tilde{y})$. Otherwise, it stays at q . If $Q_t^s(x, y, \tilde{y}) > q$, then every worker currently counted in $L_t(q; x, y)$ who receives such an offer leaves the set of workers with promised surplus below the threshold q , so the third term is an outflow from $L_t(q; x, y)$. The fourth term captures the impact of poaching inflow. Workers arrive into match (x, y) from incumbent matches $(x, \tilde{y})$ with lower surplus, entering with promised surplus $Q_t^m(x, \tilde{y}, y)$. Only workers whose arrival promise satisfies $Q_t^m(x, \tilde{y}, y) \leq q$ contribute to the cumulative mass below the threshold, so the fourth term measures the rate at which workers enter the set counted by $L_t(q; x, y)$ through successful poaching. The final term captures new matches formed from unemployment: newly hired workers enter with promised surplus $\beta S_t(x, y)$.

C.3.2 Wages

To make this analysis economically interpretable, we translate the promised values into wages by working with the HJB equations for employed workers and firms.

HJBE for the employed worker. Using $V_t^e(x, y, q) = V_t^u(x) + q$, the HJBE for the employed worker is:

$$\begin{aligned} \rho(V_t^u(x) + q) &= w_t(x, y, q) - (\delta(x, y, z_t) + \eta\alpha^b(x, y, z_t, g_t))q \\ &\quad + \mathcal{M}_t^e \int \mathbf{1}[q < Q_t^s(x, y, \tilde{y}) \leq S_t(x, y)] (Q_t^s(x, y, \tilde{y}) - q) \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y} \\ &\quad + \mathcal{M}_t^e \int \mathbf{1}[S_t(x, \tilde{y}) > S_t(x, y)] (Q_t^m(x, y, \tilde{y}) - q) \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y} \\ &\quad + \sum_{\tilde{z} \neq z_t} \lambda(z_t, \tilde{z}) (V^u(x, \tilde{z}, g_t) - V_t^u(x)) + \langle D_g V^e, \check{\mu}^g \rangle. \end{aligned}$$

The first on-the-job search integral captures the gain to the worker when the incumbent matches a credible outside offer: the promised value rises from q to $Q_t^s(x, y, \tilde{y})$. The second on-the-job search integral captures the gain when the worker departs: the worker arrives at the new match with promised value $Q_t^m(x, y, \tilde{y})$, so the gain above the current promise is $Q_t^m(x, y, \tilde{y}) - q$.

HJBE for the producing firm. Using $V_t^p(x, y, q) = S_t(x, y) - q$, the HJBE for the

producing firm is:

$$\begin{aligned}
\rho(S_t(x, y) - q) &= F(x, y, z_t) - w_t(x, y, q) \\
&\quad - (\delta(x, y, z_t) + \eta\alpha^b(x, y, z_t, g_t))(S_t(x, y) - q) \\
&\quad + \mathcal{M}_t^e \int \mathbf{1}[q < Q_t^s(x, y, \tilde{y}) \leq S_t(x, y)] (q - Q_t^s(x, y, \tilde{y})) \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y} \\
&\quad + \mathcal{M}_t^e \int \mathbf{1}[S_t(x, \tilde{y}) > S_t(x, y)] (q - Q_t^m(x, y, \tilde{y})) \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y} \\
&\quad + \sum_{\tilde{z} \neq z_t} \lambda(z_t, \tilde{z}) (S(x, y, \tilde{z}, g_t) - S_t(x, y)) + \langle D_g V^p, \check{\mu}^g \rangle.
\end{aligned}$$

The on-the-job search terms enter with opposite sign to those in the worker's HJB: every gain to the worker from a credible outside offer is an equal loss to the firm, whether the incumbent matches or the worker departs.

Recovering wages. Rearranging the employed worker's HJBE gives the wage function:

$$\begin{aligned}
w_t(x, y, q) &= \rho(V_t^u(x) + q) + (\delta(x, y, z_t) + \eta\alpha^b(x, y, z_t, g_t))q \\
&\quad - \mathcal{M}_t^e \int \mathbf{1}[q < Q_t^s(x, y, \tilde{y}) \leq S_t(x, y)] (Q_t^s(x, y, \tilde{y}) - q) \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y} \\
&\quad - \mathcal{M}_t^e \int \mathbf{1}[S_t(x, \tilde{y}) > S_t(x, y)] (Q_t^m(x, y, \tilde{y}) - q) \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y} \\
&\quad - \sum_{\tilde{z} \neq z_t} \lambda(z_t, \tilde{z}) (V^u(x, \tilde{z}, g_t) - V_t^u(x)) - \langle D_g V^e, \check{\mu}^g \rangle.
\end{aligned}$$

The dependence on $V^u(x, z, g)$ can be removed by using the unemployed worker's HJB to get

$$\begin{aligned}
w_t(x, y, q) &= b + \mathcal{M}_t^u \int \alpha(x, \tilde{y}, z_t, g_t) \beta S_t(x, \tilde{y}) \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y} \\
&\quad + (\rho + \delta(x, y, z_t) + \eta\alpha^b(x, y, z_t, g_t))q \\
&\quad - \mathcal{M}_t^e \int \mathbf{1}[q < Q_t^s(x, y, \tilde{y}) \leq S_t(x, y)] (Q_t^s(x, y, \tilde{y}) - q) \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y} \\
&\quad - \mathcal{M}_t^e \int \mathbf{1}[S_t(x, \tilde{y}) > S_t(x, y)] (Q_t^m(x, y, \tilde{y}) - q) \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y}.
\end{aligned}$$

The wage depends on the promised surplus q and therefore on the worker's history of outside offers within the match. Workers with higher promised surplus receive

higher wages because the firm must deliver a larger continuation value. Outside offers gradually raise promised surplus through incumbent matching and poaching events, reducing the firm's continuation value $S_t(x, y) - q$ and increasing the worker's share of match surplus. The pair (S_t, q) therefore provides a recursive representation of wage dynamics under on-the-job search.

C.3.3 Numerical Computation

We describe here how we solve the KFE for the distribution of promised worker surplus $L_t(q; x, y)$ derived in the previous section. We solve it for the OTJ specification used in the wage results of Section 4: the case $\chi = 1$ with worker bargaining share $\beta \rightarrow 0$, i.e. pure Bertrand competition between firms.

The grid: We discretize the promised-value coordinate q . For each match (x, y) we place a finite grid of K equally spaced points

$$q_k(x, y) = q_{\text{lo}}(x) + k\Delta_{xy}, \quad k = 0, 1, \dots, K - 1,$$

and we write $k(q) = (q - q_{\text{lo}}(x)) / \Delta_{xy}$ for the index of the grid point at which a promise q falls. The worker and firm types in the numerical surplus solution are discrete: x takes n_x values, y takes n_y values, and the aggregate state z takes two values. The surplus $S(x, y, z, g)$ is therefore defined on a finite grid, and the integrals over the firm a worker meets again become finite sums. The promised-value distribution is a separate q -distribution attached to each of the $n_x \times n_y$ cells, and the grid is built cell by cell.

We store, at each grid point, the mass $\ell_{t,k}(x, y)$ of workers whose promise sits there, rather than a density. Its cumulative sum $C_{t,k}(x, y) = \sum_{k' < k} \ell_{t,k'}(x, y)$ is the discrete counterpart of the cumulative distribution $L_t(q; x, y)$.

The discretized KFE: The discretized KFE for the cumulative mass is:

$$\begin{aligned} \partial_t L_t(q; x, y) = & - \left(\delta + \eta \alpha_t^b(x, y) \right) L_t(q; x, y) \\ & - \mathcal{M}_t^e \sum_{\tilde{y}} \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} \mathbf{1}[S_t(x, \tilde{y}) > S_t(x, y)] L_t(q; x, y) \end{aligned}$$

$$\begin{aligned}
& - \mathcal{M}_t^e \sum_{\tilde{y}} \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} \mathbf{1}[q < Q_t^s(x, y, \tilde{y}) \leq S_t(x, y)] \mathbf{1}[q < Q_t^s(x, y, \tilde{y})] L_t(q; x, y) \\
& + \mathcal{M}_t^e \frac{g_t^v(y)}{\mathcal{V}_t} \sum_{\tilde{y}} \mathbf{1}[S_t(x, y) > S_t(x, \tilde{y})] \mathbf{1}[q \geq Q_t^m(x, \tilde{y}, y)] g_t(x, \tilde{y}) \\
& + \mathcal{M}_t^u \alpha_t(x, y) \frac{g_t^v(y)}{\mathcal{V}_t} g_t^u(x) \mathbf{1}[q \geq \beta S_t(x, y)],
\end{aligned}$$

subject to the boundary condition $L_t(q; x, y) = g_t(x, y)$ for all $q \geq S_t(x, y)$. In the $\beta \rightarrow 0$ limit, $\mathbf{1}[q < Q_t^s(x, y, \tilde{y})] \rightarrow \mathbf{1}[q < S_t(x, \tilde{y})]$ and $\mathbf{1}[q \geq Q_t^m(x, \tilde{y}, y)] \rightarrow \mathbf{1}[q \geq S_t(x, \tilde{y})]$.

Soft acceptance: The indicators above are written for the binary ($\xi_e \rightarrow \infty$) acceptance limit. As discussed in the main text, the on-the-job acceptance decisions are replaced by the smoothed (logit) form of the acceptance function $\alpha^e(x, y, \tilde{y}, z_t, g_t)$, the probability that a type- x worker moves from incumbent firm y to firm $\tilde{y}$. Thus in our numerical solution of the promised-value distribution, the move indicator $\mathbf{1}[S_t(x, \tilde{y}) > S_t(x, y)]$ becomes $\alpha^e(x, y, \tilde{y}, z_t, g_t)$ and the poaching inflow indicator $\mathbf{1}[S_t(x, y) > S_t(x, \tilde{y})]$ becomes $\alpha^e(x, \tilde{y}, y, z_t, g_t)$. We use a large smoothing parameter, $\xi_e = 25$, so the logit is close to the sharp rule.

Discretized dynamics: Once $S(x, y, z, g)$ is solved, the evolution of $L_t(q; x, y)$ can be integrated forward directly on the grid: the mass at each point evolves by a balance of inflows and outflows, assembled at each time t from the current surplus, the match distribution g_t , and the aggregate state z_t .

The match measure g_t does not depend on L_t , so at each step it is advanced by its own forward-Euler update first, and L_t is then advanced using the updated g . Once S is solved for, the two evolve jointly along one simulated path of the aggregate state z_t . For all wage impulse responses we initialize L_t at its ergodic mean distribution evaluated at the ergodic mean of g . The main text shows the impulse responses for an expansion, and Figure 16 below shows the companion responses for a recession.

C.4 Additional Training Details

Here we highlight the details of the training procedure for the augmented surplus network of Section 4 that differ from those presented in Appendix B.2.1. The aug-

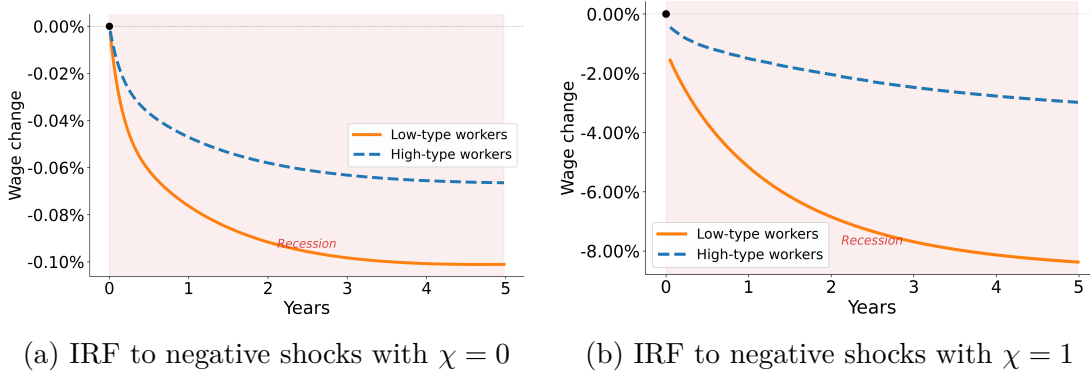


Figure 16: Wage change after negative z shocks for $\chi = 0$ and $\chi = 1$. Low, middle, and high types refer to the lowest, middle, and highest types in our $n_x = 7$ -type discretization for workers.

mented network $\hat{S}(x, y, z, g, \Psi; \Theta)$ is trained to solve the extended master equation jointly over the state space and the estimation parameter range. Relative to solving at a single calibration, augmenting the solver requires extending the sampling in two ways: across the parameter space and across the match distributions the economy visits over that range.

For the parameter space, during training we draw a set of structural parameters Ψ from fixed estimation ranges and resample them periodically. Each parameter draw is paired with training points (x, y, z, g) , so that the network learns the model's solution across the full parameter range rather than at a single point.

To sample the match distributions g , we use a modified sampling scheme compared to the one in Appendix B.2.1. There, the convex-combination sampler is the first of two training phases, followed by a principal second phase on ergodic-simulation draws at the single calibrated parameter point. The augmented network must instead solve across the whole parameter range, where an ergodic simulation at every parameter draw is too costly, so it has no per-parameter ergodic phase. Thus, more of the burden falls on the mixed-steady state sampler, which must do a better job of covering the distributions the network will face.

We therefore draw each g from a distribution centered on one of the two steady-state match distributions. For each parameter draw Ψ , let $g^{\text{ss}}(x, y; z_{\text{low}}, \Psi)$ and $g^{\text{ss}}(x, y; z_{\text{high}}, \Psi)$ denote the stationary match distributions at that parameter draw under the fixed aggregate states z_{low} and z_{high} , respectively. We assign each training point to one of these two aggregate states and draw its distribution g from a trun-

cated Gaussian around the corresponding anchor. The mean of this Gaussian is the anchor distribution, and the per-cell standard deviation is set to a tunable fraction of the gap $|g^{\text{ss}}(x, y; z_{\text{high}}, \Psi) - g^{\text{ss}}(x, y; z_{\text{low}}, \Psi)|$ between the two anchors, where we find that a fraction of 1/2 works well for our application. This yields a bimodal sample of distributions centered on the two steady states, with a controlled spread around each that reaches past the band joining them.

Finally, we flip the accompanying z -label on a large portion of these samples. This is deliberate, since along a transition path the economy is at the current state z while g still reflects the previous state, as g adjusts only gradually. Pairing a distribution drawn around one state’s anchor with the opposite z -label exposes the network to (z, g) pairs in which g still reflects the previous aggregate state rather than the current one. These pairs dominate along transition paths but would rarely be sampled if g and z came from the same aggregate state.

To approximate the augmented surplus function, we use a slightly larger feed-forward neural network than in Section 3, now with 5 hidden layers of 100 neurons each. We still use tanh activations and the Adam optimizer. Training on the mixed steady-state sample proceeds in two phases: a first phase with a higher learning rate of 5×10^{-5} and a smaller per-iteration sample size of 1024, followed by a second phase with a lower learning rate of 1×10^{-5} and a larger sample size of 2048.

After calibration, we add a final ergodic-simulation training stage at the estimated parameters to sharpen the solution near the point estimate. Holding Ψ fixed at the estimate, we follow a procedure similar to the ergodic-simulation training in Appendix B.2.1. Appendix C.5 below compares the simulated moments the model produces before and after this final stage, and shows they are near identical.

C.5 Additional Calibration Details

Table 10 reports the externally calibrated parameters we use when solving the OJS Model. Table 11 reports the estimated parameters together with the targeted moments, where columns (i)–(iii) give the model-implied values of these moments at the estimated parameter vector under three different evaluations: (i) the calibrator’s surrogate prediction; (ii) using the augmented surplus network trained on mixed steady-state samples; and (iii) the model after simulated ergodic training at the fixed estimated parameters. The three columns agree closely on the targeted moments, in-

dicating that the surrogate used in estimation is not distorting the targeted-moment fit and that the ergodic training leaves these moments essentially unchanged.

Parameter	Interpretation	Value
ρ	Discount rate	0.05
ξ, ξ^e	Extreme value for α and α^e choices	2.5, 25
$f(x, y)$	Production for match (x, y)	$0.6 + 0.4 (\sqrt{x} + \sqrt{y})^2$
$m(\mathcal{U}, \mathcal{V})$	Matching function	$\kappa \mathcal{U}^\nu \mathcal{V}^{1-\nu}$
ν	Elasticity in meeting function	0.5
A_L, A_H	TFP levels	0.985, 1.015
λ_z	Poisson transition probability	0.08
n_x	Discretization of worker types	7
n_y	Discretization of firm types	8

Table 10: Externally calibrated parameters of the OJS Model in Section 4.

Parameter	Interpretation	Value	Fitted moment	Data	(i)	(ii)	(iii)
β	Surplus division factor	0.836	$\mathbb{E}[U]$	0.058	0.057	0.057	0.057
κ	Scale for meeting function	20.0	$\mathbb{E}[V]$	0.037	0.036	0.036	0.036
c	Firm entry cost	0.698	$\mathbb{E}[EE]$	0.025	0.026	0.026	0.026
δ	Separation rates	0.159	$\mathbb{E}[UE]$	0.421	0.449	0.450	0.447
ϕ	Employed worker search intensity	0.146	$\mathbb{E}[EU]$	0.025	0.027	0.027	0.027
η	Endogenous separation intensity	0.786					
b	Unemployment benefit	0.019					

Table 11: Estimated Parameters and Targeted Moments. Columns (i)–(iii) report model-simulated moments: (i) surrogate, (ii) pre-ergodic training (mixed steady-state samples), and (iii) post-ergodic training (simulated economy paths).

Table 12 presents the specific calibrations we use after fixing the worker surplus share at $\beta = 0.5$ or $\beta \rightarrow 0$, and then re-estimating the remaining model parameters to match the target moments. These calibrations reflect the results presented in Figure 4 of the main text. To keep matching the target moments as β decreases towards 0, the model compensates by raising the meeting function scale κ and firm entry cost c .

Once the surrogate mapping from parameters to moments is learned, undertaking a systematic sensitivity analysis is also cheap because the surrogate mapping

Parameter	Interpretation	Baseline	$\beta = 0.5$	$\beta \rightarrow 0$
β	Surplus division factor	0.836		
κ	Scale for meeting function	20.0	23.98	25.28
c	Firm entry cost	0.698	1.177	2.093
δ	Separation rates	0.159	0.075	0.098
ϕ	Employed worker search rate	0.146	0.158	0.166
η	Endogenous separation intensity	0.786	0.863	0.897

Table 12: Estimated Parameters and Targeted Moments For Fixed β .

returns the model-implied moments for any parameter vector without re-solving or re-simulating the model, so we can sweep each parameter freely. Figure 17 illustrates the surrogate’s moment-distance surface over the (ϕ, δ) dimensions, holding the other parameters at their optimal values. The surface is smooth and exhibits a well-defined minimum, which is what allows DeepSAM to find the optimal parameters. Table 13 uses the surrogate model to examine local sensitivity, perturbing estimated parameters by $\pm 10\%$ and recording the resulting change in the surrogate-predicted moments, holding the other parameters at their calibrated values. The worker surplus share β has the clearest and largest impact on all the target moments.

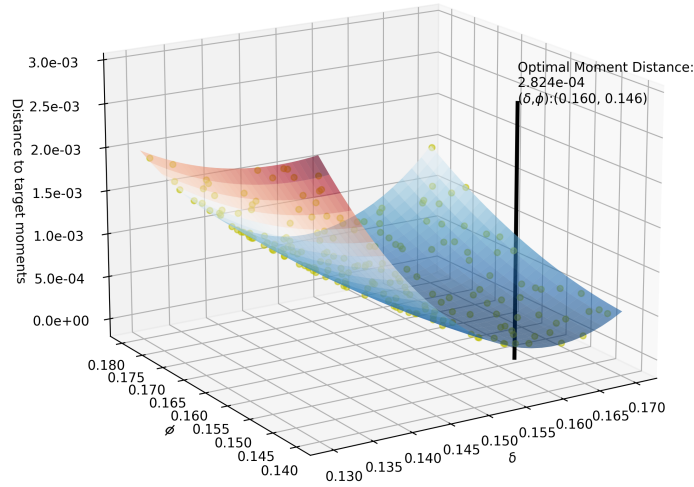


Figure 17: Surface plot of the distance to target moments across the (ϕ, δ) grid.

Parameter	Perturbation	$\mathbb{E}[U]$	$\mathbb{E}[V]$	$\mathbb{E}[e2e]$	$\mathbb{E}[e2u]$	$\mathbb{E}[u2e]$
β	+10%	+7.24%	-32.84%	-20.82%	-13.09%	-20.75%
	-10%	-15.11%	+28.25%	+9.02%	-0.38%	+17.38%
δ	+10%	+6.03%	+3.34%	+4.85%	+7.18%	+1.28%
	-10%	-5.87%	-3.50%	-5.07%	-7.15%	-1.29%
ϕ	+10%	+0.44%	+1.35%	+5.19%	-0.88%	-1.66%
	-10%	-0.44%	-1.35%	-5.19%	+0.88%	+1.66%
κ	+10%	-4.41%	-1.74%	+3.94%	-8.44%	-4.29%
	-10%	-3.51%	-4.63%	-4.77%	-4.24%	-0.93%
Surrogate at optimal parameters		0.057	0.0361	0.026	0.027	0.449
Data target		0.058	0.0368	0.025	0.025	0.421

Table 13: Surrogate-predicted moment sensitivity to $\pm 10\%$ perturbations of calibrated parameters, holding all others fixed at their calibrated values. Each entry shows the percentage deviation in the surrogate-predicted moment relative to the prediction at the optimally calibrated parameters. The $\kappa + 10\%$ row lies outside the range covered by the surrogate’s training draws and is therefore extrapolated; these values should be treated with caution.

D Idiosyncratic Productivity Shocks and Endogenous Separation

In this Appendix, we extend the on-the-job search environment from Section 4 to include idiosyncratic productivity shocks and endogenous separation. The model draws on [Krusell and Smith \(1998\)](#) and [Mortensen and Pissarides \(1994\)](#) in the way that stochastic idiosyncratic productivity is modeled.

D.1 Environment

The environment is the same as in Subsection 4.1 but with the following changes.

Match utility: As before, there is an exogenous aggregate state $z \in \mathcal{Z}$ that follows a continuous time Markov chain with transition intensities $\lambda(z, \check{z})$ for all $z, \check{z} \in \mathcal{Z}$. Now, each match also has an exogenous idiosyncratic state $\epsilon \in \mathcal{H}$ from a discrete set $\mathcal{H} = \{\epsilon_1, \dots, \epsilon_{n_\epsilon}\}$ that follows an idiosyncratic continuous time Markov chain with transition intensities $\lambda(\epsilon, \check{\epsilon})$ for all $\epsilon, \check{\epsilon} \in \mathcal{H}$. If a worker of type x is matched with a firm of type y , then they generate transferable utility $F(x, y, \epsilon, z) = A_z A_\epsilon f(x, y)$, where we use the notation that aggregate productivity, A_z , and idiosyncratic productivity, A_ϵ , are indexed by z and ϵ respectively. We assume that F is increasing in each variable and twice differentiable with uniformly bounded first partial derivatives on $\mathcal{X} \times \mathcal{Y} \times \mathcal{Z} \times \mathcal{H}$.

Distributions: As before, let $g_t^w(x)$ denote the population measure function of workers and let $g_t^f(y)$ denote the population measure function of firms. Now, let $g_t(x, y, \epsilon)$ denote the measure function of matched workers and firms with productivity ϵ , $g_t^e(x, \epsilon)$ denote the measure function of matched workers with productivity ϵ , and $g_t^u(x)$ denote the measure function of unmatched workers. Similarly on the firm side, let $g_t^p(y, \epsilon)$ denote the measure function of producing firms with idiosyncratic productivity ϵ , and $g_t^v(y)$ denote the measure function of vacant firms. The relationships between the functions are given in Table 14 below. We define the aggregate worker employment by $\mathcal{E}_t := \sum_\epsilon \int g_t^e(x, \epsilon) dx$, aggregate worker unemployment by $\mathcal{U}_t := \int g_t^u(x) dx$, aggregate producing firms by $\mathcal{P}_t := \sum_\epsilon \int g_t^p(y, \epsilon) dy$, and aggregate vacant firms by $\mathcal{V}_t := \int g_t^v(y) dy$.

Description	Measure Function	Conditional Density
Matches	$g_t(x, y, \epsilon)$	
Employed workers	$g_t^e(x, \epsilon) = \int g_t(x, y, \epsilon) dy$	$g_t^e(x, \epsilon)/\mathcal{E}_t$
Unemployed workers	$g_t^u(x) = g_t^w(x) - \sum_{\epsilon} g_t^e(x, \epsilon)$	$g_t^u(x)/\mathcal{U}_t$
Producing firms	$g_t^p(y, \epsilon) = \int g_t(x, y, \epsilon) dx$	$g_t^p(y, \epsilon)/\mathcal{P}_t$
Vacant firms	$g_t^v(y) = g_t^f(y) - \sum_{\epsilon} g_t^p(y, \epsilon)$	$g_t^v(y)/\mathcal{V}_t$

Table 14: Summary of distributions

Search and matching: All workers engage in random search. The matching function becomes $m(\mathcal{W}_t, \mathcal{V}_t)$ where $\mathcal{W}_t := \mathcal{U}_t + \phi\mathcal{E}_t$ denotes the total mass of searchers and ϕ is the exogenous relative search intensity of employed workers. The rates at which an unemployed or an employed worker meets a potential employer are given by $\mathcal{M}_t^u = \frac{m(\mathcal{W}_t, \mathcal{V}_t)}{\mathcal{W}_t}$ and $\mathcal{M}_t^e = \phi \frac{m(\mathcal{W}_t, \mathcal{V}_t)}{\mathcal{W}_t}$, while the probability that a vacant firm meets a potential hire is $\mathcal{M}_t^v = \frac{m(\mathcal{W}_t, \mathcal{V}_t)}{\mathcal{V}_t}$. Conditional on a firm meeting a worker, the probabilities that the worker is unemployed or employed are defined as $\mathcal{C}_t^u = \frac{\mathcal{U}_t}{\mathcal{U}_t + \phi\mathcal{E}_t}$ and $\mathcal{C}_t^e = \frac{\phi\mathcal{E}_t}{\mathcal{U}_t + \phi\mathcal{E}_t}$ respectively. So, the rate at which a firm meets an unemployed worker x in $X \subset \mathcal{X}$ is $\mathcal{M}_t^v \mathcal{C}_t^u \int_X \frac{g^u(x)}{\mathcal{U}_t} dx$ and the rate at which a firm meets an employed worker x in $X \subset \mathcal{X}$ is $\mathcal{M}_t^v \mathcal{C}_t^e \sum_{\epsilon} \int_X \frac{g^e(x, \epsilon)}{\mathcal{E}_t} dx$.

Bargaining between unemployed workers and firms: We impose that workers and firms see the initial draw of ϵ before deciding whether to match and that initial draws of ϵ are drawn from a multinomial distribution $\pi(\epsilon)$ on support $\mathcal{H}$. As in Section 4.1, we impose that the firm promises the worker a value q_t above their outside option $V_t^u(x)$. So, the values of an ϵ productivity match to a type x employed worker and a type y producing firm are then denoted by $V_t^e(x, y, \epsilon, q_t) = V_t^u(x) + q_t$ and $V_t^p(x, y, \epsilon, q_t)$. Denote the surplus of an ϵ productivity match between a type x worker and a type y firm as $S_t(x, y, \epsilon)$. Under Nash bargaining, $q_t = \beta S_t(x, y, \epsilon)$ at the start of the contract.

Bargaining on the job: Suppose a worker in a match (x, y) meets a firm of type $\tilde{y}$ and gets an idiosyncratic draw $\tilde{\epsilon}$ so the new surplus would be $S_t(x, \tilde{y}, \tilde{\epsilon})$. As before, the worker, incumbent firm, and poaching firm follow the parametrized bargaining protocol of Subsection 4.1, which nests several key models in the literature. Workers move to the new firm if and only if the new match has higher surplus

$S_t(x, \tilde{y}, \tilde{\epsilon}) \geq S_t(x, y, \epsilon) \geq 0$. Upon a move, the incremental surplus and the incumbent's firm-side surplus are divided according to this same protocol. The division of surplus is once again summarized by Table 4 of the main text, except that now $S_t \equiv S_t(x, y, \epsilon)$ is the incumbent's surplus, and $\tilde{S}_t \equiv S_t(x, \tilde{y}, \tilde{\epsilon})$ is the poacher's surplus.

Endogenous and exogenous separation: Workers and firms can immediately break up matches when agents draw a new idiosyncratic match state ϵ . As before, matches are destroyed at the exogenous rate $\delta(x, y, z)$.

Entry and vacancy: As in Subsection 4.1, we assume that vacancies are short-lived in the sense that at the end of each period all unfilled vacancies are destroyed and all firms leave.

D.2 Recursive Characterization of Equilibrium

We now define and characterize a recursive equilibrium. The aggregate states are (z, g) , where z is the aggregate productivity and g is the measure function over matches. The idiosyncratic states in a match are now (x, y, ϵ) .

D.2.1 Kolmogorov Forward Equation

The measure of matches over the idiosyncratic state space (x, y, ϵ) evolves by:

$$\begin{aligned}
dg_t(x, y, \epsilon) = & -\delta(x, y, z)g_t(x, y, \epsilon)dt \\
& -\mathcal{M}_t^e \sum_{\tilde{\epsilon}} \pi(\tilde{\epsilon}) \int \alpha^e(x, y, \epsilon, \tilde{y}, \tilde{\epsilon}, z, g) \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y} g_t(x, y, \epsilon) dt \\
& +\mathcal{M}_t^u \pi(\epsilon) \alpha(x, y, \epsilon, z, g) \frac{g_t^v(y)}{\mathcal{V}_t} g_t^u(x) dt \\
& +\mathcal{M}_t^e \pi(\epsilon) \sum_{\tilde{\epsilon}} \int \alpha^e(x, \tilde{y}, \tilde{\epsilon}, y, \epsilon, z, g) \frac{g_t^v(y)}{\mathcal{V}_t} g_t(x, \tilde{y}, \tilde{\epsilon}) d\tilde{y} dt \\
& -\sum_{\check{\epsilon} \neq \epsilon} \lambda(\epsilon, \check{\epsilon}) g_t(x, y, \epsilon) dt + \sum_{\check{\epsilon} \neq \epsilon} \lambda(\check{\epsilon}, \epsilon) (1 - \alpha^b(x, y, \epsilon, z, g)) g_t(x, y, \check{\epsilon}) dt \\
= &: \mu^g(x, y, \epsilon, z_t, g_t) dt
\end{aligned}$$

where now the endogenous match acceptance indicator, α , breakup indicator, α^b , and the on-the-job acceptance indicator, α^e , are:

$$\begin{aligned}\alpha(x, y, \epsilon, z, g) &:= \begin{cases} 1, & \text{if } S(x, y, \epsilon, z, g) > 0 \\ 0, & \text{otherwise} \end{cases} \\ \alpha^b(x, y, \epsilon, z, g) &:= 1 - \alpha(x, y, \epsilon, z, g) = \begin{cases} 1, & \text{if } S(x, y, \epsilon, z, g) < 0 \\ 0, & \text{otherwise} \end{cases} \\ \alpha^e(x, y, \epsilon, \tilde{y}, \tilde{\epsilon}, z, g) &:= \begin{cases} 1, & \text{if } S(x, \tilde{y}, \tilde{\epsilon}, z, g) \geq S(x, y, \epsilon, z, g) \text{ \& } S(x, \tilde{y}, \tilde{\epsilon}, z, g) \geq 0 \\ 0, & \text{otherwise.} \end{cases}\end{aligned}$$

The first term on the RHS is the exit rate due to exogenous separations, the second term is the exit rate due to workers finding better matches, the third term is the entry rate from unemployed workers finding jobs, the fourth term is the entry rate from employed workers moving on-the-job to (x, y) , the fifth term is exit from (x, y, ϵ) when ϵ switches to $\tilde{\epsilon}$, and the sixth term is the entry rate to (x, y, ϵ) from $(x, y, \tilde{\epsilon})$ when $\tilde{\epsilon}$ changes to ϵ (net of any workers who leave when ϵ changes).

D.2.2 Agent HJB Equations

Let $V^u(x, z, g)$ denote the value of an unemployed worker of type x , and $V^e(x, y, \epsilon, z, g)$ denote the value of a worker of type x employed at a firm of type y . On the firm side, let $V^v(y, z, g)$ denote the value of a vacancy for firm y , and $V^p(x, y, \epsilon, z, g)$ denote the value of a producing firm y employing a worker of type x . Finally, let $\check{\mu}^g$ denote agent's belief about the evolution of g .

Unemployed workers: Given beliefs and optimal matching decisions, the unemployed worker value function V^u satisfies the HJB equation:

$$\begin{aligned}\rho V^u(x, z, g) &= b + \mathcal{M}^u \sum_{\tilde{\epsilon}} \pi(\tilde{\epsilon}) \int \alpha(x, \tilde{y}, \tilde{\epsilon}, z, g) (V^e(x, \tilde{y}, \tilde{\epsilon}, z, g) - V^u(x, z, g)) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\ &\quad + \sum_{\check{z} \neq z} \lambda(z, \check{z}) (V^u(x, \check{z}, g) - V^u(x, z, g)) + \langle D_g V^u, \check{\mu}^g \rangle\end{aligned}$$

where $\pi(\tilde{\epsilon})$ is the probability distribution over the initial draw of $\tilde{\epsilon}$ and α is an indi-

cator function for whether workers and firms accept the match.

Vacant firms: As in the main text (Section 4.2), vacant firms are short-lived and so have no continuation value $V^v(y, z, g) = 0$.

Employed matches: To characterize on-the-job search behavior, we work with the value of a match rather than the individual value functions. We denote the combined value of an ϵ productivity match between a type x worker and a type y firm that has promised q by:

$$P(x, y, \epsilon, q, z, g) := V^e(x, y, \epsilon, q, z, g) + V^p(x, y, \epsilon, q, z, g).$$

We denote by $\tilde{P}(x, y, \tilde{y}, \tilde{\epsilon}, q, \tilde{q}, z, g)$ the combined continuation value of the worker and the incumbent firm after the worker moves to a poaching firm of type $\tilde{y}$ with match specific productivity $\tilde{\epsilon}$ to get $\tilde{q}$. Employed workers choose whether to accept matches from other firms and whether to leave their jobs to return to unemployment. So, given agent beliefs about the drift of the match density $\check{\mu}^g$, $P(x, y, \epsilon, q, z, g)$ solves the HJB equation:

$$\begin{aligned} \rho P(x, y, \epsilon, q, z, g) &= F(x, y, \epsilon, z) \\ &+ \delta(x, y, z)(V^u(x, z, g) + V^v(y, z, g) - P(x, y, \epsilon, q, z, g)) \\ &+ \mathcal{M}^e \sum_{\tilde{\epsilon}} \pi(\tilde{\epsilon}) \int \alpha^e(x, y, \epsilon, \tilde{y}, \tilde{\epsilon}, z, g)(\tilde{P}(x, y, \tilde{y}, \tilde{\epsilon}, q, \tilde{q}, z, g) - P(x, y, \epsilon, q, z, g)) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\ &+ \sum_{\check{\epsilon} \neq \epsilon} \lambda(\epsilon, \check{\epsilon}) \left((1 - \alpha^b(x, y, \check{\epsilon}, z, g)) P(x, y, \check{\epsilon}, q, z, g) \right. \\ &\quad \left. + \alpha^b(x, y, \check{\epsilon}, z, g)(V^u(x, z, g) + V^v(y, z, g)) - P(x, y, \epsilon, q, z, g) \right) \\ &+ \sum_{\check{z} \neq z} \lambda(z, \check{z}) (P(x, y, \epsilon, q, \check{z}, g) - P(x, y, \epsilon, q, z, g)) \\ &+ \langle D_g P, \check{\mu}^g \rangle. \end{aligned}$$

where, as in the main text, conditional on a move, the change in joint value is:

$$\tilde{P}(x, y, \tilde{y}, \tilde{\epsilon}, q, \tilde{q}, z, g) - P(x, y, \epsilon, q, z, g) = \beta[S(x, \tilde{y}, \tilde{\epsilon}, z, g) - S(x, y, \epsilon, z, g)].$$

Like in the main text, we can verify that $P(x, y, \epsilon, q, z, g)$ and surplus do not depend on q , and so can be written as $P(x, y, \epsilon, z, g)$ and $S(x, y, \epsilon, z, g)$.

D.2.3 Master Equation For Surplus

Under belief consistency, $\check{\mu}^g = \mu^g$, the differential “master” equation for the surplus $S(x, y, \epsilon, z, g)$ is the following:

$$\begin{aligned}
\rho S(x, y, \epsilon, z, g) &= \rho(V^p(x, y, \epsilon, z, g) - V^v(y, z, g) + V^e(x, y, \epsilon, z, g) - V^u(x, z, g)) \\
&= \rho(P(x, y, \epsilon, z, g) - V^u(x, z, g)) \\
&= F(x, y, z, \epsilon) - \delta(x, y, z)S(x, y, \epsilon, z, g) - b \\
&+ \mathcal{M}^e \sum_{\tilde{\epsilon}} \pi(\tilde{\epsilon}) \int \alpha^e(x, y, \epsilon, \tilde{y}, \tilde{\epsilon}, z, g) \beta(S(x, \tilde{y}, \tilde{\epsilon}, z, g) - S(x, y, \epsilon, z, g)) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\
&- \mathcal{M}^u \sum_{\tilde{\epsilon}} \pi(\tilde{\epsilon}) \int \alpha(x, \tilde{y}, \tilde{\epsilon}, z, g) \beta S(x, \tilde{y}, \tilde{\epsilon}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\
&+ \sum_{\epsilon \neq \check{\epsilon}} \lambda(\epsilon, \check{\epsilon}) \left((1 - \alpha^b(x, y, \check{\epsilon}, z, g)) (S(x, y, \check{\epsilon}, z, g) - S(x, y, \epsilon, z, g)) \right. \\
&\quad \left. - \alpha^b(x, y, \check{\epsilon}, z, g) S(x, y, \epsilon, z, g) \right) \\
&+ \sum_{z \neq \check{z}} \lambda(z, \check{z}) (S(x, y, \epsilon, \check{z}, g) - S(x, y, \epsilon, z, g)) + \langle D_g S, \mu^g \rangle
\end{aligned}$$

where we have used that $V^v = 0$.

D.2.4 Free Entry

The “flow” value, $J^v(y, z, g)$, of creating a vacancy with productivity y is

$$\begin{aligned}
J^v(y, z, g) &= \mathcal{M}^v \mathcal{C}^u \sum_{\tilde{\epsilon}} \pi(\tilde{\epsilon}) \int \alpha(\tilde{x}, y, \tilde{\epsilon}, z, g) (1 - \beta) S(\tilde{x}, y, \tilde{\epsilon}, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}} d\tilde{x} \\
&+ \mathcal{M}^v \mathcal{C}^e \sum_{\tilde{\epsilon}, \epsilon} \pi(\epsilon) \int \int \alpha^p(y, \tilde{x}, \epsilon, \tilde{y}, \tilde{\epsilon}, z, g) (1 - \beta) \\
&\quad \times (S(\tilde{x}, y, \epsilon, z, g) - S(\tilde{x}, \tilde{y}, \tilde{\epsilon}, z, g)) \frac{g(\tilde{x}, \tilde{y}, \tilde{\epsilon})}{\mathcal{E}} d\tilde{x} d\tilde{y}.
\end{aligned}$$

The free entry condition is:

$$0 = -c + \int_0^1 J^v(\tilde{y}, z, g) \bar{h}^v(\tilde{y}) d\tilde{y}.$$

where $\bar{h}^v(\tilde{y}) = 1$

D.3 Comparison to Mortensen and Pissarides (1994)

In both our model and Mortensen and Pissarides (1994), agents break up matches when the match-specific productivity ϵ decreases by a sufficiently large amount. In Mortensen and Pissarides (1994), agents idiosyncratically receive new draws of idiosyncratic match productivity ϵ from a fixed distribution $H(\epsilon)$ at rate λ . This means that in the steady state (or between aggregate state changes) the aggregate flow of endogenous separations is:

$$\lambda H(\epsilon_d), \quad s.t. \quad S(\epsilon_d) = 0 \quad (D.1)$$

where ϵ_d is the cutoff value of ϵ at which agents get zero surplus and so are indifferent about separation (see equation (15) in Mortensen and Pissarides (1994)).

In our model, the idiosyncratic productivity process has discrete support instead of the continuous support in Mortensen and Pissarides (1994). It also has additional complexities such as on-the-job search and a two-sided heterogeneity. This means that in our model the aggregate flow of endogenous separation from matches of type (x, y) at aggregate state (z, g) due to idiosyncratic productivity shocks is:

$$Q(x, y, z, g) = \sum_{\epsilon} g(x, y, \epsilon) \sum_{\check{\epsilon} \neq \epsilon} \lambda(\epsilon, \check{\epsilon}) \alpha^b(x, y, \check{\epsilon}, z, g), \quad (D.2)$$

$$s.t. \quad \alpha^b(x, y, \epsilon, z, g) = \begin{cases} 1, & \text{if } S(x, y, \epsilon, z, g) < 0 \\ 0, & \text{otherwise} \end{cases}$$

Comparing (D.1) and (D.2) we can see that our model provides additional characterization of how endogenous separation flows vary across the (x, y) match space and (z, g) aggregate state space.

D.4 Solution

We use the model to study how a transient aggregate downturn propagates through the cross-section of matches. To organize results across idiosyncratic states, we first introduce a decomposition of unemployment.

Calibration details: We focus on a two state idiosyncratic productivity process in which $\epsilon \in \{\epsilon_L, \epsilon_H\}$. We set the rate of transition between idiosyncratic states to match the $\lambda = 0.081$ quarterly rate of idiosyncratic shock arrival in [Mortensen and Pissarides \(1994\)](#) and the productivities to $A_{\epsilon_L} = 0.95$ and $A_{\epsilon_H} = 1.05$.

Recession experiment: Starting from the ergodic mean of the distribution with $z = z_H$, we force $z = z_L$ for one year and then let z evolve according to its continuous time Markov chain process. We simulate N worker paths and average the resulting trajectories. For each ϵ , Figure 18 plots the cumulative net aggregate outflow from matches with idiosyncratic productivity ϵ :

$$\Delta(\epsilon, t) = - \int_{t_0}^t \left(\int \int \mu^g(x, y, \epsilon, z_s, g_s) dx dy \right) ds$$

where t_0 indicates the start of the recession. Outflows from both values of ϵ rise on impact and mean-revert after the shock ends, but the rise is steeper and the recovery slower at the low idiosyncratic state ϵ_L . This is because matches at ϵ_L generate lower output and hence carry lower surplus $S(x, y, \epsilon, z, g)$, placing more of them near the breakup threshold $S = 0$.

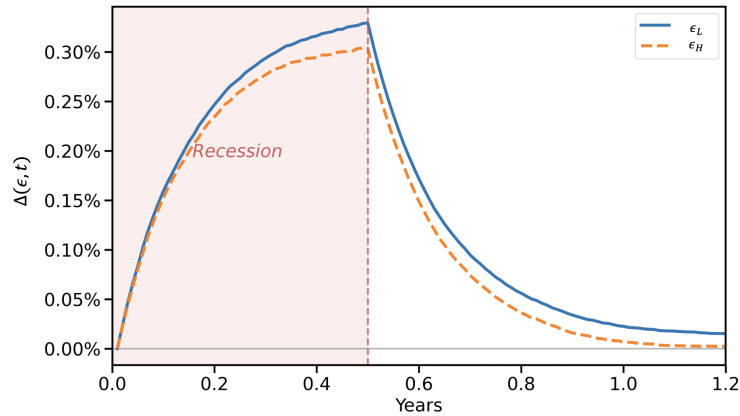


Figure 18: Decomposition of the cumulative net aggregate outflows from matches at each ϵ . The blue line is $\Delta(\epsilon_L, t)$ and the orange dashed line is $\Delta(\epsilon_H, t)$.

The cross-section of endogenous separation: The endogenous separation rate varies both across the match space (x, y) and with the aggregate state (z, g) . To illustrate this dependence, we evaluate the aggregate flow of endogenous separation, $Q(x, y, z, g)$

from equation (D.2) at the ergodic mean of the distribution g_{erg} and different $z \in \{z_H, z_L\}$. The left panel of Figure 19 plots $Q(x, y, z_H, g_{erg})$ at the ergodic mean of the distribution at z_H . The right panel plots the percentage change $(Q(x, y, z_L, g_{erg}) - Q(x, y, z_H, g_{erg}))/Q(x, y, z_H, g_{erg})$ when z_H decreases to z_L . Evidently, the recession has a larger impact on endogenous separations for high (x, y) types.

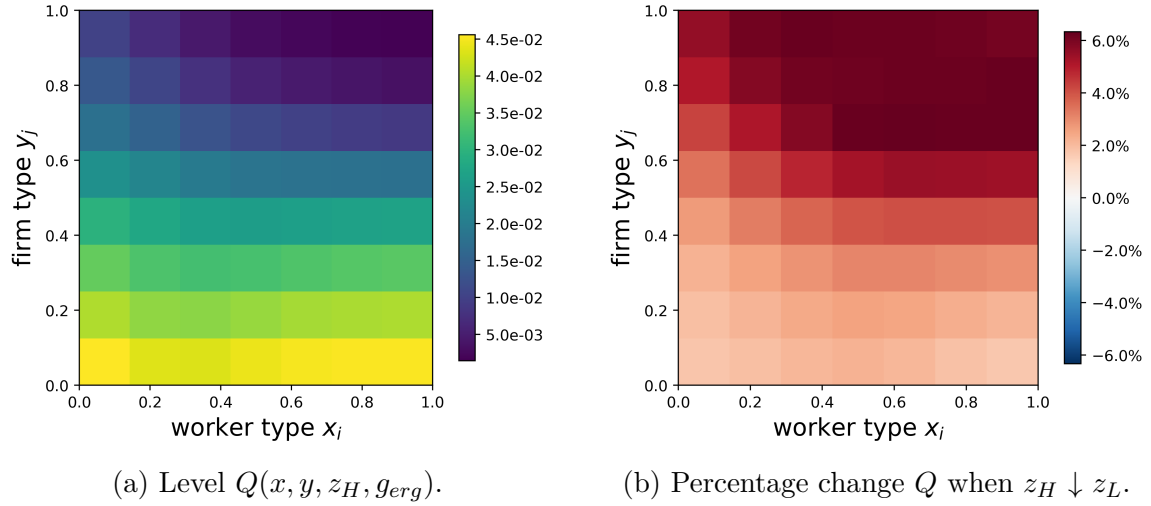


Figure 19: Aggregate flow of endogenous separation. Left: level of $Q(x, y, z_H, g_{erg})$ at high aggregate productivity. Right: Percentage change $(Q(x, y, z_L, g_{erg}) - Q(x, y, z_H, g_{erg}))/Q(x, y, z_H, g_{erg})$.

E On-The-Job-Search With Slow Entry

So far, we have always imposed that firms can enter and exit instantaneously in order to ensure that a free entry condition is satisfied. This makes the mass of firms a jump variable. In Section 2 we did this with long-lived vacancies and in Section 4 we did this with short-lived vacancies. In this Appendix, we consider a model where vacancies are long-lived and firms can only enter or exit the economy slowly. We show that the rate of firm entry has strong implications for distributional responses during a recession.

E.1 Environment Changes

The environment is the same as Section 4.1 but with the following changes to firm entry and exit. Firms exit at an exogenous flow rate ς , as in the model of Section 2. However, we remove exogenous firm exit at changes in z , so $\sigma(z, \tilde{z}) = 0$. Vacancies are now long-lived and entry is slow, rather than the firm mass being a jump variable.

When new firms enter, they draw their type y from a fixed entry density $\bar{h}^f(y)$. In our numerical application we take this density to be uniform $U(0, 1)$ so $\bar{h}^f(y) = 1$ on $y \in [0, 1]$. Entry occurs at the endogenous rate $\psi \alpha^f(z, g, \mathcal{F})$, where $\psi \geq 0$ is a parameter limiting the rate of firm entry and α^f indicates whether entry is profitable:

$$\alpha^f(z, g, \mathcal{F}) := \begin{cases} 1, & \text{if } \int V^v(\tilde{y}, z, g, \mathcal{F}) \bar{h}^f(\tilde{y}) d\tilde{y} > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Entry thus occurs only when the expected value of a new vacancy is positive. As $\psi \rightarrow 0$, there is no entry. As $\psi \rightarrow \infty$, we recover the model of Section 4.1, where the mass of firms is a jump variable.

E.2 Recursive Characterization of Equilibrium

We now define and characterize a recursive equilibrium. The aggregate states are $(z, g, \mathcal{F})$, where z is the aggregate productivity, g is the measure function over matches, and $\mathcal{F}$ is the total mass of firms. In principle, the density of firm types $g_t^f(y)$ could also be a state variable but we show that under our setup the density of firms is constant and so only the mass of firms becomes a state.

E.2.1 Distribution Evolution

We start by deriving the evolution of the different distributions in the economy. The law of motion for the measure of matches, g_t , evolves according to the same KFE (4.1) as for the case with short-lived vacancies in Section 4.2. The evolution of the distribution of unemployment is once again given by $g_t^u(x) = g_t^w(x) - \int g_t(x, y) dy$ with g_t^w fixed, so this equation is implied by the KFE for $g_t(x, y)$. What is different compared to Section 4.2 is the evolution of the measures of firms and vacancies.

Law of motion for g_t^f and $\mathcal{F}_t$. Under the new firm entry assumptions, the measure of firms of type y evolves according to:

$$dg_t^f(y) = \underbrace{\psi \alpha^f(z, g, \mathcal{F}) \bar{h}^f(y)}_{\text{entry inflow}} dt - \underbrace{\varsigma g_t^f(y)}_{\text{exit outflow}} dt$$

Integrating over y and imposing that $\bar{h}^f(y) = 1$ gives the law of motion for the total mass:

$$d\mathcal{F}_t = \psi \alpha^f(z, g, \mathcal{F}) dt - \varsigma \mathcal{F}_t dt.$$

Therefore, if $g_0^f(y) = \bar{h}^f(y) \mathcal{F}_0$, substituting the proportional form $g_t^f(y) = \bar{h}^f(y) \mathcal{F}_t$ into the exit term shows it is self-consistent,

$$dg_t^f(y) = \bar{h}^f(y) [\psi \alpha^f(z, g, \mathcal{F}) - \varsigma \mathcal{F}_t] dt = \bar{h}^f(y) d\mathcal{F}_t,$$

and so the firm density is constant at $g_t^f(y)/\mathcal{F}_t = \bar{h}^f(y)$ for all $t \geq 0$. Observe that in these expressions, we have exploited that $\bar{h}^f = 1$. If we deviated from $\bar{h}^f = 1$, then we would need to set the exogenous exit rate for firms of type y to be $\varsigma \bar{h}^f(y)$ in order to keep the density of $g_t^f/\mathcal{F}_t$ constant.

Measure functions g_t^v and $\mathcal{V}_t$: The measures of vacancies g_t^v and total vacancies $\mathcal{V}_t$ then follow from

$$\begin{aligned} g_t^v(y) &= \bar{h}^f(y) \mathcal{F}_t - g_t^p(y), \\ \mathcal{V}_t &= \mathcal{F}_t - \mathcal{P}_t = \mathcal{F}_t - \iint g_t(x, y) dx dy, \end{aligned}$$

where $g_t^p(y) = \int g_t(x, y) dx$ is the measure of producing firms of type y and $\mathcal{P}_t = \iint g_t(x, y) dx dy$ is the total mass of producing firms.

E.2.2 Agent problems

Let $V^u(x, z, g, \mathcal{F})$ denote the value of an unemployed worker of type x , $V^e(x, y, z, g, \mathcal{F})$ denote the value of a worker of type x employed at a firm of type y , $V^v(y, z, g, \mathcal{F})$ denote the value of a vacancy for firm y , and $V^p(x, y, z, g, \mathcal{F})$ denote the value of a producing firm y employing a worker of type x . Let $(\check{\mu}^g, \check{\mu}^{\mathcal{F}})$ denote agent beliefs about the evolution of g and $\mathcal{F}$ respectively.

Unemployed workers: given beliefs and optimal acceptance decision α , the unemployed worker value function V^u solves the HJB equation:

$$\begin{aligned} \rho V^u(x, z, g, \mathcal{F}) &= b + \mathcal{M}^u \int \alpha(x, \tilde{y}, z, g, \mathcal{F}) (V^e(x, \tilde{y}, z, g, \mathcal{F}) - V^u(x, z, g, \mathcal{F})) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\ &+ \sum_{\check{z} \neq z} \lambda(z, \check{z}) (V^u(x, \check{z}, g, \mathcal{F}) - V^u(x, z, g, \mathcal{F})) \\ &+ \langle D_g V^u, \check{\mu}^g \rangle + \partial_{\mathcal{F}} V^u(x, z, g, \mathcal{F}) \check{\mu}^{\mathcal{F}}(z, g, \mathcal{F}) \end{aligned}$$

where α is an indicator function for whether workers and firms accept the match under generalized Nash bargaining.

Vacant firms: given beliefs and optimal decisions, the vacant firm value function V^v solves the HJB equation:

$$\begin{aligned} \rho V^v(y, z, g, \mathcal{F}) &= -c + \mathcal{M}^v \mathcal{C}^u \int \alpha(\tilde{x}, y, z, g, \mathcal{F}) (V^p(\tilde{x}, y, z, g, \mathcal{F}) - V^v(y, z, g, \mathcal{F})) \frac{g^u(\tilde{x})}{\mathcal{U}} d\tilde{x} \\ &+ \mathcal{M}^v \mathcal{C}^e \int \int \alpha^p(\tilde{x}, y, \tilde{y}, z, g, \mathcal{F}) (1 - \beta) (S(\tilde{x}, y, z, g, \mathcal{F}) - S(\tilde{x}, \tilde{y}, z, g, \mathcal{F})) \frac{g(\tilde{x}, \tilde{y})}{\mathcal{E}} d\tilde{x} d\tilde{y} \\ &+ \sum_{\check{z} \neq z} \lambda(z, \check{z}) (V^v(y, \check{z}, g, \mathcal{F}) - V^v(y, z, g, \mathcal{F})) - \varsigma V^v(y, z, g, \mathcal{F}) \\ &+ \langle D_g V^v, \check{\mu}^g \rangle + \partial_{\mathcal{F}} V^v(y, z, g, \mathcal{F}) \check{\mu}^{\mathcal{F}}(z, g, \mathcal{F}) \end{aligned}$$

where the $-\varsigma V^v$ term reflects that the firm exits at rate ς and receives a continuation value of zero. The indicator α^p for whether a poaching firm and an employed worker accept a match satisfies:

$$\alpha^p(\tilde{x}, y, \tilde{y}, z, g, \mathcal{F}) := \begin{cases} 1, & \text{if } S(\tilde{x}, y, z, g, \mathcal{F}) \geq S(\tilde{x}, \tilde{y}, z, g, \mathcal{F}) \text{ and } S(\tilde{x}, y, z, g, \mathcal{F}) \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

Employed matches: To characterize on-the-job search behavior, we again work with the value of an employed match rather than the individual value functions. The setup is the same as Section 4.2 except that now the aggregate state space includes $\mathcal{F}$. We denote the combined value of a match between a type x worker and a type y firm that has promised q by:

$$P(x, y, q, z, g, \mathcal{F}) := V^e(x, y, q, z, g, \mathcal{F}) + V^p(x, y, q, z, g, \mathcal{F}).$$

We further denote the combined value of the incumbent firm and worker after the type x worker moves to a new firm of type $\tilde{y}$ with promise $\tilde{q}$ by $\tilde{P}(x, y, \tilde{y}, q, \tilde{q}, z, g, \mathcal{F})$. Employed workers choose whether to accept matches from other firms and whether to leave their jobs to return to unemployment. So, given agent beliefs about the drift of the match density $\tilde{\mu}^g$, $P(x, y, q, z, g, \mathcal{F})$ solves the HJB equation:

$$\begin{aligned} \rho P(x, y, q, z, g, \mathcal{F}) &= F(x, y, z) \\ &+ (\delta(x, y, z) + \eta \alpha^b(x, y, z, g, \mathcal{F}))(V^u(x, z, g, \mathcal{F}) + V^v(y, z, g, \mathcal{F}) - P(x, y, q, z, g, \mathcal{F})) \\ &+ \varsigma(V^u(x, z, g, \mathcal{F}) - P(x, y, q, z, g, \mathcal{F})) \\ &+ \mathcal{M}^e \int \alpha^e(x, y, \tilde{y}, q, \tilde{q}, z, g, \mathcal{F})(\tilde{P}(x, y, \tilde{y}, q, \tilde{q}, z, g, \mathcal{F}) - P(x, y, q, z, g, \mathcal{F})) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\ &+ \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z})(P(x, y, q, \tilde{z}, g, \mathcal{F}) - P(x, y, q, z, g, \mathcal{F})) \\ &+ \langle D_g P, \check{\mu}^g \rangle + \partial_{\mathcal{F}} P(x, y, q, z, g, \mathcal{F}) \check{\mu}^{\mathcal{F}}(z, g, \mathcal{F}). \end{aligned}$$

where, as in Section 4.2, the change in joint value conditional on a move is:

$$\tilde{P}(x, y, \tilde{y}, q, \tilde{q}, z, g, \mathcal{F}) - P(x, y, q, z, g, \mathcal{F}) = \beta[S(x, \tilde{y}, z, g, \mathcal{F}) - S(x, y, z, g, \mathcal{F})].$$

Like in the main text, we can verify that $P(x, y, q, z, g, \mathcal{F})$ and surplus do not depend on q and so can be written as $P(x, y, z, g, \mathcal{F})$ and $S(x, y, z, g)$.

E.2.3 Master Equation For Surplus

Combining the HJB equations and imposing belief consistency $(\check{\mu}^g, \check{\mu}^{\mathcal{F}}) = (\mu^g, \mu^{\mathcal{F}})$, the surplus $S(x, y, z, g)$ satisfies the Master equation:

$$\begin{aligned}
& \rho S(x, y, z, g, \mathcal{F}) \\
&= F(x, y, z) - b + c - \left(\delta(x, y, z) + \eta \alpha^b(x, y, z, g, \mathcal{F}) + \varsigma \right) S(x, y, z, g, \mathcal{F}) \\
&- \mathcal{M}^v \mathcal{C}^u \int \alpha(\tilde{x}, y, z, g, \mathcal{F}) (1 - \beta) S(\tilde{x}, y, z, g, \mathcal{F}) \frac{g^u(\tilde{x})}{\mathcal{U}} d\tilde{x} \\
&- \mathcal{M}^v \mathcal{C}^e \int \int \alpha^p(\tilde{x}, y, \tilde{y}, z, g, \mathcal{F}) (1 - \beta) (S(\tilde{x}, y, z, g, \mathcal{F}) - S(\tilde{x}, \tilde{y}, z, g, \mathcal{F})) \frac{g(\tilde{x}, \tilde{y})}{\mathcal{E}} d\tilde{x} d\tilde{y} \\
&+ \mathcal{M}^e \int \alpha^e(x, y, \tilde{y}, z, g, \mathcal{F}) \beta (S(x, \tilde{y}, z, g, \mathcal{F}) - S(x, y, z, g, \mathcal{F})) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\
&- \mathcal{M}^u \int \alpha(x, \tilde{y}, z, g, \mathcal{F}) \beta S(x, \tilde{y}, z, g, \mathcal{F}) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\
&+ \sum_{\check{z} \neq z} \lambda(z, \check{z}) (S(x, y, \check{z}, g, \mathcal{F}) - S(x, y, z, g, \mathcal{F})) \\
&+ \langle D_g S, \mu^g \rangle + \partial_{\mathcal{F}} S(x, y, z, g, \mathcal{F}) \mu^{\mathcal{F}}(z, g, \mathcal{F}).
\end{aligned}$$

The key differences to (2.11) in the main text are that $\mathcal{F}_t$ is no longer a jump variable and $V^v(y, z, g, \mathcal{F})$ is no longer zero. These changes mean that the surplus and vacancy values end up being coupled. That is, the V^v HJB equation depends on S , while S depends on V^v through the entry condition α^f , which is itself a functional of V^v through $\int V^v(\tilde{y}, z, g, \mathcal{F}) \bar{h}^f(\tilde{y}) d\tilde{y}$. As neither equation is self-contained, S and V^v must be solved jointly, which increases the complexity of the numerical solution. Notice that now there are also additional ways in which block recursivity is broken because the master equation incorporates the opportunity cost of matching to firms now that $V^v(y, z, g, \mathcal{F})$ is no longer zero.

E.3 Numerical implementation

We use our DeepSAM approach from Subsubsection 2.5.2 to estimate our model with slow entry, as in Section 4 for our model with free entry. Two features are

new. First, we use two networks, one representing S and one representing V^v , and training minimizes the sum of the residuals of the surplus master equation and the vacancy HJB on shared samples of $(z, g, \mathcal{F})$. Second, we now need to jointly sample across g and $\mathcal{F}$ to train the augmented network. Table 15 shows the results of the estimation. Figure 20 shows the impulse responses for unemployment and firm mass in our estimated model with slow entry (blue) and our estimated model with free entry from Section 4 (orange). Evidently, the mass of firms is less responsive to the business cycle in the slow entry model compared to the model with free entry. This leads to unemployment being less responsive despite the slow entry model having a lower estimated β .

Parameter	Interpretation	Value	Fitted moment	Data	Model
β	Surplus division factor	0.68	$\mathbb{E}[U]$	0.058	0.058
κ	Scale for meeting function	13.63	$\mathbb{E}[V]$	0.037	0.036
c	Firm entry cost	1.90	$\mathbb{E}[EE]$	0.025	0.025
δ	Match separation rate	0.025	$\mathbb{E}[UE]$	0.421	0.411
ϕ	Employed worker search intensity	0.083	$\mathbb{E}[EU]$	0.025	0.025
ψ	Firm entry rate	0.31			
η	Endogenous separation intensity	0.95			

Table 15: Estimated Parameters and Targeted Moments.

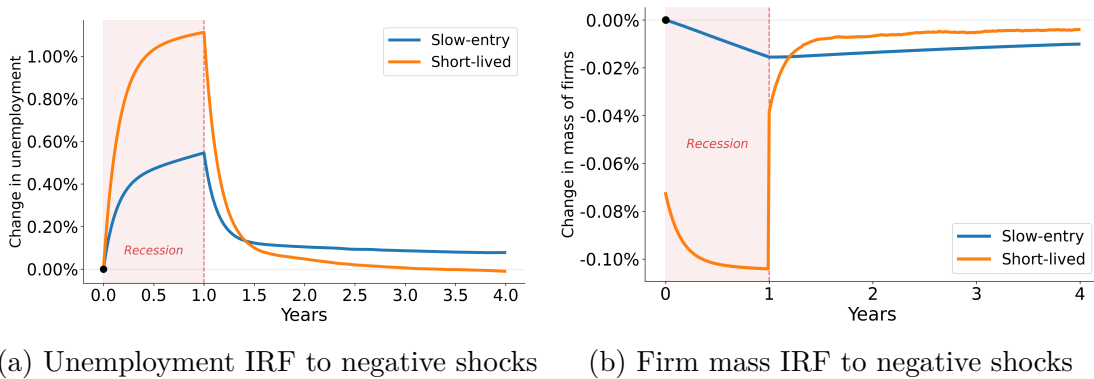


Figure 20: Impulse responses to a negative aggregate productivity z shock, comparing the ‘slow-entry’ model and ‘free-entry’ model with ‘short-lived’ vacancies.